Date: 20260701
Title: Colony PCR
Purpose: To verify bacterial colonies after a transformation step in molecular cloning
Procedure:

1. Prepare the following reaction (master mix) X 21 and aliquot solution into PCR tubes

	1 reaction	21 reactions
Forward Primer (T7 promoter)	1 µL	21 µL
Reverse Primer (SP6 promoter)	1 µL	21 µL
Q-Amp master mix	10 µL	210 µL
ddH_2O	8 µL	168 µL

2. Circle 3 colonies each from plates A to G

A: pGEM-Teasy-Csn: LEA
B: pGEM-Teasy-Csn: AmilCP
C: pGEM-Teasy-Csn: Accd
D: pGEM-Teasy-ccasm3
E: pGEM-Teasy-ccaR
F: pGEM-Teasy-Ho1: PcyA
G: pGEM-Teasy-pcpcG2-pveg

3. Pick up 3 colonies each agar plate with tip and streak the labelled grid
4. Dispense the tips with colonies into the PCR strips and wait for 5 mins
5. Take out the tips and close the lid
6. Place into PCR machine for the thermal cycle

Done by Naomi, Sarah
Recorded by Sophie H.

Date: 20260701
Title: 1.0% Agarose Gel preparation
Purpose: To prepare 1.0% Agarose Gel for further electrophoresis experiment
Procedure:

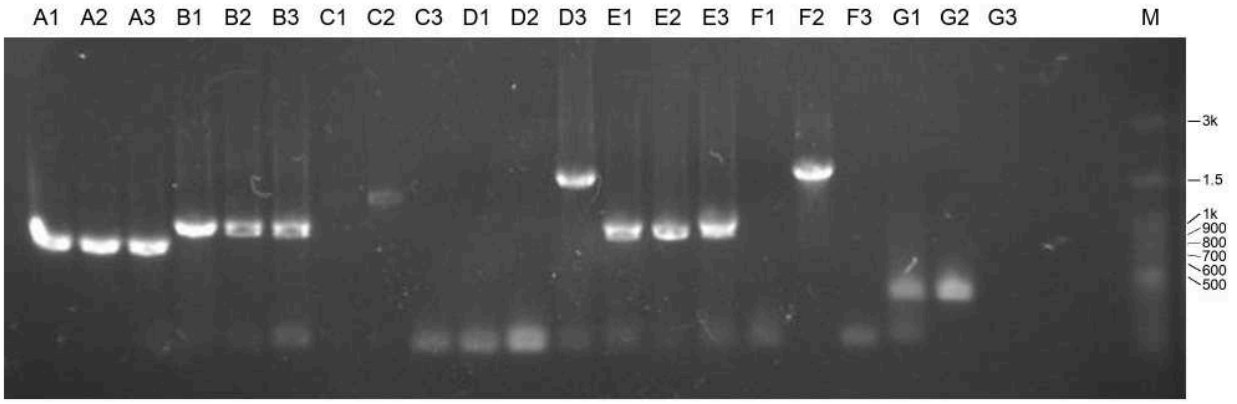
1. Prepare and assemble the mold and comb
2. Weight 0.4 g of agarose powder and add 40 ml of 1X TAE solution into flask
3. Cover the flask with plastic wrap, then use a tip to puncture holes on the plastic wrap

4. Microwave until the gel stops smoking but is still hot
5. Add 2 μL of safety dye into the gel and agitate it
6. Pour the melted gel into the tray
7. Wait until the gel cools down
8. Repeat steps 1~8 to create second copy of gel

Done by Sophie L., Ethan L.
Supervised by Chars
Recorded by Sophie H.

Date: 20260701
Title: Gel Electrophoresis
Purpose: To test the concentration of respective solutions
Procedure:

1. Load the gel in the layout shown below (5 μL each):
2. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
A: pGEM-Teasy-Csn:Lea	T7 Promoter	SP6 Promoter	811
B: pGEM-Teasy-Csn:AmilCP	T7 Promoter	SP6 Promoter	1000
C: pGEM-Teasy-Csn:Accd	T7 Promoter	SP6 Promoter	1372
D: pGEM-Teasy-CcaSm3	T7 Promoter	SP6 Promoter	1642
E: pGEM-Teasy-CcaR	T7 Promoter	SP6 Promoter	937
F: pGEM-Teasy-Ho1:pcyA	T7 Promoter	SP6 Promoter	1726
G: pGEM-Teasy-PcpcG2-Pveg	T7 Promoter	SP6 Promoter	412

Done by Sophie L., Ethan L.
Supervised by Chars
Recorded by Sophie H.

Date: 20260701
Title: Colony PCR
Purpose: To verify bacterial colonies after a transformation step in molecular cloning
Procedure:

1. Prepare the following reaction and dispense them into PCR tubes:

	B (4 rxn)	C(6 rxn)	E(6 rxn)	F(6 rxn)
Forward Primer (T7 promoter)	5 µL	7 µL	7 µL	7 µL
Reverse Primer (SP6 promoter)	5 µL	7 µL	7 µL	7 µL
Q-Amp master mix	50 µL	70 µL	70 µL	70 µL
ddH_2O	40 µL	56 µL	56 µL	56 µL

2. Circle colonies without fluorescence form B.C.E>F

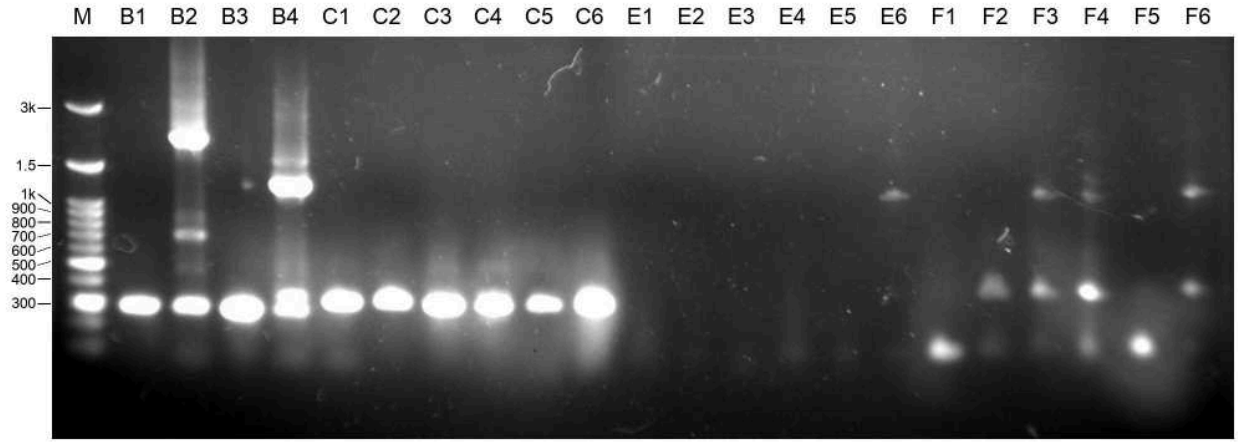
B: pSTK-1D-Pveg-EEBD-RBS0-Csn::ACCD-B0015
C: pSTK-1C-Pveg-RBS0-CcaR-B0015
E: pSTK-Exp2-Spacer x3-Pveg-EEBD-RBS0-Csn::AmilCP-B0015
F: pSTK-Exp1-Pveg-EEBD-RBS0-Spacer-GanA::Gala-histag-B0015

3. Pick up the 22 colonies with a tip and streak the labelled grid
4. Dispense the tips with colonies into PCR strip and wait 5 min
5. Take out the tips and spindown
6. Place into PCR machine and run Q-Amp thermal cycle

Done by: Chloe, Sophie L, Ethan
Supervised by: Chars

Date: 20260701
Title: Gel Electrophoresis
Purpose: verify the size of colony PCR products
Procedure:

1. Load the gel in the layout shown below (5 µL for M; 6 µL for B, C, D, F):
2. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
B: pSTK-1C-Pveg-EEBD-RBS0-Csn:AmilCP-B0015	STK-fw	STK-rv	2056
C: pSTK-Exp2-Pveg-RBS0-CcaR-B0015	STK-fw	STK-rv	1442
E: pSTK-Exp2-Pveg-spacer x3 - Pveg-EEBD-RBS0-Csn: AmilCP-B0015	EXP-fw	EXP-rv	1833
F: pSTK-Exp1-Pveg-EEBD-RBS0-spacer GanA-Gala-histag-B0015	EXP-fw	EXP-rv	2993

Done by Sophie H.

Date: 20260701
Title: Transformation of TA cloning
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:

1. Turn on water bath and set it to 42 °C and prepare an ice bucket
2. Mix 100 µL of competent cell with 10 µL plasmid
3. Incubate on ice for 20 min
4. Prepare reaction for blue-white screening → 1000 µL of ddH_2O and 4 µL of x-gal & 1µL of 1M IPTG, then spread on LBA plate
5. Heat shock at 42 °C for 45 sec
6. Incubate on ice for 3 min

7. Add 1ml of SOC medium to each microtube and incubate at 37 °C for 1 hour
8. Centrifuge at 5000 g for 5 min and remove 1 ml of supernatant
9. Resuspend the cell pallet in the remaining supernatant & spread onto LBA plate

A: Csn:LEA
B: Csn:ACCD
C: PcpcG2-veg
D: Csn:AmilCP
E: Ho1:pcyA
F: CcaR
G: Csn:AmilCP
H: Ccasm3

Done by Sophie L, Chloe, Ethan L
Supervised by Chars

Date: 20260702
Title: Chlorophyll Extraction
Purpose: To isolate the green photosynthetic pigment from arabidopsis
Procedure:

1. Weight the fresh weight of arabidopsis (9 at a time)
2. Take the plant from the agar plate, careful not to hurt it by squeezing the tweezers.
3. Put the 9 plants on a tissue paper, lightly press to remove the agar striking to the plates
4. Pick up the plants and put them on a weighting plate to weight
5. Weight the plants
6. Pick up the plants and put them in a microtube, take care that the roots need to go in too
7. Repeat these steps for as many times as the sets of arabidopsis you are testing
8. Place 3 glass bits in each of the microtubes, place in -80 ℃ for 10 min
9. Dilute acetone to 80 %, add 1 mL to each tube (10 mL in total)
10. Wait for the chlorophyll to extract, and can use a vortex to crush the plants
11. Add the liquid to the circus well, 200 µL in each well, 3 wells for each sample
12. ELISA reader insert the 96 well, run the result through the equation below:

*Total chlorophyll content equation:
(20.2*OD645+*.02*OD663)V/(1000*fw)

Done by Phoebe, Abby T.
Recorded by Sophie H.

Date: 20260702
Title: Colony PCR
Purpose: To verify bacterial colonies after a transformation step in molecular cloning
Procedure:

1. Prepare the following reactions and dispense them into PCR tubes

	Forward Primer	Reverse Primer	Q-Amp master mix	dd H_2O
C (4 rxn)	T7 promoter 5 µL	Sp6 promoter 5 µL	50 µL	40 µL
D (4 rxn)	T7 promoter 5 µL	Sp6 promoter 5 µL	50 µL	40 µL
F1 (4 rxn)	T7 promoter 5 µL	Sp6 promoter 5 µL	50 µL	40 µL
G (4 rxn)	T7 promoter 5 µL	Sp6 promoter 5 µL	50 µL	40 µL
E (6 rxn)	pSTK-Exp fw 7 µL	pSTK-Exp rv: 7 µL	70 µL	56 µL

F2-1 (6 rxn)	pSTK-Exp fw: 7 µL	pSTK-Exp rv: 7 µL	70 µL	56 µL
F2-2 (11 rxn)	pSTK-Exp fw: 7 µL	pSTK-Exp rv: 12 µL	100 µL	96 µL

2. Circe colonies without fluorescence from C, D, F, G, E, F2-1, F2-2

C: pGEM-Teasy-Csn: Accd
D: pGEM-Teasy-ccasm3
F1: pGEM-Teasy-Ho1:PcyA
G: pGEM-Teasy-pcpcG2-pveg
E: spacer*3-pveg-EEBD-RBS0-Csn:AmilCP-B0015 pExp2
F2-1: 20260701pveg-EEBD-RBS0-spacer-Gana-Gala-histag-B0015 pExp1
F2-2: 20260630pveg-EEBD-RBS0-spacer-Gana-Gala-histag-B0015 pExp1

3. Pick up the colonies with a tip and streak the labeled grid
4. Dispense the tip with colonies into PCR strip and wait for 5 min
5. Take out the tips and spindown
6. Place into PCR machine and run Q-amp thermal cycle

Done by Chloe, Alex, Ryan Wei
Supervised by Ryan Yuan, Olivia
Record by Sophie H.

Date: 20260702
Title: Liquid Culture
Purpose: To grow bacteria inside liquid culture from agar plate
Procedure:

1. Label 9 liquid culture tubes
2. Turn on bunsen burner
3. Add 5 µL of TB medium and 5 µL of ampicillin into each tube
4. Dip a pipette tip into the colonies
5. Dispense tips in their respective tubes
6. Place the tubes in the incubator at a slanted angle

Done by Alex
Supervised by Chars
Record by Sophie H.

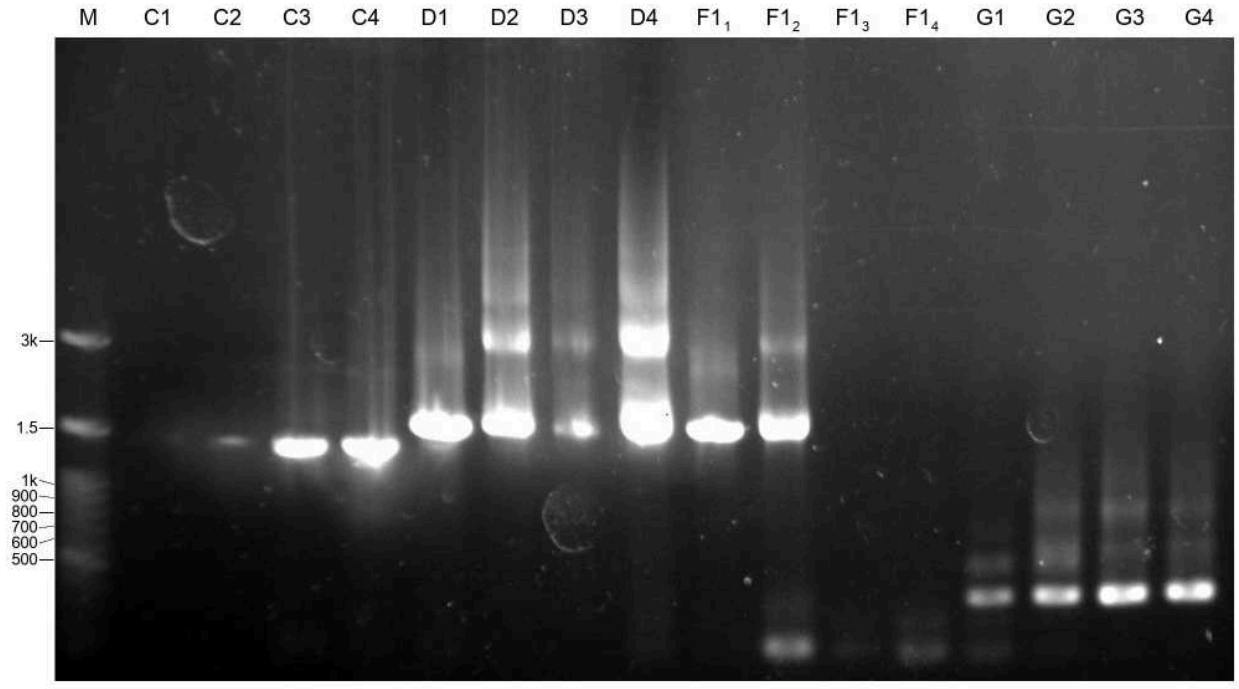
Date: 20260703
Title: Prepare 0.8% Agarose Gel
Purpose: Make gel for electrophoresis
Procedure:

1. Prepare and assemble mold and comb
2. Weigh 0.16 g agarose powder and add 80 mL of 1X TAE solution into flask
3. Cover the flask with plastic wrap, then use a tip to puncture holes on the plastic wrap.
4. Microwave until all particles are dissolved
5. Wait until the melted gel stops smoking but is still hot.
6. Add 4 µL of safety dye and agitate it
7. Pour the gel two trays with equal amount
8. Wait until the gel cools down

Done by Alex
Supervised by Chars

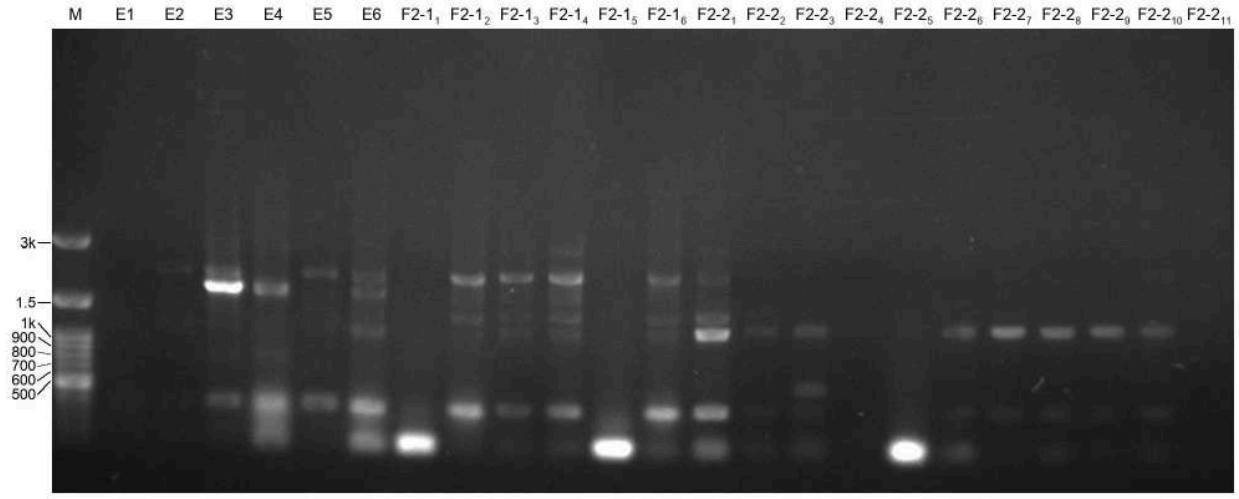
Date: 20260703
Title: Gel Electrophoresis
Purpose: To test the concentration of respective solutions
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)

C: pGEM-Teasy-Csn:Accd	T7 Promoter	SP6 Promoter	1372
D: pGEM-Teasy-CcaSm3	T7 Promoter	SP6 Promoter	1642
F1: pGEM-Teasy-Ho1:PcyA	T7 Promoter	SP6 Promoter	1726
G: pGEM-Teasy-PcpcG2-veg	T7 Promoter	SP6 Promoter	412



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
E: spacer * 3 - pveg-EEBD-RBS0-Csn:AmilCP -B0015 pExp2	EXP-fw	EXP-rv	1833
F2-1: 20260701 pveg-EEBD-RBS0-spacer-Gan a-Gala-histag-B0015 pExp1	EXP-fw	EXP-rv	2993
F2-2: 20260630 pveg-EEBD-RBS0-spacer-Gan a-Gala-histag-B0015 pExp1	EXP-fw	EXP-rv	2993

Done by Alex
Supervised by Chars

Date: 20260703
Title: Miniprep
Purpose: Plasmid purification

Procedure:

A1: pGEM-Teasy-Csn:LEA
B1: pGEM-Teasy-Csn:AmilCP
E1: pGEM-Teasy-CcaR
A2: pGEM-Teasy-Csn:LEA
B2: pGEM-Teasy-Csn:AmilCP
E2: pGEM-Teasy-CcaR
A3: pGEM-Teasy-Csn:LEA
B3: pGEM-Teasy-Csn:AmilCP
E3: pGEM-Teasy-CcaR

1. Transfer 1.8 mL of medium to 2 mL microtube
2. Centrifuge at 13444 g 30 seconds
3. Discard the supernatant
4. Add 200 µL of MX1 buffer into each microtube
5. Resuspend the pellet through pipetting
6. Add 250 µL of MX2 buffer
7. Gently mix by inverting microtube 8-10 times
8. Incubate at room temp for 2 mins
9. Add 350 µL of MX3 buffer
10. Gently mix by inverting microtube 4-6 times
11. centrifuge for 5 minutes at 16000 g
12. Transfer supernatant (~800µL) into assembled column
13. Centrifuge for 30 seconds at 7000 g and discard flow through
14. Wash the column with 500µL WN buffer and centrifuge for 30 seconds at 7000 g and discard the ft
15. Wash the column with 700 µL WS buffer and centrifuge for 30 seconds at 7000 g and discard the ft
16. Place the column into new 1.5 mL microtube
17. Add 50 µL of ddH_2O to the center of membrane and let it stand for 2-3 min and centrifuge for 3 min at 10,000 g

Result:

Biodrop Results	**Concentration ng/µL**	**A260/230**	**A260/280**
A1	77	1.167	1.925
A2	60	1.500	1.935
A3	65	1.083	1.857
B1	55	1.375	1.897
B2	67	1.241	1.861
B3	49	1.195	1.885
E1	71	1.511	1.972

E2	77	1.481	2.026
E3	66	1.245	1.886

Done by: Abigail and Chloe

Date: 20260703
Title: Liquid culture
Purpose: Let bacteria grow in liquid medium
Procedure:

1. Label 7 liquid culture tubes
2. Turn on bunsen burner
3. Add 5 mL of TB medium and 5 µL of ampicillin into 6 tubes
4. Add 5 mL of TB medium and 5 µL of chloramphenicol into 1 tubes (E3)
5. Dip a pipette tip into the colonies
6. Dispense tips into their respective tubes
7. Place tubes in the incubator at a slanted angle

Done by: Alex & Sophie C
Supervised by: Chars

Date: 20260704
Title: Miniprep
Purpose: extract plasmid DNA from liquid culture of colony PCR products
Procedure:
0701 E3: pSTK-Exp2- spacer ×3 - pveg - EEBD - RBS0 - Csn:AmilCP - B0015
0701 F2: pGEM -Teasy-Ho1:pcyA
0701 F4: pSTK-Exp1-Pveg-EEBD-RBS0-spacer-GanA-Gala-histag-B0015
0702 C3: pGEM-Teasy-Csn:ACCD
0702 D1: pGEM-Teasy-Ccasm3
0702 F1: pGEM-Teasy-Ho1:pcyA
0702 G3: pGEM-Teasy-PcpcG2-veg

1. Transfer 1.8 µL TB medium to 2 mL microtube.
2. Centrifuge at 16.000 g for 1 min.
3. Discard the supernatant.
4. Repeat steps 2-4 until collected all medium.
5. Add 200 µL MX1 buffer.
6. Resuspend the pellet by vortexing or pipetting.
7. The cells should completely resuspend.
8. Add 250 µL MX2 buffer.
9. Gently mix the buffer well.
10. Do not vortex. Invert microtube 8-10 times.
11. Incubate the mixture at room temperature for 2 min.
12. Add 350 µL MX3 buffer.
13. Immediately and gently mix the solution. (Do not vertex.)
14. Centrifuge at 16,000 g for 5 min.
15. While centrifuging, assemble the column and collection tube.
16. Transfer the supernatant, around 800 µL, to the column.
17. Centrifuge at 7,000 g for 30 sec.
18. Discard the flow through.
19. Add 500 µL WN buffer.
20. Centrifuge at 7,000 g for 30 sec.
21. Discard the flow through.
22. Add 500 µL WS buffer.
23. Centrifuge at 7,000 g for 30 sec.
24. Discard the flow-through.
25. Centrifuge the empty column at 10,000 g for 3 min to remove residual ethanol.
26. Place the column into a new 1.5 mL microtube.
27. Add 50 µL sterile distilled water on the center of the membrane.
28. Incubate the column at room temperature for 3 min.
29. Centrifuge at 10,000 g for 3 min.

	Concentration (ng/μL)	A260/230	A260/280
E3	130	0.802	1.831
F2	77	0.987	1.878
F4	93	0.959	1.860
C3	142	0/940	1.844
D1	128	0.921	1.910
F1	136	0.845	1.838
G3	129	0.816	1.817

Done by Sarah, Sophie H

Date: 20260704
Title: Liquid Culture
Purpose: Bacteria culture
Procedure:

1. Label 4 liquid culture tubes
2. Turn on bunsen burner
3. Add 5mL of TB medium and 5 μL of ampicillin into each tubes
4. Dip a pipette tip into the colonies
5. Dispense tips into their respective tubes
6. Place tubes in the incubator at a slanted angle

Done by Chloe, Jacquelyn
Supervised by Alex

Date: 20260704
Title: Making salinity agar plates
Purpose: prepare salinity agar plates for plant model Experimental Set #5 0, 1, 5, 10 mM trehalose vs. 150, 250 mM NaCl
Procedure:
The following protocol is for 600 ml of agar

- 12 agar plates
    - Control (no trehalose) x control (no NaCl)
    - Control (no trehalose) x 150 mM NaCl
    - Control (no trehalose) x 250 mM NaCl
    - 1 mM trehalose x control (no NaCl)
    - 1 mM trehalose x 150 mM NaCl
    - 1 mM trehalose x 250 mM NaCl
    - 5 mM trehalose x control (no NaCl)
    - 5 mM trehalose x 150 mM NaCl

- 5 mM trehalose x 250 mM NaCl
- 10 mM trehalose x control (no NaCl)
- 10 mM trehalose x 150 mM NaCl
- 10 mM trehalose x 250 mM NaCl

1. Prepare the following:

1/2 MS medium powder (w/o sucrose)	1.38 g
MES buffer	0.3 g
Distilled water	600 ml
Sucrose(1%)	6 g
Agar powder(1%)	6 g

2. Add the contents into a 1000 ml flask
3. Use a hot plate to stir (with a magnetic stirring rod) until no sediments are left in the solution
4. Use the pH probe and measure the pH of the solution, make sure the pH is 5.7-5.8
    a. If the pH is too low, add 0.1 M KOH slowly to increase it
5. Pour 200 ml of mixture out into another flask to separate into three flasks, label with [0 mM], [150 mM] and [250 mM]
6. Add 2.192 g of NaCl into the flask labelled [150 mM NaCl]
7. Add 3.652 g of NaCl into the flask labelled [250 mM NaCl]
8. Prepare 4 flasks for different trehalose concentrations(0, 1, 5, 10 mM)
9. Prepare 12 flasks and pour the respective trehalose and NaCl concentration in each of the flasks
    a. making sure there is 50 ml in each of the flasks (12)
    b. label each flask clearly according to their trehalose concentrations
10. To their respective trehalose concentrations, add

Trehalose concentration	Amount of trehalose (for 50ml of medium)
0 mM	0 µL
1 mM	500 µL
5 mM	2500 µL
10 mM	5000 µL

11. Wrap the necks of the flasks with aluminum foil and stick a piece of indicator tape on top
12. Autoclave for 20 minutes at 127 °C

    a. Wait for the pressure to reach 0 before retrieving the flask
13. During the autoclave, label 8 of the petri dishes with a marker in the biosafety cabinet
    a. right top: date and arrow pointing up to show direction for pictures
    b. bottom lid: different trehalose concentrations and NaCl concentration or control
14. Transfer half of the flasks to BSC
    a. Store the other half of the flasks in the hot bath while pouring so that it does not solidify
15. Make sure its surface is dry by wiping the flasks with a paper towel before spraying 70 % EtOH on them
16. Pour each of the medium mixtures into their square petri dishes (around 50ml)
17. Transfer the other half of the flasks into the BSC
18. Pour each of the medium mixtures into their square petri dishes
19. Let the petri dishes solidify at room temperature under the BSC

Done by Abby K, Abby T, Audrey H

Date: 20260705
Title: Colony PCR
Purpose: Amplify part from 6/29 and screen bacterias for successful DNA inserts
Procedure:

1. Prepare the following reaction (master mix) x22 reactions and dispense solutions

Into PCR tubes	1 rxn	22 rxn
Forward primer (Exp seq-fw 10 mM)	1 µL	22 µL
Reverse primer (Exp-rv 10 mM)	1 µL	22 µL
Q-Amp master mix	10 µL	220 µL
ddH_2O	8 µL	176 µL

2. Circle 21 colonies each from plate C (Exp-spacer-spacer-spacer-Pveg-EEBD-RBS0-Csn:AmilCP-B0015)
3. Pick up 21 colonies from each agar plate with tip and streak the labelled grid
4. Dispense the tips with colonies into the PCR strips and wait 5 minutes
5. Take out the tips and close the lid
6. Place into PCR machine for thermal cycle

95 °C	2 min	1 cycle
95 °C	30 secs	30 cycles
65 °C	4 secs	30 cycles
72 °C	2 min (bp 1800)	30 cycles
72 °C	1 min	1 cycle

Done by Chloe, Abigail, Abby K
Supervised by Gabriel

Date: 20260705
Title: Miniprep
Purpose: plasmid purification
Procedure:
A: Pveg-EEBD-RBS0-Csn:AmilCP-B0015
B: PcpcG2-veg-EEBD-RBS0-Csn:AmilCP-B0015
C: Pveg-RBS0-CcaR-B0015 (B2)

1. Transfer 1.8 mL of medium to 2 mL microtube
2. Centrifuge for 30 sec at 23444 g

3. Discard the supernatant
4. Repeat steps 2-3 until all medium is collected.Remember to empty the medium in the pipette tip from the medium tube
5. Add 200 µL of MX1 buffer
6. Resuspend by vortexing or pipetting
7. Add 250 µL of MX2 buffer
8. Invert slowly 8-10 times. Incubate at room temp for 2 minutes
9. Add 350 µL of MX3 buffer
10. Invert slowly 4-5 times
11. Centrifuge for 5 min at 16000 g
12. Transfer around 800 µL of supernatant into the assembled column
13. Centrifuge at 7000 g for 30 seconds and discard the flow through
14. Wash the column with 500 microliters of WN buffer. Centrifuge at 7000 g for 30 seconds and discard the flow through
15. Wash the column with 700 µL of WS buffer. Centrifuge at 7000 g for 30 seconds and discard the flow through
16. Centrifuge for 8 minutes at 10,000 g
17. Place column into new 1.5 mL microtube
18. Add 50 microliters ddH_2O to the center of membrane
19. Incubate for 3 min at room temp
20. Centrifuge at 10,000 g for 3 min

Results:

Samples	Concentration (ng/microliter)	A260/A230	A260/A280
A	331.0	1.623	2.122
B	473.0	1.708	2.131
C	147.0	1.157	1.986

Done by Chloe, Jacquely, Noah

Date: 20260705
Title: Make 1.0 % agarose gel
Purpose: making gel for electrophoresis
Procedure:

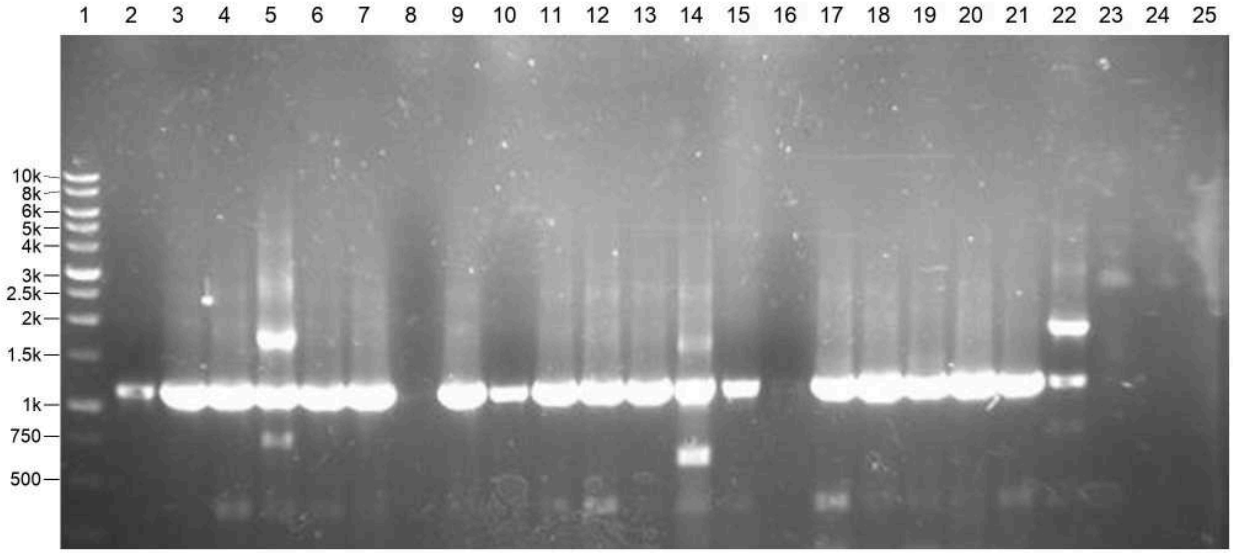
1. Assemble the tray and tray and mold and prepare the comb
2. Weigh 0.4 g agarose powder and 40 mL 1x TAE solution
3. Add 1x TAE and powder to the 100 mL conical flask
4. Cover the flask with plastic wrap using a tip to puncture holes on it
5. Microwave for 30 sec then take out and agitate it
6. Microwave for 10 sec and agitate it. Repeat this step until there is no particles left in the gel
7. Wait until the melted gel stops smoking but is still hot

8. Add 2 microliters of safety dye and agitate it
9. Pour the melted gel into the tray and assemble the comb
10. Leave until solidify in room temp

Done by Abby K, Abigail

Date: 20260705
Title: Gel electrophoresis
Purpose: verify colony PCR product size and miniprep product size
Procedure:

1. Load the gel in the layout shown below (3 µL for M; 5 µL for E; 5 µL plus 1 µl dye for A, B, C):
2. Resolve at 100 v for 40 min



M: 1KB DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
E: EXP2-spacer-spacer-spacer-Pveg-EEBD-RBS0-Csn::AmilCP-B0015	Exp-seq-fw	Exp-seq-rv	1833
A: Pveg-EEBD-RBS0-Csn:AmilCP-B0015			3585
B: PcpcG2-veg-EEBD-EBS0-csn::AmilCP-B0015-Amp			3485
C: Pveg-RBS0-CcaR-B0015			3243

Done by Abby K, Sophie H
Supervised by Gabriel

Date: 20260705
Title: Liquid Culture
Purpose: bacteria culture
Procedure:

1. Label 7 liquid culture tubes
2. Turn on bunsen burner
3. Add 3 mL of TB medium and 3 µL of ampicillin into 4 tubes; 3 µL of chloramphenicol into 3 tubes
4. Dip a pipette tip into the colonies
5. Dispense tips into their respective tubes
6. Place tubes in the incubator at a slanted angle

Done by Jacquelyn and Chloe
Supervised by Chars

Date: 20260706
Title: Colony pcr(redone)
Purpose: amplify parts from 6/29 and screen bacterias for successful DNA inserts
Procedure:

1. Prepare the following reaction(master mix) x 12 reactions and dispense solutions into PCR tubes

For A construct(Pveg-RBS0-Ho1:pcyA-B0015)

	1 reaction	13 reactions(for pipetting error)
Forward primer: pSTK-seq-fw 10mM	1 µL	13 µL
Reverse primer: pSTK-seq-rv 10mM	1 µL	13 µL
Q amp master mix	10 µL	130 µL
ddH_2O	8 µL	104 µL

For F construct(Pveg-EEBD-RBS0-GanA-Gala-B0015)

	1 reaction	13 reactions(for pipetting error)
Forward primer: pEXP-seq-fw 10mM	1 µL	13 µL
Reverse primer: pEXP-seq-rv 10mM	1 µL	13 µL
Q-Amp master mix	10 µL	130 µL
ddH_2O	8 µL	104 µL

2. Circle 12 colonies each from plate A and 3 colonies (F3, F4, F7) from plate F (chose the 3 colonies from F because others are empty)
    a. A: Pveg-RBS0-Ho1:pcyA-B0015
    b. F: Pveg-EEBD-RBS0-GanA-Gala-B0015
3. Pick up 12 colonies from A and 3 from F with tip and dispense in the PCR tubes(take colonies from incubated plates)
4. Take out tips and close the lid
5. Place into PCR machine for thermal cycle

95 °C	2 min	1 cycle
95 °C	30 sec	30 cycles
65 °C	40 sec	

72 °C	2.5 min	
72 °C	1 min	1 cycles

Done by Chloe, Abby K, Abigail

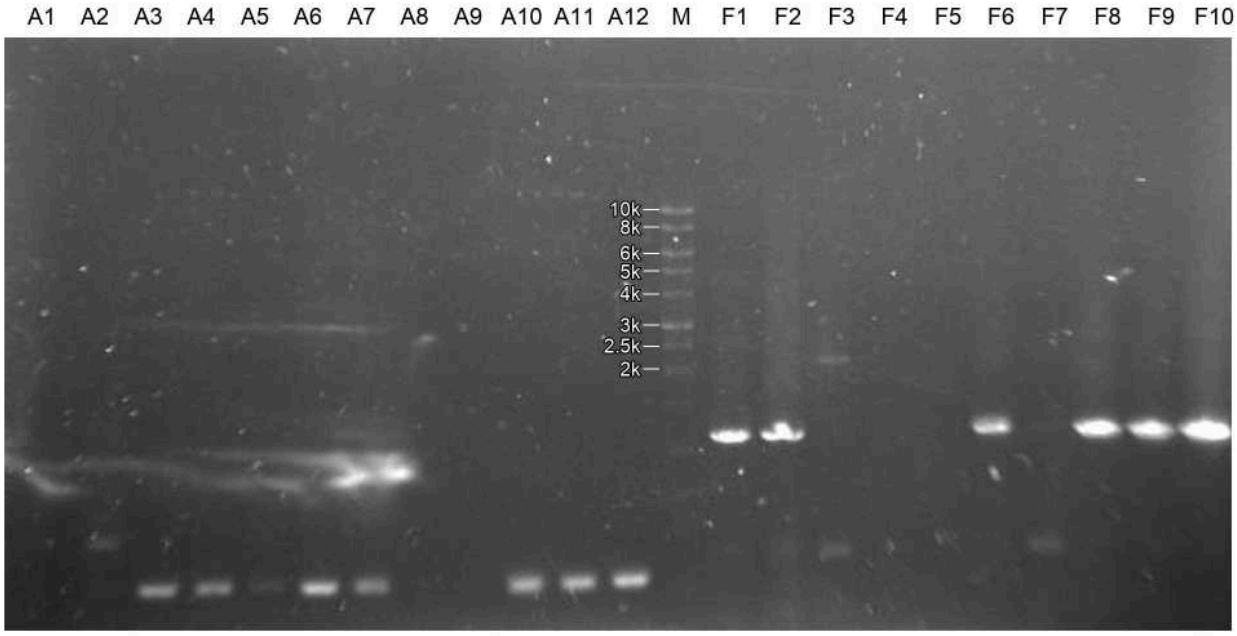
Date: 20260706
Title: Prepare 0.8 % Agarose gel for DNA electrophoresis (x2; one redone)
Purpose: DNA electrophoresis for A:Pveg–RBS0-Ho1:pcyA-B0015 and F:Pveg-EEBD-RBS0-GanA-Gala-B0015
Procedure:

1. Assemble the tray and tray and mold and prepare the comb
2. Weigh 0.32 g agarose powder and 40 mL 1x TAE solution
3. Add 1x TAE and powder to the 100 mL conical flask
4. Cover the flask with plastic wrap using a tip to puncture holes on it
5. Microwave for 30 sec then take out and agitate it
6. Microwave for 10 sec and agitate it. Repeat this step until there is no particles left in the gel
7. Wait until the melted gel stops smoking but is still hot
8. Add 2 microliters of safety dye and agitate it
9. Pour the melted gel into the tray and assemble the comb
10. Leave until solidify in room temperature

Done by Abigail, Abby T

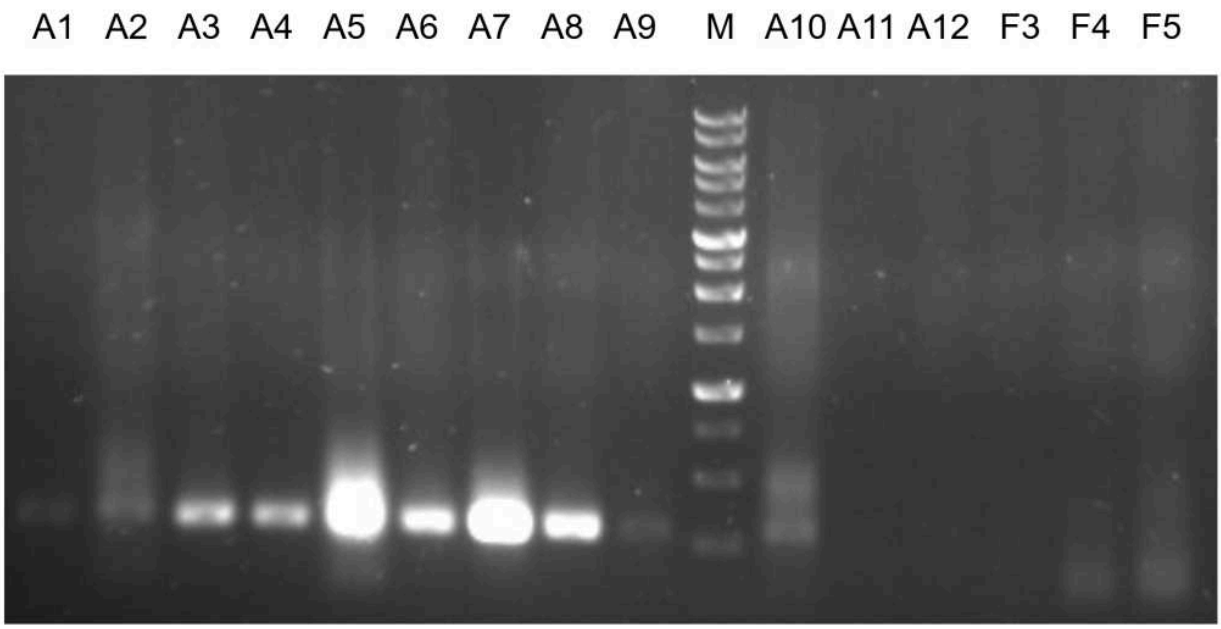
Date: 20260706
Title: Gel electrophoresis for 0705 colony pcr
Purpose: validate the bps of pcr results
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 40 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
A: Pveg–RBS0-Ho1:pcyA-B0015	pSTK-seq-fw	pSTK-seq-rv	2231
F: Pveg-EEBD-RBS0-GanA-Gala-B0015	EXP-seq-fw	EXP-seq-rv	2993

3. Redid the previous gel electrophoresis



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
A: Pveg–RBS0-Ho1:pcyA-B0015	pSTK-seq-fw	pSTK-seq-rv	2231
F: Pveg-EEBD-RBS0-GanA-Gala-B0015	EXP-seq-fw	EXP-seq-rv	2993

Done by Abigail, Abby T

Date: 20260706
Title: Miniprep
Purpose:
Procedure:
C4: Exp2-spacer3x Pveg-EEBD-RBS0-Csn:AmilCP-B0015
C21: Exp2-spacer3x Pveg-EEBD-RBS0-Csn:AmilCP-B0015
C13: Exp2-spacer3x Pveg-EEBD-RBS0-Csn:AmilCP-B0015
B2: Pveg-RBS0-CcaR-B00153
G3: PcpcG2-veg-EEBD-RBS0-Csn:AmilCP-B0015
H10: Pveg-EEBD-RBS0-Csn:AmilCP-B0015

1. Transfer 1.8 mL of medium to 2 mL microtube
2. Centrifuge for 30 sec at 23444 g
3. Discard the supernatant

4. Repeat steps 2-3 until all medium is collected.Remember to empty the medium in the pipette tip from the medium tube
5. Add 200 microliters of Mx1 buffer
6. Resuspend by vortexing or pipetting
7. Add 250 microliters of Mx2 buffer
8. Invert slowly 8-10 times. Incubate at room temp for 2 minutes
9. Add 350 microliters of Mx3 buffer
10. Invert slowly 4-5 times
11. Centrifuge for 5 min at 16000 g
12. Transfer around 800 microliters of supernatant into the assembled column
13. Centrifuge at 7000 g for 30 seconds and discard the flow through
14. Wash the column with 500 microliters of WN buffer. Centrifuge at 7000 g for 30 seconds and discard the flow through
15. Wash the column with 700 microliters of WS buffer. Centrifuge at 7000 g for 30 seconds and discard the flow through
16. Centrifuge for 8 minutes at 10,000 g
17. Place column into new 1.5 mL microtube
18. Add 50 microliters ddH_2O to the center of membrane
19. Incubate for 3 min at room temp
20. Centrifuge at 10,000 g for 3 min

The results are measured by Biodrop

Samples	Concentration(ng/microliter)	A260/A280
C4	88	1.833
C13	90	1.837
C21	58	1.871
B2	134	1.861
G3	168	1.846
H10	184	min

Done by Sophie C and Sophia Y

Date: 20260706
Title: making gel for gel electrophoresis (0.8% agarose gel)
Purpose: Validate the miniprep results
Procedure:

1. Assemble tray and mold, prepare the comb
2. Weigh 0.16 g agarose powder and 20 mL of 1x TAE solution
3. Add 1x TAE and powder to 100 mL conical flask
4. Cover flask with the plastic wrap and puncture holes on top with a tip

5. Microwave for 30 seconds, take it out and agitate it
6. Microwave 1o seconds and agitate and repeat until no particles are left in the melted gel
7. Wait until gel stops smoking and add 1 microliters of safety dye and agitate it
8. Pour the melted gel in the tray and assemble the comb
9. Leave to solidify in room temperature

Done by Sophie C

Date: 20260706
Title: Golden gate assembly
Purpose: assemble level 1 genetic circuit
Procedure:

1. prepare the reaction as follow:

A: (STK) Module 1(1A): Pveg–RBS0-Ho1:pcyA-B0015

Part	Original Amount	Dilution
Vector (1A) 115 ng/μL, 3120 bp	0.176348 μL	1.8 μL (dilute 1/10)
Insert promoter (Pveg) 280 ng/μL, 2361 bp	0.109618 μL	1.1 μL (dilute 1/10)
Insert RBS(RBS0) 188 ng/μL, 2113 bp	0.1461117 μL	1.5 μL (dilute 1/10)
Insert CDS(Ho1::PcyA) 238 ng/μL, 1556 bp	0.0849916 μL	1.7 μL (dilute 1/20)
Insert Terminator (B0015) 107 ng/μL, 2213 bp	0.268869 μL	1.3 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water	15.2140617 μL	8.6 μL
T4 DNA ligase	1 μL	1 μL
BsaI-HFv2	1 μL	1 μL

Total	20 µL	20 µL

Use the thermocycler to run the program below:

Temperature	Time	Cycles
37°C (Restriction enzyme digestion)	5 minutes	30 cycles
16°C (Ligation)	5 minutes	
37°C (Restriction enzyme digestion)	10 minutes	1 cycle
80°C (Heat inactivation of BsaI restriction enzyme)	20 minutes	1 cycle

B: (STK) Module 4-Pveg - EEBD- RBS0 - CSN::ACCD - B0015

Vector (Exp1) ● concentration (ng/µL): 162 ● bp: 6929	0.278015 µL	1.4 µL (dilute 1/5)
Insert promoter (Pveg) ● concentration(ng/µL): 280 ● bp: 2361	0.109618 µL	1.1 µL (dilute 1/10)
Insert RBS(EEBD-RBS0) ● concentration(ng/µL): 150 ● bp: 2392	0.207306 µL	1.0 µL (dilute 1/5)
Insert CDS(Csn::ACCD) ● concentration(ng/µL): 105 ● bp: 1195	0.147952 µL	1.5 µL(dilute 1/10)
Insert Terminator (B0015) ● concentration(ng/µL): 107 ● bp: 2213	0.268869 µL	1.3 µL (dilute 1/5)
T4 DNA ligase buffer	2 µL	2 µL
Sterile water		8.6 µL

T4 DNA ligase	1 µL	1 µL
BsaI-HFv2	1 µL	1 µL
Total	20 µL	20 µL

Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	5 minutes	30 cycles
16 °C (Ligation)	5 minutes	
37 °C (Restriction enzyme digestion)	10 minutes	1 cycle
80 °C (Heat inactivation of BsaI restriction enzyme)	20 minutes	1 cycle

C: (STK) Module 4-Pveg - EEBD- RBS0 - Csn::LEA - B0015

Vector (Exp1) ● Concentration (ng/µL): 162 ● bp: 6929	0.278015 µL	1.4 µL (dilute 1/5)
Insert promoter(Pveg) ● concentration (ng/µL): 280 ● bp: 2361	0.109618 µL	1.1 µL (dilute 1/10)
Insert RBS (EEBD-RBS0) ● Concentration (ng/µL): 150 ● bp: 2392	0.207306 µL	1.0 µL (dilute 1/5)
Insert CDS (Csn::LEA) ● Concentration (ng/µL): 77 ● bp: 634	0.05351948051 µL	1.1 µL (dilute 1/20)
Insert Terminator (B0015) ● Concentration (ng/µL): 107 ● bp: 2213	0.268869 µL	1.3 µL (dilute 1/5)

T4 DNA ligase buffer	2 µL	2 µL
Sterile water		10.0 µL
T4 DNA ligase	1 µL	1 µL
BsaI-HFv2	1 µL	1 µL
Total	20 µL	20 µL

Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	5 minutes	30 cycles
16 °C (Ligation)	5 minutes	
37 °C (Restriction enzyme digestion)	10 minutes	1 cycle
80 °C (Heat inactivation of BsaI restriction enzyme)	20 minutes	1 cycle

D: (STK-EXP1) Pveg-EEBD-RBS0-SamyQ::TAT-HA::AmilCP-B0015

Part	Original Amount	Dilution
Vector (shuttle vector): pSTK-EXP-1-sfGFP ● concentration(ng/µL): 162 ● bp: 6929	0.278015 µL	1.4 µL (dilute 1/5)
Insert promoter(Pveg) ● concentration(ng/µL): 280 ● bp: 2361	0.109618 µL	1.1 µL (dilute 1/10)

Insert RBS(EEBD-RBS0) ● concentration(ng/μL): 150 ● bp: 2392	0.207306 μL	1.0 μL (dilute 1/5)
Insert CDS(SamyQ::TAT-Ha::AmilCP) ● concentration(ng/μL): 52 ● bp: 899	0.22475 μL	1.1 μL (dilute 1/5)
Insert Terminator (B0015) ● concentration(ng/μL): 107 ● bp: 2213	0.268869 μL	1.3 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water	14.9114418	10.0 μL
T4 DNA ligase	1 μL	1 μL
BsaI-HFv2	1 μL	1 μL
Total	20 μL	20 μL

Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	5 minutes	30 cycles
16 °C (Ligation)	5 minutes	
37 °C (Restriction enzyme digestion)	10 minutes	1 cycle
80 °C (Heat inactivation of BsaI restriction enzyme)	20 minutes	1 cycle

Done by: Abigail, Chloe, Abby T

Date: 20260707
Title: Arabidopsis germination rate
Purpose: To verify and transform the healthy Arabidopsis from the rest
Procedure:

1. Place the cap holder in 2000ml of 10% bleach for 10 min (200ml bleach and 1800ml of water)
2. After bleached for 10 min, take the cap holder out and cleanse it under ddH_2O tab(slightly tilt the holder to let the water flow through every corner of it) until no bleach smell left(at least 3 min)
3. Placed the cleansed holder in biosafety cabinet along with the plants you want to transform and tip box (blue)
4. Assemble the cap holder with tip box, than pour MS medium inside the tip holders to almost full, but never pour over the front side of the Arabidopsis holder
5. Select and place the desired plants on the cap holder

	1	2	3	4	5	6	7	8	9	10	11	12
A	DT	SPA	PA	PA	SPA	SPA	PA	PA	PA	PA	PA	PA
B	DT	PA	SPA	DT	PA	PA	DT	PA	SPA	SPA	DT	SPA
C	DT	DT	SPA	DT	PA	DT	PA	PA	DT	DT	DT	SPA
D	PA	PA	DT	DT	DT	DT	DT	SPA	DT	DT	DT	DT

*DT: Didn't germinate *PA(17 caps): Prioritize Arabatopsis *SPA(10 caps): Second Prioritize Arabatopsis

Done by Sarah, Sophie H, Sophie C, Alex, Naomi
Supervised by Alex, Gabriel
Recorded by Sophie H

Date: 20260707
Title: Making 1/2 MS medium
Purpose: Make a total of 1 liter of hydroponic medium for hydroponic box

1. Measure and add to a flask:

1/2 MS medium (without sucrose)	2.3 g
MES buffer	0.5 g
ddH_2O	1 L

2. Use the hot plate to stir(with a stirring rod) until no sediments are left in the solution
3. Determine pH of the solution and adjust the pH using a pH probe to around 5.6-5.7 using 1M KOH
    a. Record original and final pH
        i. original: 4.0
        ii. final: 5.6
4. Wrap the neck with aluminum foil and place the indicator tape on top
5. Autoclave at 127 °C for 20 minutes
6. Wait for the pressure to reach 0 before opening the autoclave
7. Pour 300 ml of medium into each of the hydroponic boxes

Done by Abby K/T

Date: 20260707
Title: Transfer arabidopsis to the salinity plates
Purpose: transfer the germinated seedlings from neutral plates to salinity plates to conduct salinity stress tests (experimental set 5).
Procedure:

1. Draw a standardized baseline on each of the salinity plates to signal where the roots will be lined up after transferring
2. Place all equipments into the BSC after sanitizing
3. Get the neutral plate inside the BSC one at a time
4. Lay a template sheet of paper underneath the salinity plate so that it is clear where to place the seedlings
5. Use the tweezer to slowly lift the arabidopsis plant from the top of the roots
6. Place the seedling softly onto the salinity plate, with the bottom of the root precisely onto the baseline ; 9 seedlings per row, per plate.
    a. make sure to pick seedlings that have around the same root length
7. Repeat step 3-6 until all seedlings transferred into all of the salinity plates
8. Place into the incubator

Done by Abby K/T

Date: 20260707
Topic: Remake the salinity plates

Purpose: prepare salinity plates for experimental set 5

1. Prepare the following in a 500 ml flask:

1/2 MS medium powder(w/o sucrose)	0.69 g
MES buffer	0.15 g
Distilled water	300 ml
Sucrose (1 %)	3 g

2. Use a hot plate to stir (with a magnetic stirring rod) until no sediments are left in the solution
3. Use the pH probe and measure the pH of the solution, make sure the pH is 5.7-5.8
    a. If the pH is too low, add 0.1 M KOH slowly to increase it
4. Pour 50ml of mixture into the 6 flasks
5. Prepare 6 flasks and pour the respective trehalose and NaCl concentration in each of the flasks
    a. making sure there is 50 ml in each of the flasks (6)
    b. label each flask clearly according to their trehalose concentrations
6. Add:

150 mM NaCl, 5 mM trehalose

trehalose (µL)	NaCl (g)	Agar powder (g)
2500	0.548	0.5

150 mM NaCl, 10 mM trehalose

trehalose (µL)	NaCl (g)	Agar powder (g)
5000	0.548	0.5

250 mM NaCl, 5 mM trehalose

trehalose (µL)	NaCl (g)	Agar powder (g)
2500	0.913	0.5

250 mM NaCl, 0 mM trehalose

trehalose (µL)	NaCl (g)	Agar powder (g)
0	0.913	0.5

0 mM NaCl, 0 mM trehalose

trehalose (µL)	NaCl (g)	Agar powder (g)
0	0	0.5

0 mM NaCl, 1 mM trehalose

trehalose (µL)	NaCl (g)	Agar powder (g)
500	0	0.5

7. Wrap the necks of the flasks with aluminum foil and stick a piece of indicator tape on top
8. Autoclave for 20 minutes at 127 °C
    a. Wait for the pressure to reach 0 before retrieving the flask

During the autoclave, label 6 of the petri dishes with a marker in the biosafety cabinet

    b. right top: date and arrow pointing up to show direction for pictures
    c. bottom lid: different trehalose concentrations and NaCl concentration or control
2. Transfer half of the flasks to BSC
    a. Store the other half of the flasks in the hot bath while pouring so that it does not solidify
3. Make sure its surface is dry by wiping the flasks with a paper towel before spraying 70 % EtOH on them
4. Pour each of the medium mixtures into their square petri dishes (around 50 ml)
5. Transfer the other half of the flasks into the BSC
6. Pour each of the medium mixtures into their square petri dishes
7. Let the petri dishes solidify at room temperature under the BSC

Done by Abby K/T

Date: 20260707
Topic: Transformation
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:

A	Pveg-RBS0-CcaR-B0015 (3243 bp)
B	PcpcG2-veg-EEBD-RBS0-Csn:AmilCP-B0015 (3485 bp)
C	PliaG-RBS0-CcaSm3-B0015 (3792 bp)
D	Pveg-EEBD-RBS0-Csn:AmilCP-B0015 (7524 bp)
E	pGEM-Teasy-Csn:LEA
F	pGEM-Teasy-ccaR
G	pGEM-Teasy-csn:amilCP

1. Mix 15 μL competent cell with 1 μL plasmid
2. Incubate in ice for 20 min
3. Heat shock at 42 ℃ for 45 sec
4. Place on ice for 3 min to recover
5. Spread the mixture on TBA plate
6. Place in incubator at 37 ℃ overnight

Done by Audrey, Felix, Noah, Chloe
Supervised by Gabriel

Date: 20260707
Topic: Transformation
Purpose: Transform and cultivate engineered bacteria on agar plate
A: Pveg-RBS0-CcaR-B0015
B: PcpcG2-veg-EEBD0RBS0-Csn:AmilCP-B0015
C: PliaG-RBS0-CcaSm3-B0015
D: Pveg-EEBD-RBS0-Csn:AmilCP-B0015
E: pGEM-Teasy-Csn:LEA
F: pGEM-Teasy-CcaR
G: pGEM-Teasy-Csn:AmilCP

1. Mix 15 μL competent cell with 1 μL plasmid
2. Incubate in ice for 20 min
3. Heat shock at 42 ℃ for 45 sec
4. Place on ice for 3min to recover
5. Spread the mixture on TBA plate
6. Place in incubator at 25 ℃ overnight

Done by Sophie H, Naomi, Alex, Sophie C
Supervised by Chars

Date: 20260708
Topic: Colony PCR
Purpose: To verify bacterial colonies after a transformation step in molecular cloning
Procedure:

1. Prepare the following reaction (master mix) and dispense solutions into PCR tubes

For samples A, B, C, D, and A1

Component	1 reaction	14 reactions
Forward primer: T7 promoter	1 µL	14 µL
Reverse primer: SP6 promoter	1 µL	14 µL
Qamp master mix	10 µL	140 µL
ddH_2O	8 µL	112 µL
Total	20 µL	280 µL

For samples E, F, and G

Component	1 reaction	10 reactions
Forward primer: T7 promoter	1 µL	10 µL
Reverse primer: SP6 promoter	1 µL	10 µL
Qamp master mix	10 µL	100 µL
ddH_2O	8 µL	80 µL
Total	20 µL	200 µL

For samples *B, *C, *D

Component	1 reaction	8 reactions
Forward primer: T7 promoter	1 µL	8 µL
Reverse primer: SP6 promoter	1 µL	8 µL
Qamp master mix	10 µL	80 µL
ddH_2O	8 µL	64 µL
Total	20 µL	160 µL

2.

a. Circle 3 colonies from each of the 7 plates plus 1 from A1

A: Pveg – RBS0 – CCaR – B0015
B: PcpcG2 – veg – EEBD – RBS0 – Csn::AmiCP – B0015
C: PliaG – RBS0 – CcaSm3 – B0015
D: Pveg – EEBD – RBS0 – Csn::AmiCP – B0015
E: pGEM-TEASY – Csn::LEA
F: pGEM-TEASY – CCaR
G: pGEM-TEASY – Csn:: AmiCP
A1: Pveg – RBS0 – Ho1-pcyA-B0015

b. Circle 1 colony each on plate *B and *C and 5 on plate *D

*B:Pveg-EEBD-RBS0-Csn::ACCD-B0015
*C: Pveg-EEBD-RBS0-Csn::LEA-B0015
*D: Pveg-EEBD-RBS0-SamyQ::TAT-HA:::AmilCP-B0015

3. Pick up colonies from each of the plates with a tip and dispense into PCR tubes (colonies from incubated plates).
4. Remove tips and close the lid
5. Place into PCR machine for thermal cycle

95 ℃	2 min	1 cycle
95 ℃	30 secs	30 cycles
55 ℃	4 secs	30 cycles
72 ℃	2 min (bp 1800)	30 cycles
72 ℃	1 min	1 cycle

Done by: Sophie C, Abigail, Naomi
Supervised by: Gabriel

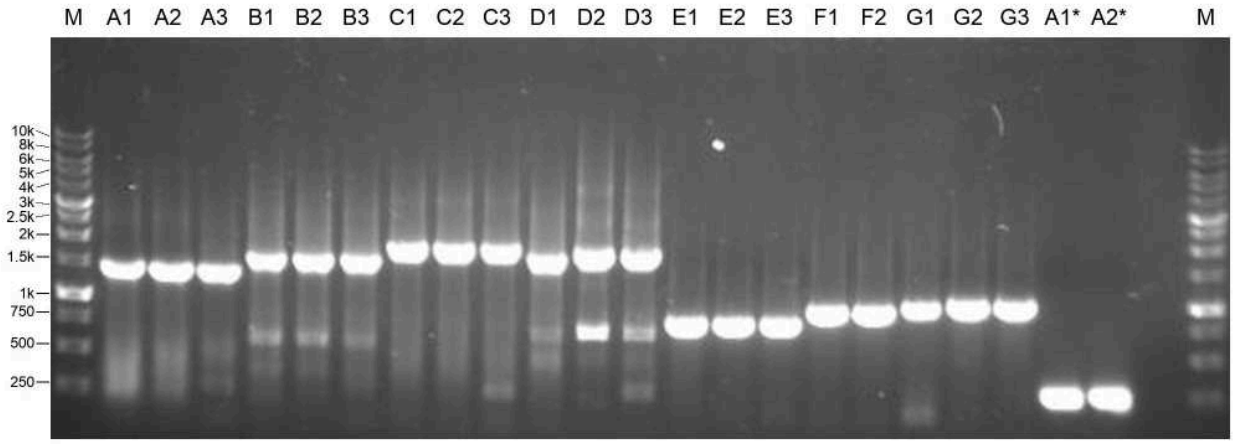
Date: 20260708
Title: 1.0% Agarese Gel preparation
Purpose: To prepare 1.0% Agarese Gel for further electrophoresis experiment
Procedure:

1. Prepare and assemble the mold and comb
2. Weight 0.4 g of agarose powder and add 40 ml of 1X TAE solution into flask
3. Cover the flask with plastic wrap, then use a tip to puncture holes on the plastic wrap
4. Microwave until the gel stops smoking but is still hot
5. Add 2 µL of safety dye into the gel and agitate it
6. Pour the melted gel into the tray
7. Wait until the gel cools down
8. Repeat steps 1~8 to create second copy of gel

Done by Sophie L, Abby T
Supervised by Gabriel

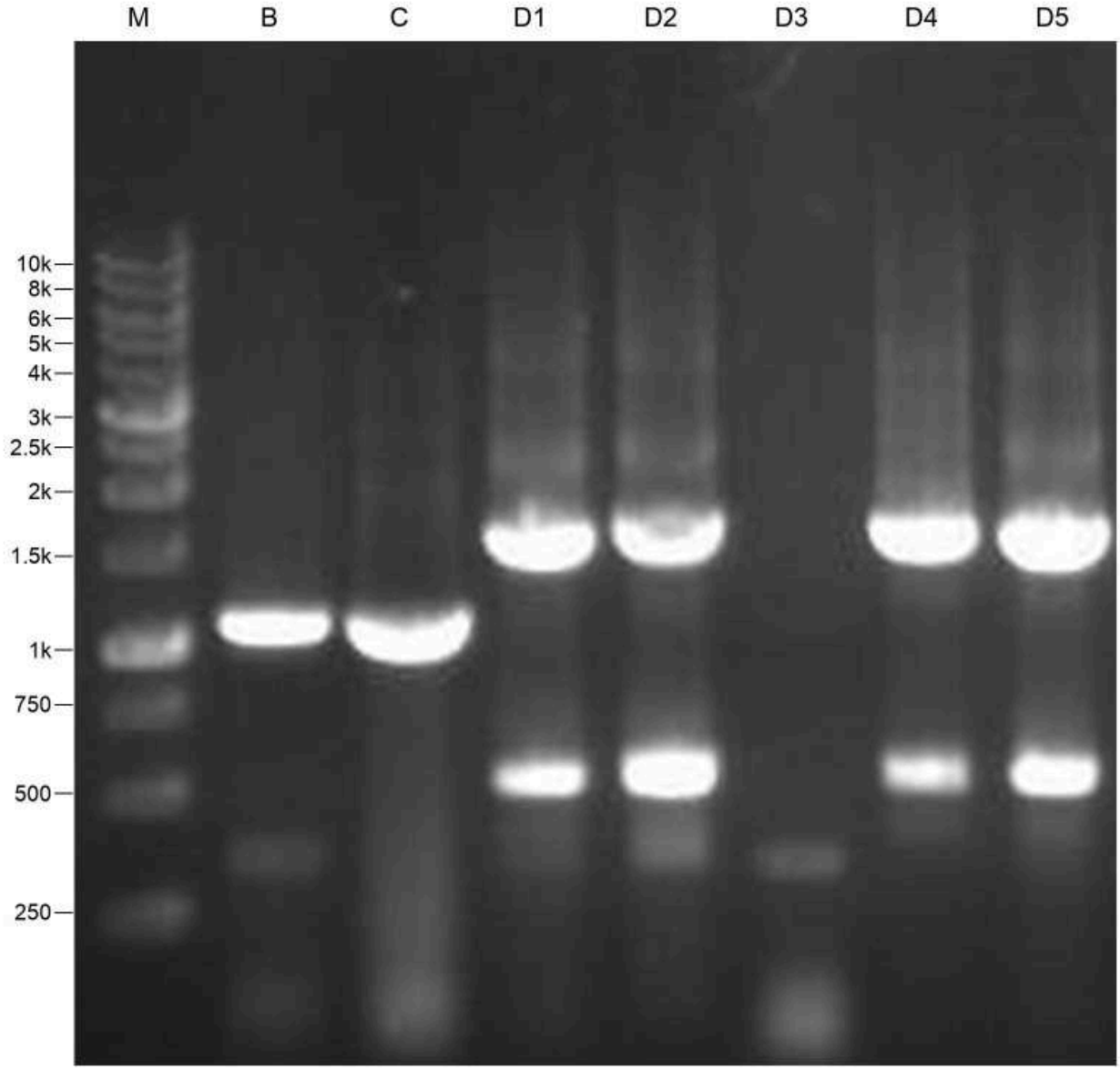
Date: 20260708
Topic: Gel Electrophoresis
Purpose: verify the sizes of the colony PCR products
Procedure:

1. Load the gel in the layout shown below (5 µL for M; 6 µL for A, B, C, D, E, F, G, *A):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
A: Pveg-RBS0-CcaR-B0015	pSTK-seq-fw	pSTK-seq-rv	1442
B: PcpcG2-veg-EEBD-RBS0-csn::AmilCP-B0015	pSTK-seq-fw	pSTK-seq-rv	1684
C: PilaG-RBS0-Ccasm3-B0015	pSTK-seq-fw	pSTK-seq-rv	1991
D: Pveg-EEBD-RBS0-csn:: AmilCP-B0015	pSTK-seq-fw	pSTK-seq-rv	1784
E: pGEM-Teasy-csn::LEA	T7 promoter fw	SP6 promoter rv	811
F: pGEM-Teasy-CcaR	T7 promoter fw	SP6 promoter rv	937
G: pGEM-Teasy-csn::AmilCP	T7 promoter fw	SP6 promoter rv	1000
A*: Pveg-RBS0-Ho1::pcyA-B0015	pSTK-seq-fw	pSTK-seq-rv	2231

3. Load the gel in the layout shown below (5 µL for M; 6 µL for B, C, D):
4. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
B: Pveg-EEBD-RBS0-csn::ACCD-B0015	Exp-seq-fw	Exp-seq-rv	1999

C: Pveg-EEBD-RBS0-csn::LEA-B0015	Exp-seq-fw	Exp-seq-rv	1262
D: Pveg-EEBD-RBS0-SamyQ::TAT-HA::AmilCP-B0015	Exp-seq-fw	Exp-seq-rv	1696

Done by: Abby T/K
Supervised by: Gabriel

Date: 20260708
Topic: Liquid Culture
Purpose: bacteria culture from successful colony PCR product
Procedure:

1. Label 9 liquid culture tubes
2. Turn on bunsen burner
3. Add 3 ml of TB medium and 3 µL of chloramphenicol into C4; 3 ml of TB medium and 3 µL ampicillin into rest of the tubes
4. Dip a pipette tip into the colonies
5. Dispense tips in their respective tubes
6. Place the tubes in the incubator at a slanted angle around 45-50 °

A1: Pveg-RBS0-CCaR-B0015
B1: PcpcG2-veg-EEBD-RBS0-Csn::AmilCP-B0015
C1: PliaG-RBS0-CcaSm3-B0015
D2: Pveg-EEBD-RBS0-Csn::AmilCP-B0015
E1: pGEM-TEASY-Csn::LEA
F1: pGEM-TEASY-CCaR
G1: pGEM-TEASY-Csn:: AmiCP
*D2: Pveg-EEBD-RBS0-SamyQ::TAT-HA::AmilCP-B0015
C4: Exp2-spacer-spacer-spacer-Pveg-EEBD-RBS0-csn::AmilCP-B0015

Done by: Abby T
Supervised by: Gabriel

Date: 20260709
Title: Miniprep
Purpose: Extract plasmid DNA from the liquid culture of colony PCR products for amplification
Procedure:

1. Prepare Glycerol stock, formula as follows:

	Exp2-spacer-spacer-spacer-Pveg-EEBD-RBS0-csn::AmilCP-B0015
50% Glycerol Stock	500 μL
Bacteria Culture Medium	500 μL

2. Then store in -80 ℃

A1: Pveg-RBS0-CCaR-B0015
B1: PcpcG2-veg-EEBD-RBS0-Csn::AmiCP-B0015
C1: PliaG-RBS0-CcaSm3-B0015
D2: Pveg-EEBD-RBS0-Csn::AmiCP-B0015
E1: pGEM-TEASY-Csn::LEA
F1: pGEM-TEASY-CCaR
G1: pGEM-TEASY-Csn:: AmiCP
*D2: Pveg-EEBD-RBS0-SamyQ::TAT-HA::AmilCP-B0015
C4: Exp2-spacer-spacer-spacer-Pveg-EEBD-RBS0-Csn::AmilCp-B0015

Results**:**

	Concentration (ng/μL)	A260/230	A260/230
A. pSTK-1C-Pveg-RBS0-CcaR-B0015	126	1.145	2.000
B. pSTK - PcpcG2 - v eg - EEBD - RBS0 - Csn::AmilCP - B0015	186	1.094	1.958
C. pSTK - PliaG - RBS0 - Ccasm3 - B0015	134	1.411	2.062
D. pSTK - Preg - EEBD - RBS0 - Csn::AmilCP - B0015	11	1.181	1.375
E. pGEM-TEASY - Csn::LEA	150	0.848	1.948
F. pGEM-TEASY - CcaR	112	1.008	1.800
G. pGEM-TEASY - Csn::AmilCP	128	1.356	1.910

*D2. Pveg - EEBD - RBS0 - SamyQ::TAT-HA::AmilCP - B0015	160	1.552	2.000
C4. Exp2 - spacer - spacer - spacer - Pveg - EEBD - RBS0 - csn::AmilCP - B0015	72	0.524	1.940

Done by Abby T/K, Naomi, Sarah C
Supervised by Chars

Date: 20260709
Title: Colony PCR
Purpose: To amplify bacterial sequences after a transformation step in molecular cloning
Procedure
D: Pveg-EEBD-RBS0-SamyQ-TAT-HA-AmilCP-B0015 (Exp1)
A: 1A-Pveg-RBS0-Ho1:pcyA-B0015

1. Prepare the following reaction (master mix) and dispense solutions into PCR tubes

A	1 rxn	13 rxn
Fw Primer (pSTK-seq-fw, 10mM)	1 µL	13 µL
Rv Primer (STK-seq-rv, 10mM)	1 µL	13 µL
Q-Amp master mix	10 µL	130 µL
ddH_2O	8 µL	104 µL

D	1 rxn	13 rxn
Fw primer (STK-Exp1-seq-fw, 10mM)	1 µL	13 µL
Rv Primer (STK-Exp1-seq-rv, 10mM)	1 µL	13 µL
Q-Amp master mix	10 µL	130 µL
ddH_2O	8 µL	104 µL

2. Circle 12 colonies with no fluorescent signal under UV light from plates A and D

A: 1A- Pveg – RBS0 – Ho1:pcyA – B0015
D: Pveg – EEBD – RBS0 – SamyQ – TAT – HA – AmilCP – B0015 (Exp1)

3. Pick up colonies from each of the plates with a tip and dispense into PCR tubes (colonies from incubated plates).
4. Remove tips and close the lid
5. Place into PCR machine for thermal cycle

95 °C	5 min	1 cycle
95 °C	20 secs	35 cycles
55 °C	20 secs	
72 °C	2.5 min (bp 2200)	
72 °C	5 min	1 cycle

Done by: Sophie C., Audrey C. , Sophie H.
Supervised by: Chars
Recorded by Sophie H.

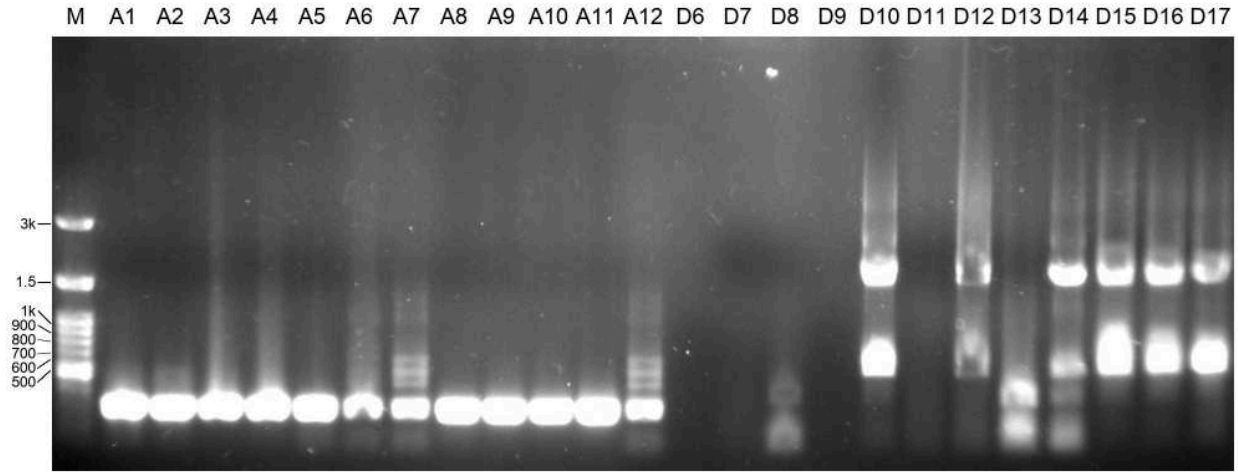
Date: 20260709
Title: 1.0% Agarese Gel preparation (x2 gels)
Purpose: To prepare 1.0% Agarese Gel for electrophoresis of miniprep
Procedure:

1. Prepare and assemble the mold and comb
2. Weight 0.4 g of agarose powder and add 40 ml of 1X TAE solution into flask
3. Cover the flask with plastic wrap, then use a tip to puncture holes on the plastic wrap
4. Microwave until the gel stops smoking but is still hot
5. Add 2 µL of safety dye into the gel and agitate it
6. Pour the melted gel into the tray
7. Wait until the gel cools down
8. Repeat steps 1~8 to create second copy of gel

Done by Abby T/K, Sophie C, Audrey C, Sophie H
Supervised by Chars

Date: 20260709
Title: Gel Electrophoresis
Purpose: Verify the sizes of the cloning PCR product of bacterial colony A and D
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min

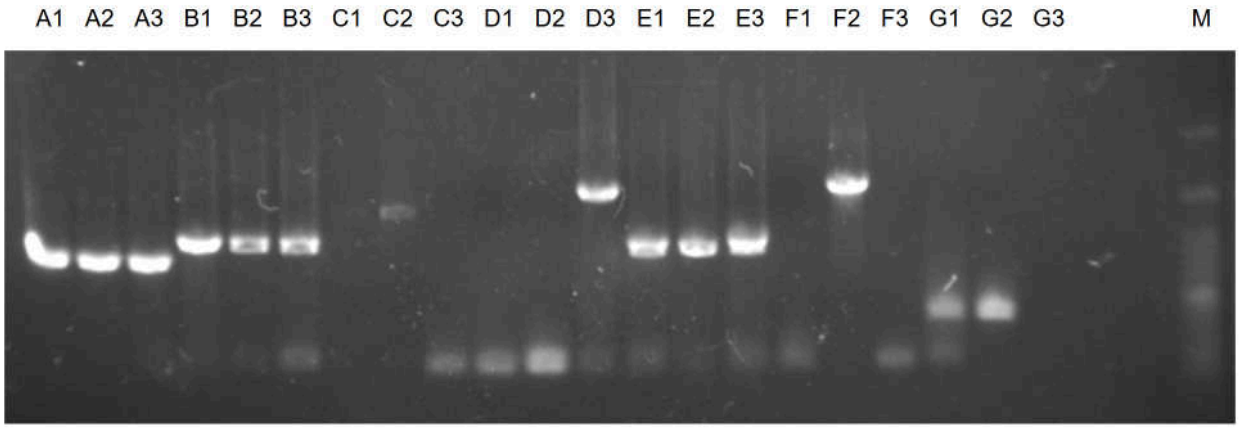


M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder 5 uL			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
A: 1A-Pveg-RBS0-H01:PcyA-B0015	Exp-seq-fw	Exp-seq-rv	2231
D: Pveg-EEBD-RBS0-SamyQ-TAT-HA-AmilCP-B0015(Exp1)	Exp-seq-fw	Exp-seq-rv	1696
Note: D10, D14 positive clone			

Done by Sophie C, Sarah, Sophie H, Naomi

Date: 20260709
Topic: Gel Electrophoresis
Purpose: verify the sizes of the miniprep product
Procedure:

1. Load the gel in the layout shown below (3 μL for M; 6 μL for all samples):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder		
Sample	Plasmid Construct	Expected Size (bp)
A1	pSTK-1C-Pveg-RBS0-CcaR-B0015	3243
C1	pSTK-PilaG-RBS0-CcaSm3-B0015	3792
C4	Exp2-spacer-spacer-spacer-Pveg-EEBD-RBS0-csn::AmilCP-B0015	5775
D2	pSTK-Pveg-EEBD-csn::AmilCP-B0015	3585
E1	pGEM-TEASY-csn:LEA	3651
F1	pGEM-TEASY-CcaR	3777
G1	pGEM-TEASY-csn:AmilCP	3840
D2*	Pveg-EEBD-RBS0-SamyQ::TAT-HA::AmilCP-B0015	7464
B1	pSTK-PcpcG2-veg-EEBD-RBS0-csn:AmilCP-B0015	3485

Done by Abby K, Abby T, Audrey Chu
Supervised by: Chars

Date: 20260709
Title: Golden gate assembly
Purpose:
Procedure:
A: pSTK-1A-Pveg-RBS0-Ho1:pcyA-B0015

1. Prepare the reaction as follow:

Part	Original Amount	Dilution
Insert vector (pSTK-1A) 115 ng/µl (bp:3120)	0.176348 µl	1.76 µl (1/10)
Insert Promoter (Pveg)	0.109618 µl	1.1 µl (1/10)
Insert RBS (RBS0)	0.1461117 µl	1.46 µl (1/10)
Insert CDS (Ho1::pcyA)	0.436455824 µl	1.31 µl (1/3)
Insert terminator (B0015)	0.268869 µl	1.34 µl (1/5)
T4 DNA ligase buffer	2 µl	2 µl
Sterile water	15.2040617 µl	9.03 µl
T4 DNA ligase	1 µl	1 µl
BsaI-HFv2	1 µl	1 µl
Total	20 µl	20 µl

2. Use the thermal cycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	5 minutes	30 cycles
16 °C (Ligation)	5 minutes	
37 °C (Restriction enzyme digestion)	10 minutes	1 cycle
80 °C (Heat inactivation of BsaI restriction enzyme)	20 minutes	1 cycle

Done by Abby T/K, Audrey Chu, Sophie H, Sophie C, Naomi, Sarah
Supervised by Chars

Date: 20260709
Title: Liquid Culture
Purpose: bacteria culture
Procedure:

1. Label 4 liquid culture tubes
2. Turn on Bunsen Burner
3. Add 3 mL of TB medium into each tube
4. Add 3 µl of chloramphenicol into C4 tube
5. Add 3 µl of ampicillin into the rest of the 3 tubes(D10, D12, D14)
6. Dip a pipette tip into the colonies
7. Dispense tips in their respective tubes
8. Place the tubes in the incubator at a slanted angle

C4: spacer*3 - pveg - EEBD - RBS0 - csn:AmilCP - B0015 Exp2
D10, D12, D14: Pveg - EEBD - RBS0 - SamyQ - TAT - HA - AmilCP - B0015 Exp1

Done by Alex
Supervised by Chars

Date: 20260710
Title: Miniprep
Purpose: Isolate plasmid from DH5x for sequencing
Procedure:
C4: pSTK-EXP2-Spacerx3-Pveg-EEBD-RBS0-Csn:AmilCP-B0015
D10: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ: TAT-HA:AmilCP-B0015
D12: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ: TAT-HA:AmilCP-B0015
D14: pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ: TAT-HA:AmilCP-B0015

1. Collect liquid culture 1.8 mL in 2 mL microtubes
2. Centrifuge at 23444 g for 30 seconds. Discard the supernatant
3. Repeat step 1 until collect all of the liquid culture
4. Resuspend pellet in 250 microliters of P1 buffer(Qiagene)
5. Add 250 microliters of P2 buffer.
6. Invert the microtubes for 10 times. Incubate at RT for 2 min
7. Add 350 microliters of N3 buffer. Invert the microtubes for 10 times
8. Centrifuge at 13000 g for 10 mins
9. Transfer supernatant 800 µL to column and discard the pellet
10. Centrifuge at 17900g for 1 min. Discard the flowthrough.
11. Add 500 µL PB buffer. Centrifuge at 17900 g for 1 min and discard the flowthrough
12. Add 750 µL PE buffer. Centrifuge at 17900 g for 1 min and discard the flowthrough
13. Centrifuge the empty column at 17900 g for 3 min
14. Place the column on the 1.5 mL microtube
15. Add 50 µl sterile deionized water at 65°C
16. Incubate at RT for 5 min
17. Centrifuge at 17900 g for 3 min. Discard the column
18. Measure the concentration using Biodrop.

Biodrop results:

	Concentration (ng/µl)	A260/280	A260/230
C4	94.0	1.880	2.186
D10	277.0	1.847	2.347
D12	306.0	1.913	2.354
D14	354.0	1.914	2.376

Done by Chars

Date: 20260712
Topic: Prepare 1x 1/2 MS medium agar
Purpose: To prepare agar for seed vernalization
Procedure:

1. The following protocol is for 250 ml of 1x agar.

1/2 MS medium powder (w/o sucrose)	0.575 g
MES buffer	0.125 g
Distilled water	250 ml
Sucrose(1%)	2.5 g
Agar powder(1%)	2.5 g

2. Add the 1/2 MS medium powder, MES buffer, Distilled water, and Sucrose into a 500 ml flask.
3. Use a hot plate to stir (with a magnetic stirring rod) until no sediments are left in the solution.
4. Use the pH probe and measure the pH of the solution, make sure the pH is 5.7-5.8.
    a. If the pH is too low, add 0.1 M KOH slowly to increase it.
5. Add Agar powder into the solution.
6. Wrap the neck of the flask with aluminum foil and stick a piece of indicator tape on top.
7. Autoclave for 20 minutes at 127 °C.
    a. Wait for the pressure to reach 0 before retrieving the flask.
8. Prepare 48 eppendorf microtube lids.
9. Add 250 µL of agar into each lid.
10. Wait until it solidifies.

Done by: Alex
Supervised by: Gina

Date: 20260712
Topic: Prepare 2x 1/2 MS medium agar
Purpose: To prepare agar for seed vernalization for later use
Procedure:

1. The following protocol is for 250 ml of 2x agar.

1/2 MS medium powder (w/o sucrose)	1.15 g
MES buffer	0.25 g
Distilled water	250 ml

Sucrose(1%)	5.0 g
Agar powder(1%)	5.0 g

2. Add the 1/2 MS medium powder, MES buffer, Distilled water, and Sucrose into a 500ml flask.
3. Use a hot plate to stir (with a magnetic stirring rod) until no sediments are left in the solution.
4. Use the pH probe and measure the pH of the solution, make sure the pH is 5.7-5.8.
    a. If the pH is too low, add 0.1 M KOH slowly to increase it.
5. Add Agar powder into the solution.
6. Wrap the neck of the flask with aluminum foil and stick a piece of indicator tape on top.
7. Autoclave for 20 minutes at 127 °C.
8. Wait for the pressure to reach 0 before retrieving the flask.
9. Aliquote 25 mL agar into 50 mL centrifuge tubes.
10. Storage at room temperature.

Done by: Alex
Supervised by: Gina

Date: 20260712
Title: Prepare 1/2 MS medium (hydroponic medium)
Purpose: To prepare medium for hydroponics testing
Procedure:

1. The following protocol is for 1000 ml of 1/2 MS medium

1/2 MS medium powder (w/o sucrose)	2.3 g
MES buffer	0.5 g
Distilled water	1000 ml

2. Add the contents into a 1000 ml flask
3. Use a hot plate to stir (with a magnetic stirring rod) until no sediments are left in the solution
4. Use the pH probe and measure the pH of the solution, make sure the pH is 5.7-5.8
5. If the pH is too low, add 0.1 M KOH slowly to increase it
6. Wrap the neck of the flask with aluminum foil and stick a piece of indicator tape on top
7. Autoclave for 20 minutes at 127 °C
8. Wait for the pressure to reach 0 before retrieving the flask
9. Add 2.2 mL of medium into each tube rack.

Done by Naomi
Supervised by: Gina

Date: 20260712
Title: Golden Gate Assembly
Purpose: Assemble multiple DNA fragments into a construct
Procedure:
A: (STK) Module 2-PlepA-RBS0-CcaSm3-B0015

1. Prepare the following:

Vector(1B)	1.3 μL(dilute 1/10)
Insert promoter(PlepA)	1.9 μL(dilute 1/10)
Insert RBS(RBS0)	1.5 μL(dilute 1/10)
Insert CDS(CcaSm3)	1.5 μL(dilute 1/10)
Insert Terminator(B0015)	1.3 μL (dilute 1/5)
T4 DNA ligase Buffer	2 μL
Sterile water	8.5 μL
T4 DNA ligase	1 μL
BsaI-HFv2	1 μL
Total	20 μL

2. Use thermocycler to run

37 ℃	5 min	30 cycles
16 ℃	5 min	
37 ℃	10 min	
80 ℃	20 min	

B: (STK) Module 4-Pveg-EEBD-RBS0-CSN::ACCD-B0015

1. Prepare the following:

Vector(Exp1)	1.4 μL(dilute 1/5)
Insert promoter(Pveg)	1.0 μL(dilute 1/5)
Insert RBS(EEBD-RBS0)	1.1 μL(dilute 1/10)
Insert CDS(CSN::Accd)	0.6 μL(dilute 1/10)
Insert Terminator(B0015)	1.3 μL (dilute 1/5)
T4 DNA ligase Buffer	2 μL
Sterile water	10.58 μL
T4 DNA ligase	1 μL
BsaI-HFv2	1 μL

Total 20 µL

2. Use thermocycler to run

37 ℃	5 min	30 cycles
16 ℃	5 min	
37 ℃	10 min	
80 ℃	20 min	

C: (STK) Module 4-PcpcG2-veg-EEBD-RBS0-Csn::ACCD-B0015

1. Prepare the following:

Vector(1D) 1.5 µL(dilute 1/5)
Insert promoter(Pveg) 1.2 µL(dilute 1/50)
Insert RBS(PcpcG2-veg) 1.0 µL(dilute 1/5)
Insert CDS(Csn:Accd) 0.6 µL(dilute 1/10)
Insert Terminator(B0015) 1.3 µL (dilute 1/5)
T4 DNA ligase Buffer 2 µL
Sterile water 10.4 µL
T4 DNA ligase 1 µL
BsaI-HFv2 1 µL

Total 20 µL

2. Use thermocycler to run

37 ℃	5 min	30 cycles
16 ℃	5 min	
37 ℃	10 min	
80 ℃	20 min	

D: (STK) Module 4-Pveg-EEBD-RBS0-Csn::LEA-B0015

1. Prepare the following:

Vector(Exp1) 1.4 µL(dilute 1/5)
Insert promoter(Pveg) 1.1 µL(dilute 1/10)
Insert RBS(EEBD-RBS0) 1.0 µL(dilute 1/5)

Insert CDS(CSN::LEA)	1.1 μL(dilute 1/10)
Insert Terminator(B0015)	1.3 μL(dilute 1/5)
T4 DNA ligase Buffer	2 μL
Sterile water	10.0 μL
T4 DNA ligase	1 μL
Bsal-HFv2	1 μL
Total	20 μL

2. Use thermocycler to run

37 ℃	5 min	30 cycles
16 ℃	5 min	
37 ℃	10 min	
80 ℃	20 min	

E: (STK) Module 4-PcpcG2-veg-EEBD-RBS0-CSN::LEA-B0015

1. Prepare the following:

Vector(1D)	1.5 μL(dilute 1/10)
Insert promoter(PcpeG2-veg)	1.2 μL(dilute 1/50)
Insert RBS(EEBD-RBS0)	1.0 μL(dilute 1/5)
Insert CDS(Csn:LEA)	1.1 μL(dilute 1/10)
Insert Terminator(B0015)	1.3 μL(dilute 1/5)
T4 DNA ligase Buffer	2 μL
Sterile water	10.05 μL
T4 DNA ligase	1 μL
Bsal-HFv2	1 μL
Total	20 μL

2. Use thermocycler to run

37 ℃	5 min	30 cycles
16 ℃	5 min	
37 ℃	10 min	
80 ℃	20 min	

F: Pveg-EEBD-RBS0-GanA-Gala-B0015

1. Prepare the following:

Vector(Shuttle vector) (pSTK-Exp-1sfGFP)	1.5 μL(dilute 1/5)
Insert promoter(Pveg)	1.2 μL(dilute 1/50)
Insert RBS(EEBD-RBS0)	1.0 μL(dilute 1/5)
Insert CDS(Gana-Gala)	0.6 μL(dilute 1/10)
Insert Terminator(B0015)	1.3 μL (dilute 1/5)
T4 DNA ligase Buffer	2 μL
Sterile water	10.4 μL
T4 DNA ligase	1 μL
BsaI-HFv2	1 μL
Total	20 μL

2. Use thermocycler to run

37 ℃	5 min	30 cycles
16 ℃	5 min	
37 ℃	10 min	
80 ℃	20 min	

Done by Abby K
Supervised by Chars

Date: 20260712
Title: Fresh weight of Arabidopsis
Purpose: To measure stress's impact to the plant's fresh weight
Procedure:

Control-1(contains 3 plants) 0.0154 g
Control-2 // 0.0111 g
Control-3 // 0.014 g

1 mM trehalose // -1 0.0095 g
1 mM trehalose // -2 0.0014 g
1 mM trehalose // -3 0.0087 g

5 mM trehaslose // -1 0.0101 g
5 mM trehaslose // -2 0.1050 g
5 mM trehaslose // -3 0.0112 g

10 mM trehaslose // -1 0.0114 g
10 mM trehaslose // -2 0.0134 g
10 mM trehaslose // -3 0.0024 g

150 mM NaCl // -1 0.0090 g
150 mM NaCl // -2 0.0013 g
150 mM NaCl // -3 0.0024 g

150 mM NaCl+1 mM trehaslose // -1 0.0020 g
150 mM NaCl+1 mM trehaslose // -2 0.0015 g
150 mM NaCl+1 mM trehaslose // -3 0.0025 g

150 mM NaCl+5 mM trehaslose // -1 0.0029 g
150 mM NaCl+5 mM trehaslose // -2 0.0023 g
150 mM NaCl+5 mM trehaslose // -3 0.0035 g

150 mM NaCl+10 mM trehaslose // -1 0.0040 g
150 mM NaCl+10 mM trehaslose // -2 0.0036 g
150 mM NaCl+10 mM trehaslose // -3 0.0022 g

250 mM NaCl (contains 7 plants) 0.0013 g

250 mM NaCl+1 mM trehaslose // -1 0.0008 g
250 mM NaCl+1 mM trehaslose // -2 0.0009 g
250 mM NaCl+1 mM trehaslose // -3 0.0014 g

250 mM NaCl+5 mM trehaslose // -1 0.0008 g
250 mM NaCl+5 mM trehaslose // -2 0.0009 g
250 mM NaCl+5 mM trehaslose // -3 0.0011 g

250 mM NaCl+10 mM trehaslose // -1 0.0008 g
250 mM NaCl+10 mM trehaslose // -2 0.0004 g
250 mM NaCl+10 mM trehaslose // -3 0.0008 g

Done by Sophie H, Sophie Yeh, Naomi
Supervised by Gina

Date: 20260713
Title: Chlorophyll Extraction
Purpose: To isolate the green photosynthetic pigment from arabidopsis
Procedure:

1. Take out the microtubes with arabidopsis from -80 ℃ (34 microtubes)
2. Dilute acetone to 80 %, add 200 µL to each tube
3. Use a crusher the plants in the acetone, crush until no visible green plants/leaves
4. Add 800 µL of acetone to the tubes
5. Vortex and store in a dark place
6. Centrifuge at 7000 g for 30 sec
7. Add the liquid to the circus 96 well, 200 µL in each well. 3 wells for each sample

	1	2	3	4	5	6	7	8	9	10	11	12
A	Control 1	1T 1	5T 1	10T 1	150 NaCl 1	150 NaCl 1T 1	150 NaCl 5T 1	150 NaCl 10T 1	250NaCl 1	250 NaCl 1T 1	250 NaCl 5T 1	250 NaCl 10T 1
B	Control 2	1T 2	5T 2	10T 2	150 NaCl 2	150 NaCl 1T 2	150 NaCl 5T 2	150 NaCl 10T 2		250 NaCl 1T 2	250 NaCl 5T 2	250 NaCl 10T 2
C	Control 3	1T 3	5T 3	10T 3	150 NaCl 3	150 NaCl 1T 3	150 NaCl 5T 3	150 NaCl 10T 3		250 NaCl 1T 3	250 NaCl 5T 3	250 NaCl 10T 3

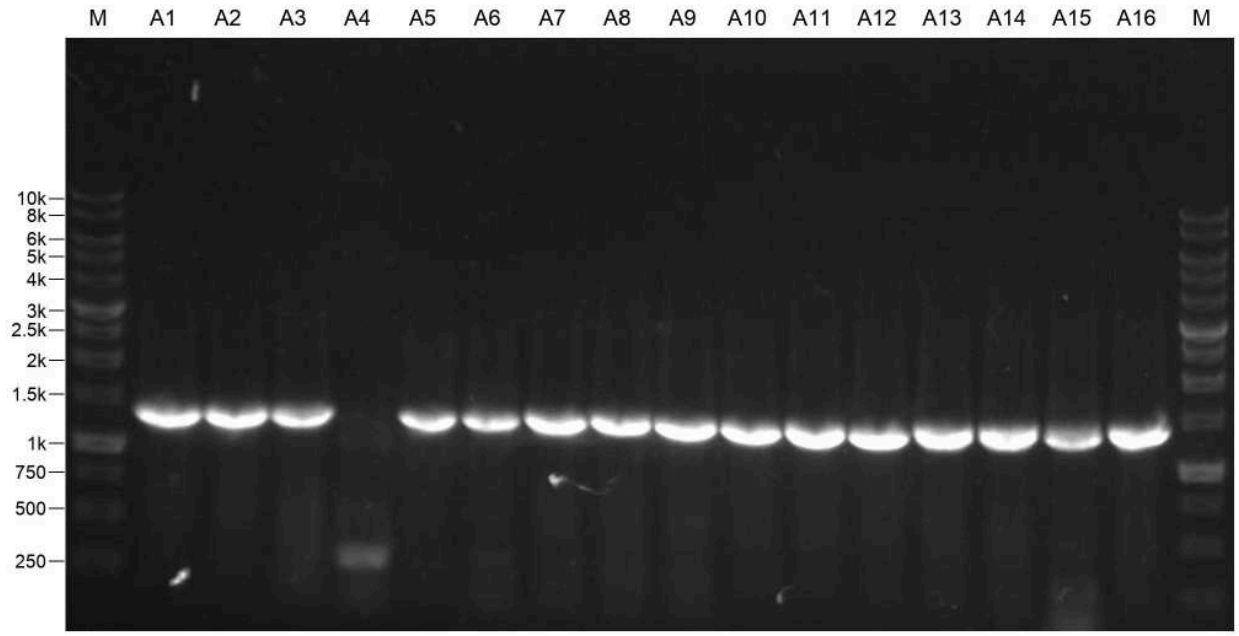
8. ELISA reader, insert 96 well, run the results through the equation below, make sure the select the correct wavelength (A663 and A645)

*Total chlorophyll content equation (mg/g): (20.2*0D645+8.02*0D663) V / (1000*fw), V = 1 ml
*250 NaCl 1T 3 ml use 200 µL acetone

Done by Abby T, Sophie C, Audrey H, Abby K.
Supervised by Gabriel

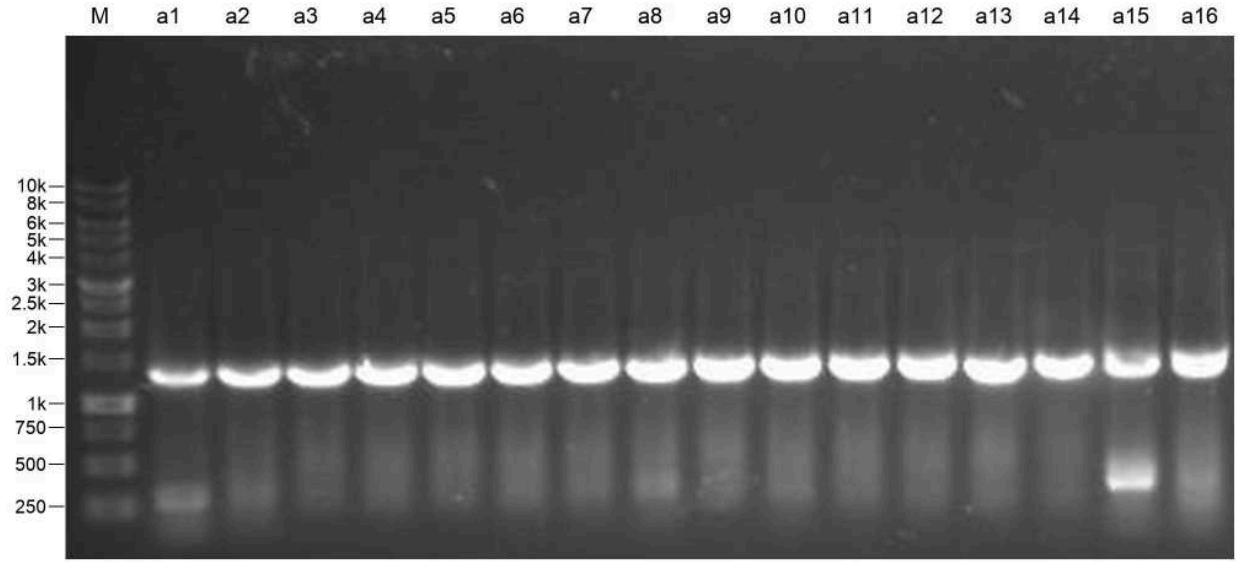
Date: 20260713
Title: Gel Electrophoresis
Purpose: To run the gel to verify the length of amplified DNA sequence
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
A1-A16: pSTK-1A-Pveg-RBS0-ho1::pcyA-B0015 (w/ promegaligase)	pSTK-seq-fw	pSTK-seq-rv	2231
Note: fail			

3. Load the gel in the layout shown below (5 µL each):
4. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
a1-a16: pSTK-1A-Pveg-RBS0-ho1::pcyA-B0015 (w/ new ligase MEBT4)	pSTK-seq-fw	pSTK-seq-rv	2231
Note: fail			

Done by Sophie H.

Date: 20260713
Title: Transformation
Purpose: Transform bacterial liquid on agar plate
Procedure:
B: (STK) Module 4-Pveg-EEBD-RBS0-Csn:ACCD-B0015
D: (STK) Module 4-Pveg-EEBD-RBS)-Csn:LEA- B0015
F: Pveg-EEBD-RBS0-GanA-Gala-B0015
Z: SamyQ-TAT-HA-AmilCP
a: pSTK-1A-Pveg-RBS0-Ho1:pcyA-B0015

1. Turn on the water bath and set it to 42°C
2. Prepare an ice bucket
3. Thaw the competent cells on ice until 50% thawed
4. Add 15 µl of plasmid to 100 µl of competent cell(do not pipette)
5. Place the mixture on ice for 20 minutes

6. Place the tubes in a 42 °C at water bath for 45 seconds - 1 minute
7. Place tubes back on ice for 3 minutes to recover
8. Add 1 mL of soc medium under a bunsen burner
9. Incubate at 37 °C for 1 hour
10. Centrifuge for 5 minutes at 5000 g
11. Remove 1 mL of supernatant
12. Resuspend the cell pellet in the remaining supernatant
13. Spread the mixture on LBA plate
14. Incubate at 37 °C for 16 hours

Done by Chloe, Abby K, Alex, Abby T, Sophie C, Olivia

Date: 20260713
Title: TA cloning
Purpose: Insert PCR-amplified DNA fragments into a plasmid vector without relying on restriction enzymes
Procedure:

1. **Prepare the following reaction:**

Z: samyQ-TAT-HA-AmilCP (277 ng/μL; 892 bp)

Reagents	Standard Rxn
2x rapid ligation buffer, T4 DNA ligase	5 μL
pGEM Teasy vector	1 μL
pSTK SamyQ-TAT-HA-AmilCP (Z)	1.6 μL
T4 DNA ligase	1 μL
ddH_2O	1.4 μL
Total:	10 μL

Done by: Chloe

Date: 20260714
Title: Colony PCR
Purpose: To amplify bacterial sequences after a transformation step in molecular cloning
Procedure:
A: pSTK-1A-Pveg-RBS0-Ho1:pcyA-B0015

1. Prepare the following reaction (master mix) x24 reaction and dispense solution into PCR tubes

	1 reaction	26 reactions
Fw Premier (pSTK-seq-fw, 10mM)	1μL	26 μL
Rv Premier (STK-seq-rv, 10mM)	1μL	26 μL
Q-amp master mix	10μL	260 μL
ddH_2O	8μL	208 μL
Total	20μL	520 μL

2. Circle 24 colonies from the plate
3. Pick up colonies from the plate with tip and dispense on PCR tube respectfully
4. Take out tips and close the lid after incubating in room temperature for 5 mins(can also shake the tubes gently)
5. Place into PCR machine for thermal cycle

95 ℃	5 min
95 ℃	30 sec
55 ℃	30 sec
72 ℃	2.5 min
72 ℃	5 min

Done by Sarah, Sophie C, Sophie H.
Supervised by Sophie C.

Date: 20260714
Title: Colony PCR
Purpose: To amplify bacterial sequences after a transformation step in molecular cloning

Procedure:

1. Prepare the following reaction (master mix) x24 reactions and dispense solution into PCR tubes

	1 reaction	26 reactions
Fw Premier (Exp-seq-fw)	1 μL	26 μL
Rv Premier (Exp-seq-rv)	1 μL	26 μL
Q-Amp master mix	10 μL	260 μL
ddH_2O	8 μL	208 μL
Total	20 μL	520 μL

A:Pveg-EEBD-RBS0-Csn:ACCD-B0015
B: Pveg-EEBD-RBS0-GanA-Gala-B0015
C: Pveg-EEBD-RBS0-Csn:LEA-B0015

2. Circle colonies from each plate (A: 3 B: 18 C: 13)
3. Pick up colonies from each plate with tip and dispense on PCR tubes respectfully
4. Take out plate and close the lid after incubating in room temperature for 5 min (or gently shake the PCR tubes horizontally)
5. Place into PCR machine for thermal cycle

95 ℃	5 min
95 ℃	30 sec
55 ℃	30 sec
72 ℃	2.5 min
72 ℃	5 min

Done by Alex, Ryan Wei
Supervised by Chloe

Date: 20260714
Title: Prepare 0.8 % Gel for Gel Electrophoresis (for 2)
Purpose: Make gel for electrophoresis
Procedure

1. Assemble tray and mold, prepare comb
2. Weigh 0.64 g agarose powder and 80 mL 1XTAE buffer

3. Add 1XTAE and powder to flask
4. Cover flask with plastic wrap and puncture holes
5. Microwave for 30 seconds and agitate it. Repeat this step until all particles dissolved
6. Wait until the gel stops smoking, add 2 µL safety dye and agitate it.
7. Pour the melted gel in the tray and assemble the comb with
8. Leave to solidify in room temperature

Done by: Sarah, Sophie C, Sophie H

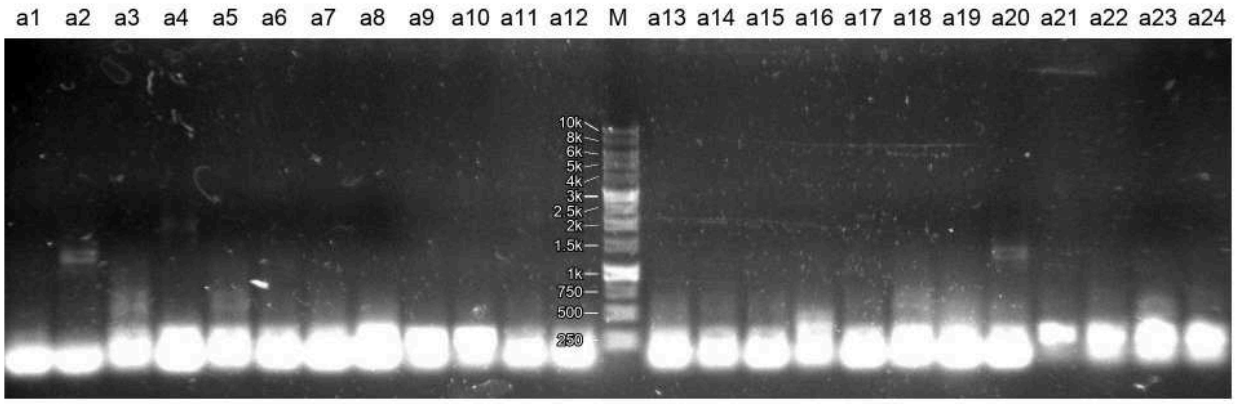
Date: 20260714
Title: LB Plate (LBA and LBA with glycerol)
Purpose: prepare agar plate
Procedure

1. Formula as below:
    a. Broth powder: 6.25 g + (Glycerol 2 mL)
    b. Agar powder: 3.75 g
    c. ddH_2O: 250 mL
2. Add one magnetic stirring bar to the flask
3. Place the flask containing the mixture on the hot plate, and stir until no sediments
4. Remove the stirring bar from the flask
5. Cover the flask with aluminum foil and apply an indicator tape to ensure the mixture is successfully autoclaved
6. Autoclave for 20 minutes at 127 °C.
7. Place the new petri dishes in BSC and turn on the UVC
8. Set up water bath at 55
9. Spray the working surface of the BSC and equipments with 70% ethanol
10. Add 500µL of antibiotics (ampicillin) into each corresponding flask
11. Use serological pipette to add 10-15 mL of melting LB agar into the petri dish, avoid overfilling (pipette first to mix the antibiotics)
12. Arrange the petri dishes in stacks within the cabinet
13. Let the agar plates dry for at least 1 hour in the BSC

Done by: Sophie C, Sarah, Naomi

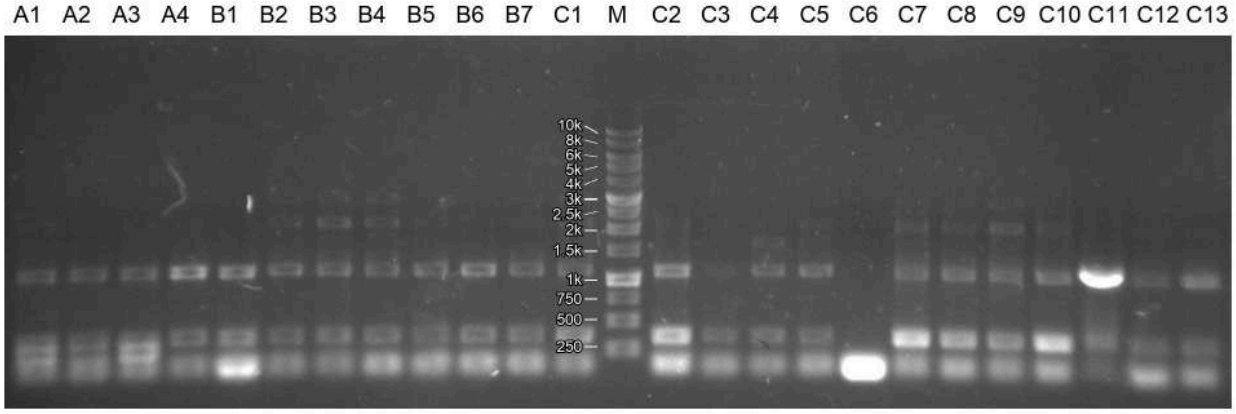
Date: 20260714
Title: Gel electrophoresis
Purpose: Check base pair of colony pcr product
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
a1-a24: pSTK-1A-Pveg-RBS0-ho1::pcyA-B0015	pSTK-seq-fw	pSTK-seq-rw	2231

3. Load the gel in the layout shown below (5 µL each):
4. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
A: Pveg-EEBD-RBS0-Csn:AccD-B0015	Exp-seq-fw	Exp-seq-rw	1999

B: Pveg-EEBD-RBS0-GanA-Gala-B0015	Exp-seq-fw	Exp-seq-rw	2993
C: Pveg-EEBD-RBS0-Csn:LEA-B0015	Exp-seq-fw	Exp-seq-rw	1262

Done by: Sophie C, Alex, Chloe, Audrey C, Ryan W.

Date: 20260714
Title: Liquid Culture
Purpose: Transfer colony to medium
Procedure

1. Label 7 centrifuge tubes and 8 liquid culture tubes
2. Turn on Bunsen Burner
3. Add 3 µL of ampicillin and 3 mL of TB medium into every tube
4. Dip a pipette tip into the colonies
5. Disperse tips in their respective tubes
6. Place the tubes in the incubator at a slanted angle

A1: pSTK-1A-Pveg-RBS0-Ho1:PcyA-B0015
B2, B3, B4: pSTK-Exp1-Pveg-EEBD-RBS0-GanA-Gala-B0015
C1, C2, C11: pSTK-Exp1-Pveg-EEBD-RBS0-Csn:LEA-B0015
D2: pSTK-Exp1-Pveg-EEBD-RBS0-SamyQ-TAT-HA-Csn:AmilCP-B0015

Done by: Chars

Date: 20260715
Title: Colony PCR
Purpose: Verify DNA insert of Ho1:pcyA for light circuit experiment
Procedure:

1. Prepare the following master mix x 22 reactions and dispense solutions into pcr tubes

	1 reaction	22 reactions
Forward primer(biobrick prefix)	1 µl	22 µl
Reverse primer(pcy A reverse primer: ATA TAT GAA GAC ATG AGG GGT CTC ATG TCT TAC TGG ATG ACA TCA AA)	1 µl	22 µl
Q-Amp master mix	10 µl	220 µl
dd water	8 µl	176 µl

2. Circle 20 colonies from plate: a) pSTK-1A-Pveg-RBS0-Ho1:pcyA-B0015
3. Pick up 20 colonies from the agar plate with tip and streak the labelled grid
4. Dispense tips with colonies into PCR strips and wait 5 mins
5. Take out tips and close lids
6. Place in PCR machine for thermal cycle

95 °C	2 min	1 cycle
95 °C	30 sec	30 cycles
65 °C	40 sec	
72 °C	2 min (bp 1800)	
72 °C	1 min	1 cycle

Done by Ryan Y, Joshua
Supervised by Chars

Date: 20260715
Title: Enzyme Digestion
Purpose: verify the effectiveness of BsaI-HFv2 RE
Procedure:

1. Calculate reaction formula

Master mix: 10X cutsmart buffer 2*8=16
BsaI:0.5*8=4 μL

Dilution	DNA	ddH_2O
B2	3.6 μL	13.7 μL
B3	1.6 μL	15.9 μL
B4	3 μL	14.5 μL
C1	3.05 μL	14.45 μL
C2	3.72 μL	13.77 μL
SamyQ	3.55 μL	13.95 μL
#4	2.9 μL	14.6 μL

2. Prepare the reaction according to the formula:dilute the DNA with ddH_2O first and make the master mix (8 reactions for pipetting error). Distribute the master mix into 7 of the tubes (seal tubes)
3. Incubate formula in water bath (seal microtubes with parafilm)

Done by Sophie C, Abigail
Supervised by Chars

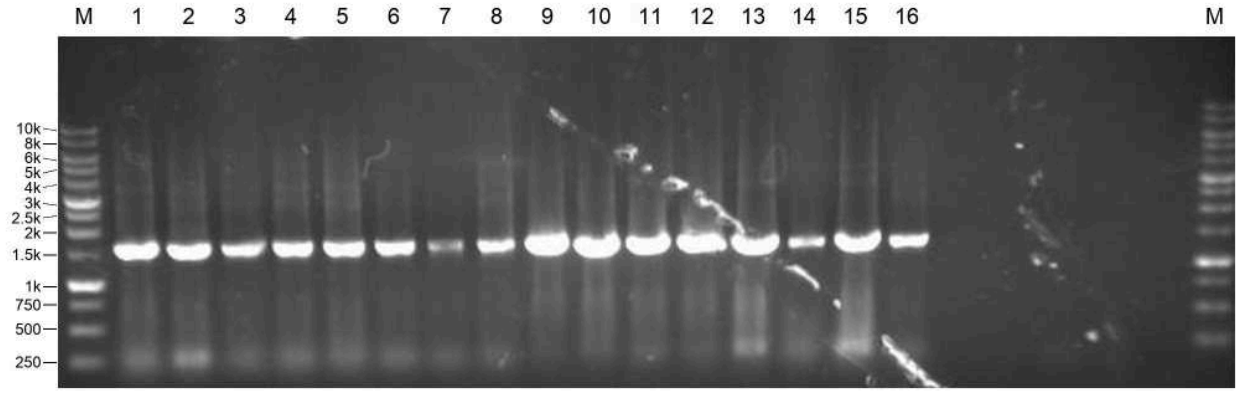
Date: 20260715
Title: Make 0.8 % Gel(x2)
Purpose: make agarose gel for electrophoresis
Procedure:

1. Assemble tray and mold, prepare comb
2. Weigh 0.48 g agarose powder and 60 ml 1xTAE solution
3. Add 1xTAE and powder to 250 ml conical flask
4. Cover flask with plastic wrap and puncture holes
5. Microwave for 30 seconds and agitate it. Repeat this step until all particles dissolved
6. Wait until the gel stops smoking, add 2 μl safety dye and agitate it
7. Pour the melted gel in the tray and assemble the comb
8. Leave to solidify in room temperature

Done by Abby T/K

Date: 20260715
Title: Gel electrophoresis
Purpose: verify colony PCR products
Procedure:

1. Load the gel in the layout shown below (5 μL each):
2. Resolve at 100 v for 30 min

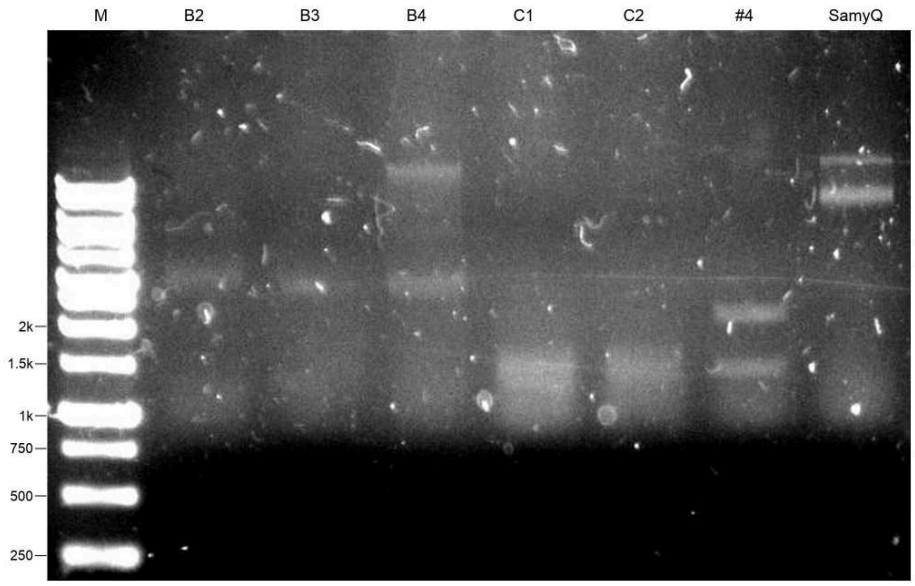


M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
1-16: pSTK-1A-Pveg-RBS0-Ho1:pcyA-B0015	biobrick prefix	pcyA reverse primer	1871
Note: 17-20 fail due to collected agar			

Done by: Abby K, Ryan Y
Supervised by: Chars

Date: 20260715
Title: Gel electrophoresis
Purpose: verify enzyme digestion products
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder		
Sample	Plasmid Construct	Expected Size (bp)
B	pSTK-EXP1-Pveg-EEBD-RBS0-GanA-Gala-B0015	8761
C	pSTK-EXP2-Pveg-EEBD-RBS0-Csn::LEA-B0015	
#4	pSTK-1A-Pveg-RBS0-ho1::pcyA-B0015	4032
SamyQ	Pveg-EEBD-RBS0-SamyQ-TATHA-Csn::AmilCP-B0015	7464

Done by: Sophie C, Ryan Y
Supervised by: Chars

Date: 20260715
Title: Transformation
Purpose: Transfer bacterial liquid on agar plate
Procedure
A: PlepA-RBS0-ccaSm3-B0015
C: pcpcG2-veg-EEBD-RBS0-Csn:ACCD-B0015
E: pcpcG2-veg-EEBD-RBS0-Csn:LEA-B0015

1. Turn on waterbath and set it to 42 °C and prepare ice bucket

2. Get LBA plate
3. Thaw competent cells on ice until they're 50% thawed and add 15 µL of plasmid
4. Incubate on ice for 20 min
5. Heat shock for 45 sec - 1 min
6. Recover on ice for 3 min
7. Add 1 mL soc medium & incubate at 37°C for 1 hour
8. Centrifuge at 5000 x 9 for 5 min and remove 1 mL supernatant
9. Resuspend pellet in remaining supernatant & spread on LBA plate
10. Incubate at 37 °C for 16 hours

Done by: Abby K, Chloe

Date: 20260715
Title: Transformation
Purpose: amplification of arrived level 0 DNA parts (plasmids)
Procedure:

A. pTwist-Amp-High-SamyQ-LEA-2.2
B. pTwist-Amp-High-SamyQ-AmilCP-2.1
C. pTwist-Amp-High-SamyQ-LEA-2.1
D. pTwist-Amp-High-SamyQ-ACCD-v3t1
E. pTwist-Amp-High-SamyQ-ACCD-2.1
F. pTwist-Amp-High-SamyQ-AmilCP-v3t1

1. Turn on water bath and set it to 42 °C
2. Get LBA plates and label corresponding DNA parts
3. Add 15 µL competent cells and 1 µL plasmid into 1.5 ml microtube
4. Incubate on ice for 45 sec
5. Recover on ice for 3 min

Done by Sophie H.

Date: 20260715
Title: Miniprep
Purpose: Collect concentrated bacterial liquid
Procedure:
B2, B3, B4: pSTK-Exp1 Pveg-EEBD-RBS0-GanA-Gala-B0015
C1, C2: pSTK-Exp1 Pveg-EEBD-RBS0-Csn:LEA-B0015
#4: pSTK-1A-Pveg-RBS0-Ho1:pcyA-B0015
SamyQ: Pveg-EEBD-RBS0-SamyQ-TAT-HA-Csn:AmilCP-B0015

1. Transfer 1.8mL of TB medium to each 2mL microtube
2. Centrifuge at 23444 g for 30 seconds. Discard the supernatant
3. Repeat steps 1-2 until all medium is collected. Remember to empty the medium in the pipette tips too.
4. Add 200 µl of Mx1 buffer

5. Vortex or pipette to resuspend
6. Add 250 µl of MX2 buffer
7. Invert slowly 8-10 times. Incubate at RT for 2 mins
8. Add 350 µl of MX3 buffer
9. Invert slowly 4-5 times
10. Centrifuge for 5 mins at 16000 g
11. Transfer ~900 µl of supernatant into assembled column(until all transferred) using a 200 µl pipette
12. Centrifuge at 7000 g for 30 seconds and discard flow through
13. Wash the column with 500 µl of WN buffer. Centrifuge at 7000 g for 30 seconds. Discard the flow through
14. Wash the column with 700 µl of WS buffer. Centrifuge at 7000 g for 30 seconds. Discard the flow through
15. Centrifuge for 8 mins at 10000 g
16. Place column into new 1.5mL microtube
17. Add 50 µl of EB buffer to the center of the membrane
18. Incubate for 3 mins at RT
19. Centrifuge at 10000 g for 3 mins

Biodrop results:

	Concentration (ng/µl)	A260/280
B2	139	1.958
B3	312	2.026
B4	168	1.976
C1	164	1.843
C2	134	1.836
SamyQ	141	1.958
#4	174	2.000

Done by Abigail and Sophie C

Date: 20260716
Topic: Prepare LB plates
Purpose: Prepare plates for bacteria growing
Procedure:

1. Formula as below
    a. LB broth powder: 6.25 g
    b. Agar Powder: 3.75 g
    c. ddH_2O: 250 mL
2. Add one magnetic stirring bar to the flask
3. Place the flask containing the mixture on the hot plate and stir until no sediments
4. Remove the stirring bar from the flask
5. Cover the flask with aluminum foil and apply an indicator tape to make sure the mixture is successfully autoclaved
6. Autoclave for 20 minutes at 127 °C.
7. Place the new petri dishes in BSC(adjust sash)
8. Set the water bath at 55 °C

Done by Sophie C/H, Sarah, Naomi

Date: 20260716
Topic: Prepare 1/2 MS medium(hydroponics medium)
Purpose: prepare medium for hydroponic testing
Procedure:

1. The following protocol is for 1000mL of 1/2 MS medium
    a. 1/2 MS medium powder(w/o) sucrose: 2.3 g
    b. MES buffer: 0.5 g
    c. Distilled water: 1000 mL
2. Add the contents in the 2000 mL beaker
3. Use a hot plate to stir(with a magnetic stir bar) until no sediments are left in the solution
4. Use the pH probe and measure the pH of the solution, ensure the pH is 5.7-5.8
    a. pH before adjustment: 4.1
    b. pH after adjustment: 5.7
5. If the pH is too low, add 0.1 M KOH slowly to increase it
6. Wrap the beaker with aluminum foil and stick a piece of autoclave indicator tape on top
7. Autoclave for 20 minutes at 127 °C

Done by Sophie C/H, Sarah, Naomi, Alex

Date: 20260716
Topic: Colony PCR
Purpose: screen for successful cloning from 0715 transformation
Procedure:
A: pSTK-1B-PlepA-RBS0-CcaSm3-B0015 (pcr product bp:1667)
E:pSTK-1D-PcpcG2-veg-EEBD-RBS0-Csn:LEA-B0015 (pcr product bp:1135)
C: pSTK-1D-PcpcG2-veg-EEBD-RBS0-Csn:ACCD-B0015 (pcr product bp:1596)

1. Prepare the following reaction (3 master mix for each plate) and dispense solutions into PCR tubes

A: pSTK-1D-PlepA-RBS0-CcaSm3-B0015 (pcr product bp:1667)

	1 reaction	12 reactions
Forward primer (Biobrick Prefix)	1 μL	12 μL
CDS primer (CcaSm3 R, ATA TAT GAA GAC ATG AGG GGT CTC ATG TCT TAA GCT CGA GGC AAA TGG T)	1 μL	12 μL
Q-amp master mix	10 μL	120 μL
ddH_2O	8 μL	96 μL

E:pSTK-1D-PcpcG2-veg-EEBD-RBS0-Csn:LEA-B0015 (pcr product bp:1135)

	1 reaction	12 reactions
Forward primer (Biobrick Prefix)	1 μL	12 μL
CDS Reverse primer (LEA R, ATA TAT GAA GAC ATG AGG GGT CTC ATG TCT TAA TGA TGA TGA TGA TGA T)	1 μL	12 μL
Q-amp master mix	10 μL	120 μL
ddH_2O	8 μL	96 μL

C: pSTK-1D-PcpcG2-veg-EEBD-RBS0-Csn:ACCD-B0015 (pcr product bp:1596)

	1 reaction	12 reactions
Forward primer (Biobrick Prefix)	1 μL	12 μL
CDS reverse primer (ACCD R, ATA TAT GAA GAC ATG AGG GGT CTC ATG TCT TAA TGA TGA TGA TGA TGA T)	1 μL	12 μL
Q-amp master mix	10 μL	120 μL

ddH_2O	8 µL	96 µL

2. Circle 10 colonies each from plate
3. Pick up 10 colonies from each agar plate with tip and streak the labelled grid
4. Dispense the tips with colonies into the PCR strips and wait 5 minutes
5. Take out the tips and close the lid
6. Place into PCR machine for thermal cycle

95 ℃	2 min	1 cycle
95 ℃	30 sec	30 cycles
65 ℃	40 sec	
72 ℃	2 min	
72 ℃	1 min	1 cycle

Done by Abby T/K, Abigail, Mia, Audrey C, Olivia D
Supervised by Chars

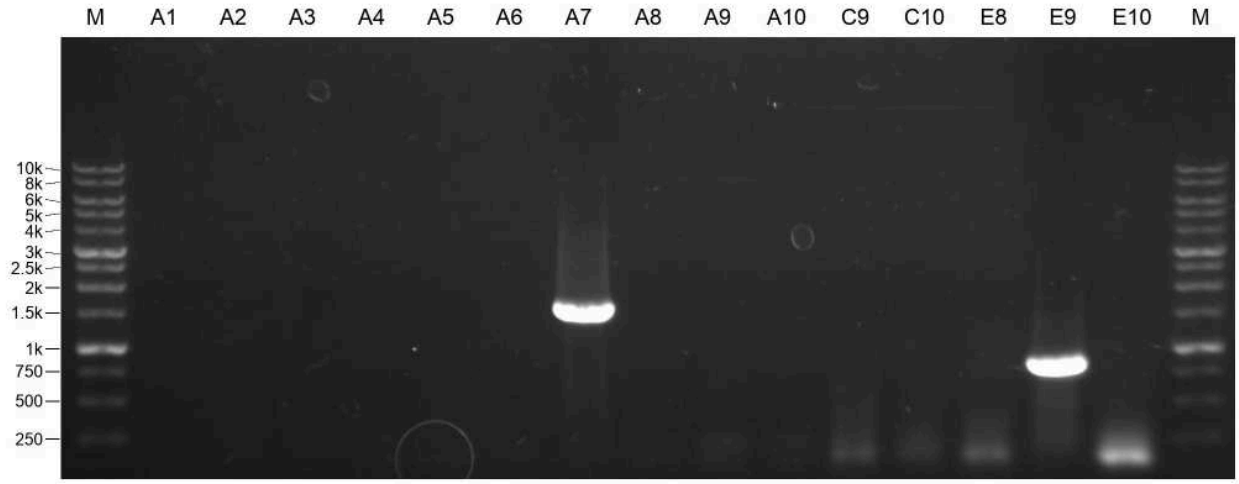
Date: 20260716
Topic: Prepare 0.8 % Gel for electrophoresis
Purpose: prepare gel for electrophoresis for colony PCR
Procedure:
A: pSTK-1B-PlepA-RBS0-CcaSm3-B0015 (pcr product bp:1667)
E:pSTK-1D-PcpcG2-veg-EEBD-RBS0-Csn:LEA-B0015(pcr product bp:1135)
C: pSTK-1D-PcpcG2-veg-EEBD-RBS0-Csn:ACCD-B0015(pcr product bp:1596)

1. Assemble the tray, the mold and the comb (x2)
2. Weigh 0.32 g of agarose powder and add to the first flask
3. Add 40 mL of 1x TAE buffer to the flask
4. Repeat step 2 and 3 for another flask
5. Cover the flasks with plastic wrap and puncture holes on the wrap
6. Microwave until all particles are dissolved
7. Wait until the gel stops smoking but is still hot
8. Add 2 µl of safety dye to each gel
9. Pour the gel to the assembled trays and use a tip to squeeze out the bubbles
10. Leave it to solidify

Done by Naomi, Sophie C/H, Sarah C

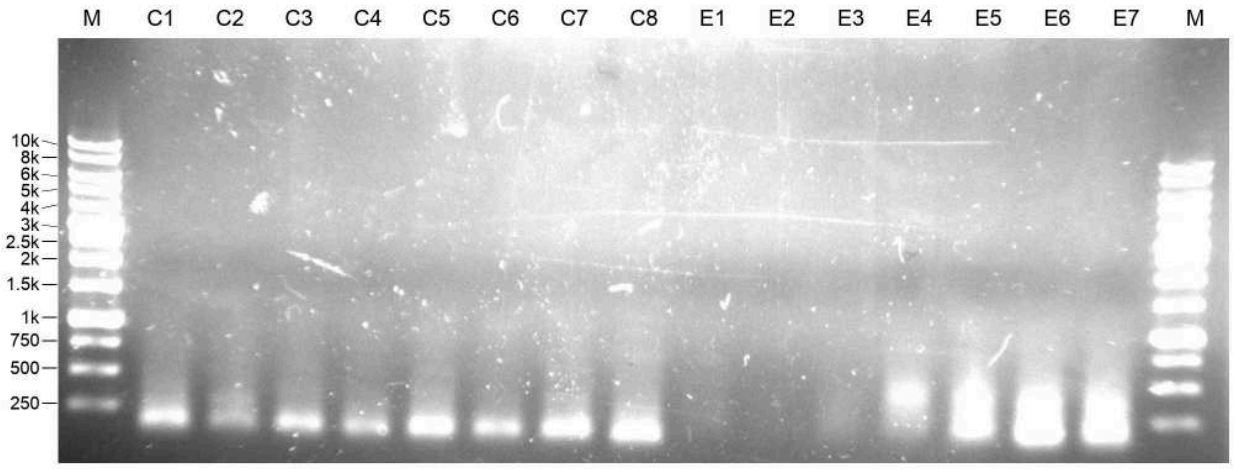
Date: 20260716
Topic: Gel electrophoresis of colony PCR products
Purpose: verify plasmid sizes
Procedure:

1. Load the gel in the layout shown below (5 μL for M; 6 μL for A, C, E):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
A1-A10: pSTK-1B-PlepA-RBS0-Ccasm3-B0015	BioBrick Prefix	Ccasm3 Reverse	1667
C9-C10: pSTK-1D-PcpcG2-veg-EEBD-RBS0-csn:ACCD-B0015	BioBrick Prefix	ACCD Reverse	1696
E8-E10: pSTK-1D-PcpcG2-veg-EEBD-RBS0-csn:LEA-B0015	BioBrick Prefix	LEA Reverse	1135

3. Load the gel in the layout shown below (5 μL for M; 6 μL for C, E):
4. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
C1-C8: pSTK-1D-PcpcG2-veg-EEBD-RBS0-csn:ACCD-B0015	BioBrick Prefix	ACCD Reverse	1696

Done by Sarah C, Naomi, Sophie H, Audrey C

Date: 20260716
Topic: Golden Gate assembly
Purpose: redo of golden gate assembly of A, C, E constructs
Procedure:
C: Module 2-PlepA-RBS0-CcaSm3-B0015
I: Module 4- PcpcG2-veg-EEBD-RBS0-Csn:ACCD-B0015
G: Module 4-PcpcG2-veg-EEBD-RBS0-Csn:Lea-B0015

C: Module 2-PlepA-RBS0-CcaSm3-B0015

1. Prepare the reaction as follow

Vector (1B) ● Concentration (ng/μL): 151 ● bp: 3120	1.3 μl (dilute 1/10)
Insert promoter (PlepA) ● Concentration (ng/μL): 155 ● bp: 2241	1.9 μl (dilute 1/10)
Insert RBS (RBS0)	1.5 μl (dilute 1/10)

• Concentration (ng/μL): 188 • bp: 2113	
Insert CDS (CcaSm3) • Concentration (ng/μL): 128 • bp: 1465	1.5 μL(dilute 1/10)
Insert Terminator (B0015) • Concentration (ng/μL): 107 • Bp: 2213	1.3 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL
Sterile water	8.5 μL
T4 DNA ligase	1 μL
BsaI-HFv2	1 μL
Total	20 μL

2. Use the thermocycler to run the cycle below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	5 minutes	45 cycles
16 °C (Ligation)	5 minutes	
37 °C (Restriction enzyme digestion)	10 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	20 minutes	1 cycle

I: Module 4-PcpcG2-veg - EEBD- RBS0 - CSN::ACCD - B0015

1. Prepare the reaction as follow:

Vector (1D) ● concentration(ng/μL): 132 ● bp: 3120	1.5 μL (dilute 1/10)
Insert promoter(PcpcG2-veg) ● concentration(ng/μL): 65 ● bp: 235	1.2 (dilute 1/50)
Insert RBS(EEBD-RBS0) ● concentration(ng/μL): 150 ● bp: 2392	1.0 μL (Dilute 1/5)
Insert CDS(Csn::ACCD) ● concentration(ng/μL): 142 ● bp: 1195	1.0 μL (dilute 1/10)
Insert Terminator (B0015) ● concentration(ng/μL): 107 ● bp: 2213	1.3 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL
Sterile water	10.4 μL
T4 DNA ligase	1 μL
BsaI-HFv2	1 μL
Total	20 μL

2. Use the thermocycler to run the cycle below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	5 minutes	45 cycles
16 °C (Ligation)	5 minutes	

37 °C (Restriction enzyme digestion)	10 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	20 minutes	1 cycle

G: Module 4-PcpcG2-veg - EEBD- RBS0 - Csn::LEA - B0015

1. Prepare the reaction as follow:

Vector (1D) ● concentration(ng/μL): 132 ● bp: 3120	1.5 μL (dilute 1/10)
Insert promote r(PcpcG2-veg) ● concentration(ng/μL): 65 ● bp: 235	1.2 μL (dilute 1/50)
Insert RBS (EEBD-RBS0) ● concentration(ng/μL): 150 ● bp: 2392	1.0 μL (Dilute 1/5)
Insert CDS (Csn::LEA) ● concentration(ng/μL): 77 ● bp: 634 bp	1.1 μL (dilute 1/20)
Insert Terminator (B0015) ● concentration(ng/μL): 107 ● bp: 2213	1.3 μL (dilute 1/5)
T4 DNA ligase buffer	2 μL
Sterile water	10.1 μL
T4 DNA ligase	1 μL
BsaI-HFv2	1 μL
Total	20 μL

2. Use the thermocycler to run the cycle below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	5 minutes	45 cycles
16 °C (Ligation)	5 minutes	
37 °C (Restriction enzyme digestion)	10 minutes	1 cycle
60 °C (Heat inactivation of BsaI restriction enzyme)	20 minutes	1 cycle

Done by Sophie C, Abby T/K

Date: 20260717
Title: Miniprep
Purpose: To isolate and purify small amounts of plasmid DNA from bacterial cultures done in 20260716
Procedure:
A: pTwist-SamyQ-LEA-2.2
B: pTwist-SamyQ-AmilCP-2.1
C: pTwist-SamyQ-LEA-2.1
D: pTwist-SamyQ-ACCD-v3t1
E: pTwist-SamyQ-ACCD-2.1
F: pTwist-SamyQ-AmilCP-v3t1

1. Transfer 1.8 mL of TB medium to each 2 mL microtube
2. Centrifuge at 23444 g for 30 seconds. Discard the supernatant
3. Repeat steps 1-2 until all medium is collected. Remember to empty the medium in the pipette tips too.
4. Add 200 µl of Mx1 buffer
5. Vortex or pipette to resuspend
6. Add 250 µl of Mx2 buffer
7. Invert slowly 8-10 times. Incubate at RT for 2 mins
8. Add 350 µl of Mx3 buffer
9. Invert slowly 4-5 times
10. Centrifuge for 5 mins at 16000 g
11. Transfer ~900 µl of supernatant into assembled column(until all transferred) using a 200 µl pipette
12. Centrifuge at 7000 g for 30 seconds and discard flow through
13. Wash the column with 500 µl of WN buffer. Centrifuge at 7000 g for 30 seconds. Discard the flow through
14. Wash the column with 700 µl of WS buffer. Centrifuge at 7000 g for 30 seconds. Discard the flow through
15. Centrifuge for 8 mins at 10000 g
16. Place column into new 1.5mL microtube
17. Add 50 µl of EB buffer to the center of the membrane
18. Incubate for 3 mins at RT
19. Centrifuge at 10000 g for 3 mins

Results:

BioDrop Results	Concentration (ng/µL)	A260/230	A260/280
A	111.0	1.982	1.121
B	138.0	2.091	1.366
C	141.0	1.932	1.014

D	142.0	1.945	1.068
E	125.0	1.923	1.608
F	150.0	1.829	0.789
5-2	147.0	1.909	0.987
8-2	216.0	2.038	1.350
14-2	143.0	1.882	0.935

Date: 20260717
Title: Golden Gate Assembly
Purpose: assemble level 1 genetic circuit
Procedure:

1. Prepare the reaction as follows:

F: Pveg-EEBD-RBS0-GanA-Gala-B0015

Part	Original	Dilution
Vector (shuttle vector): pSTK-EXP-1-sfGFP ● concentration(ng/μL): 162 ● bp: 6929	0.278015 μL	1.39 μL (dilute 1/5)
Insert promoter (Pveg) ● concentration(ng/μL): 280 ● bp: 2361	0.109618 μL	1.1 μL (dilute 1/10)
Insert RBS (EEBD-RBS0) ● concentration(ng/μL): 150 ● bp: 2113	0.207306 μL	2.1 μL(dilute 1/10)
Insert CDS (GanA-Gala) ● concentration(ng/μL): 177 ● bp: 2080	0.1527683616 μL	1.53 (dilute 1/10) μL
Insert Terminator (B0015) ● concentration(ng/μL): 107 ● bp: 2213	0.268869 μL	1.34 μL (dilute 1/5)
Spacer (pSTK-OCX-spacer) ● concentration(ng/μL): 127 ● bp: 2095	0.10722440 μL	1.07 μL (dilute 1/10)

Histag (pSTK-OCZ-HisGX) ● concentration(ng/μL): 135 ● bp: 2402	0.1156185 μL	1.16 μL (dilute 1/10)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water	μL	8.54 μL
T4 DNA ligase	1 μL	1 μL
BsaI-HFv2	1 μL	1 μL
Total	20 μL	20 μL

2. Run the thermal cycle as following:

Temperature	Time	Cycles
37°C (Restriction enzyme digestion)	5 minutes	45 cycles
16°C (Ligation)	5 minutes	
37°C (Restriction enzyme digestion)	10 minutes	1 cycle
60°C (Heat inactivation of BsaI restriction enzyme)	20 minutes	1 cycle

Done by Abby T, Abby K

Date: 20260717
Title: Transformation
Purpose: Transform golden gate assembly parts from 7/16 on agar plate with antibiotics to verify if the plasmids are successfully taken up by the bacteria through antibiotic selection
Procedure:

C: (STK) M2-PlepA-RBS0-CcaSm3-B0015
I: (STK) M4-PcpcG2-veg-EEBD-RBS0-Csn:ACCD-B0015

G: (STK) M4-PcpcG2-veg-EEBD-RBS0-Csn:Lea-B0015

1. Turn on the water bath, set it to 42 °C and prepare the ice.
2. Prepare LBA plates without glycerol
1. Thaw competent cells on ice until they're 50 % thawed and add 10 µL of plasmid into 100 µL competent cells (don't pipette)
2. Incubate on ice for 20 minutes.
3. Heat shock at 42 °C for 45 seconds.
4. Incubate on ice for 3 minutes.
5. Add 1 mL soc medium & incubate at 37 °C for 1 hour
6. Centrifuge 5000 g for 5 minutes.
7. Take out 1000 ml and resuspend the cell.
8. Spread on the agar plate with Ampicillin.
9. Place in incubator for 37 ℃ overnight.

Done by Chloe, Abigail
Supervised by Chars

Date: 20260717
Title: Transformation
Purpose: transform single construct and culture bacteria to verify if the plasmids are successfully taken up by the bacteria through antibiotic selection
Procedure:
pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-TAT-HA-AmilCP-B0015

1. Turn on the water bath, set it to 42 °C and prepare the ice.
2. Prepare LBA plates with glycerol
3. Thaw competent cells on ice until they're 100 % thawed
4. Take out 1 µL of plasmid into 15 µL competent cells (don't pipette)
5. Incubate on ice for 20 minutes.
6. Heat shock at 42 °C for 45 seconds.
7. Incubate on ice for 3 minutes.

Done by Abby K, Sophia Yeh
Supervised by Chars

Date: 20260718
Title: miniprep
Purpose: plasmid purification
Procedure:
A: pSTK-0A-Pveg
B: pSTK-0B-EEBD
C: pSTK-0D-B0015

1. Transfer 1.8 mL of TB medium to each 2 mL microtube
2. Centrifuge at 23444 g for 30 seconds. Discard the supernatant
3. Repeat steps 1-2 until all medium is collected. Remember to empty the medium in the pipette tips too.
4. Add 200 µl of Mx1 buffer
5. Vortex or pipette to resuspend
6. Add 250 µl of MX2 buffer
7. Invert slowly 8-10 times. Incubate at RT for 2 mins
8. Add 350 µl of MX3 buffer
9. Invert slowly 4-5 times
10. Centrifuge for 5 mins at 16000 g
11. Transfer ~900 µl of supernatant into assembled column(until all transferred) using a 200 µl pipette
12. Centrifuge at 7000 g for 30 seconds and discard flow through
13. Wash the column with 500 µl of WN buffer. Centrifuge at 7000 g for 30 seconds. Discard the flow through
14. Wash the column with 700 µl of WS buffer. Centrifuge at 7000 g for 30 seconds. Discard the flow through
15. Centrifuge for 8 mins at 10000 g
16. Place column into new 1.5 mL microtube
17. Add 50 µl of EB buffer to the center of the membrane
18. Incubate for 3 mins at RT
19. Centrifuge at 10000 g for 3 mins

Biodrop Results	Concentration ng/µL	A260/230	A260/280
A	168	2.012	1.263
B	151	1.936	0.950
C	148	1.922	1.050

Done by Sarah, Naomi, Sophie C.

Date: 20260718
Title: Colony PCR
Purpose: screen for successful cloning from 0715 transformation

Procedure:

1. Prepare the following reaction (3 master mix for each plate) and dispense solutions into PCR tubes

C: pSTK-1B-PlepA-RBS0-CcaSm3-B0015 (pcr product bp:1667)

	1 reaction	10 reactions
Forward primer (Biobrick Prefix)	1 µL	10 µL
CDS primer (CcaSm3 R, ATA TAT GAA GAC ATG AGG GGT CTC ATG TCT TAA GCT CGA GGC AAA TGG T)	1 µL	10µL
Q-Amp master mix	10 µL	100 µL
ddH_2O	8 µL	80 µL

G: pSTK-1D-PcpcG2-veg-EEBD-RBS0-Csn:LEA-B0015 (pcr product bp:1135)

	1 reaction	10 reactions
Forward primer (Biobrick Prefix)	1 µL	10 µL
CDS Reverse primer (LEA R, ATA TAT GAA GAC ATG AGG GGT CTC ATG TCT TAA TGA TGA TGA TGA TGA T)	1 µL	10 µL
Q-Amp master mix	10 µL	100µL
ddH_2O	8 µL	80 µL

I. pSTK-1D-PcpcG2-veg-EEBD-RBS0-Csn:ACCD-B0015 (pcr product bp:1596)

	1 reaction	10 reactions
Forward primer (Biobrick Prefix)	1 µL	10 µL
CDS reverse primer (ACCD R, ATA TAT GAA GAC ATG AGG GGT CTC ATG TCT TAA TGA TGA TGA TGA TGA T)	1 µL	10 µL

Q-Amp master mix	10 µL	100 µL
ddH_2O	8 µL	80 µL

2. Circle 8 colonies each from plate
3. Pick up 8 colonies from each agar plate with tip and streak the labelled grid
4. Dispense the tips with colonies into the PCR strips and wait 5 minutes
5. Take out the tips and close the lid
6. Place into PCR machine for thermal cycle

95 ℃	2 min	1 cycle
95 ℃	3 sec	30 cycles
65 ℃	40 sec	
72 ℃	1 min 45 sec	
72 ℃	1 min	1 cycle

Done by Abby T/K, Audrey H

Date: 20260718
Title: prepare 2 plates of 0.8 % agarose gel (25 holes each)
Purpose: for electrophoresis running for 20260717 colony PCR
Procedure:

1. Weight 0.32 g of agarose powder and 40 ml of TAE buffer into the flask and mix by shaking (x2)
2. Cover the flask with plastic wrap and poke holes on it by using tip
3. Put it into the microwave and heat until the mixture become clear and with no
4. Remove the plastic wrap and wait until the mixture becomes less hot
5. Add 4 µL of safety dye and mix by shaking
6. Pour the gel evenly into the two plates and wait for them to solidify

Done by Sophie C Sophie H
ELN by Sophie H

Date: 20260718
Title: prepare 2 plates of 0.8 % agarose gel (25 holes each)
Purpose: for electrophoresis running for 0718 colony PCR
Procedure:

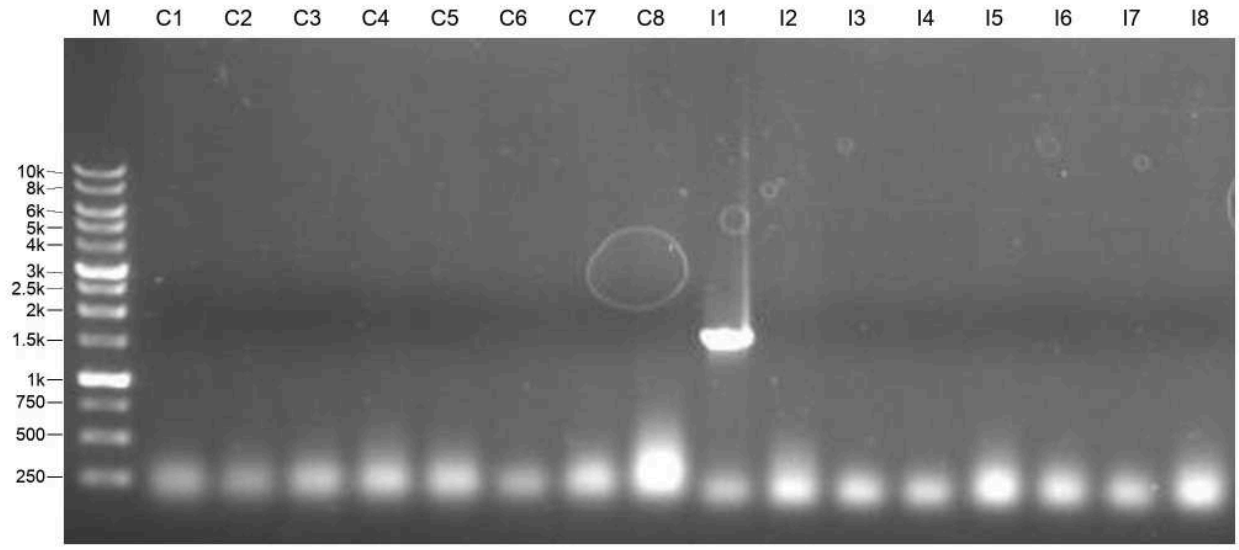
1. Weight 0.32 g of agarose powder and 40 ml of TAE buffer into the flask and mix by shaking (x2)
2. Cover the flask with plastic wrap and poke holes on it by using tip
3. Put it into the microwave and heat until the mixture become clear and with no

4. Remove the plastic wrap and wait until the mixture becomes less hot
5. Add 4 μL of safety dye and mix by shaking
6. Pour the gel evenly into the two plates and wait for them to solidify

Done by Audrey H
Supervised by Abby T

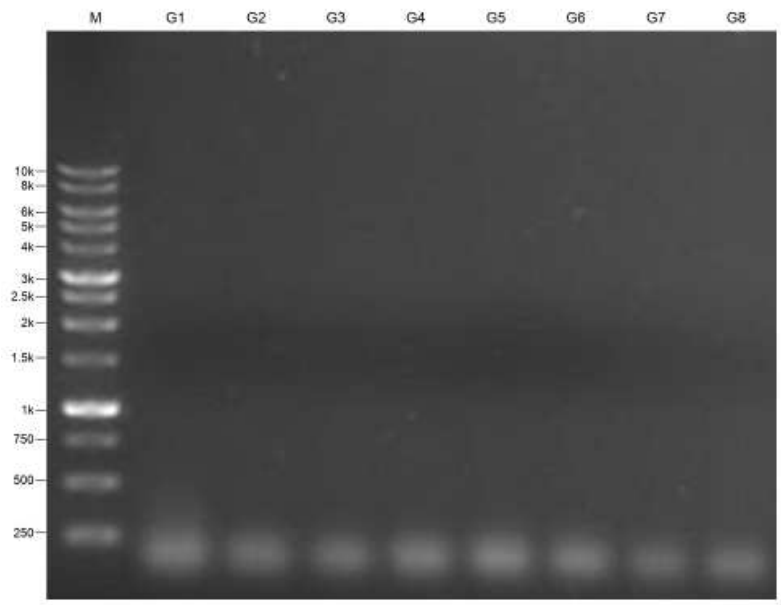
Date: 20260718
Title: Gel electrophoresis
Purpose: verify the size of the 0718 colony PCR product
Procedure:

1. Load the gel in the layout shown below (5 μL for M; 6 μL for C, I):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
C1-C8: pSTK-1B-PlepA-Ccasm3-B0015	biobrick prefix	CcaSm3 R	1596
I1-I8: pSTK-1D-PcpcG2-veg-csn:ACCD-B0015	biobrick prefix	ACCD R	1667

3. Load the gel in the layout shown below (5 μL for M; 6 μL for G):
4. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
G1-G8: pSTK-1D-PcpcG2-veg-csn:LEA-B0015	biobrick prefix	LEA R	1135

Done by Sophie H.

Date: 20260718
Title: Refilling LBA plates
Purpose: prepare agar plates for culturing bacteria
Procedure:

1. Formula as below:
    a. Broth powder: 6.25 g
    b. Agar powder: 3.75 g
    c. ddH_2O: 250 mL
2. Add one magnetic stirring bar to the flask
3. Place the flask containing the mixture on the hot plate, and stir until no sediments
4. Remove the stirring bar from the flask
5. Cover the flask with aluminum foil and apply an indicator tape to ensure the mixture is successfully autoclaved
6. Autoclave for 20 minutes at 127 °C.
7. Place the new petri dishes in BSC and turn on the UVC
8. Set up water bath at 55 ℃
9. Spray the working surface of the BSC and equipments with 70 % ethanol

10. Add 500µL of antibiotics (ampicillin) into each corresponding flask
11. Use serological pipette to add 10-15 mL of melting LB agar into the petri dish, avoid overfilling (pipette first to mix the antibiotics)
12. Arrange the petri dishes in stacks within the cabinet
13. Let the agar plates dry for at least 1 hour in the BSC

Done by Sophie C, Naomi, Sophie H

Date: 20260719
Title: Colony PCR (redo 0718 colony PCR)
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:

1. Prepare the following reaction (3 master mix for each plate) and dispense solutions into PCR tubes

C: pSTK-1B-PlepA-RBS0-CcaSm3-B0015 (PCR product bp:1667)

	1 reaction	10 reactions
Forward primer (Biobrick Prefix)	1 μL	10 μL
CDS primer (CcaSm3 R, ATA TAT GAA GAC ATG AGG GGT CTC ATG TCT TAA GCT CGA GGC AAA TGG T)	1 μL	10 μL
Q-amp master mix	10 μL	100 μL
ddH_2O	8 μL	80 μL

G: pSTK-1D-PcpcG2-veg-EEBD-RBS0-Csn:LEA-B0015 (PCR product bp:1135)

	1 reaction	10 reactions
Forward primer (Biobrick Prefix)	1 μL	10 μL
CDS Reverse primer (LEA R, ATA TAT GAA GAC ATG AGG GGT CTC ATG TCT TAA TGA TGA TGA TGA T)	1 μL	10 μL
Q-amp master mix	10 μL	100 μL
ddH_2O	8 μL	80 μL

I. pSTK-1D-PcpcG2-veg-Pveg-EEBD-RBS0-Csn:ACCD-B0015 (PCR product bp:1596)

	1 reaction	10 reactions
Forward primer (Biobrick Prefix)	1 μL	10 μL
CDS reverse primer (ACCD R, ATA TAT GAA GAC ATG	1 μL	10 μL

AGG GGT CTC ATG TCT TAA TGA TGA TGA TGA TGA T)		
Q-amp master mix	10 µL	100 µL
ddH_2O	8 µL	80 µL

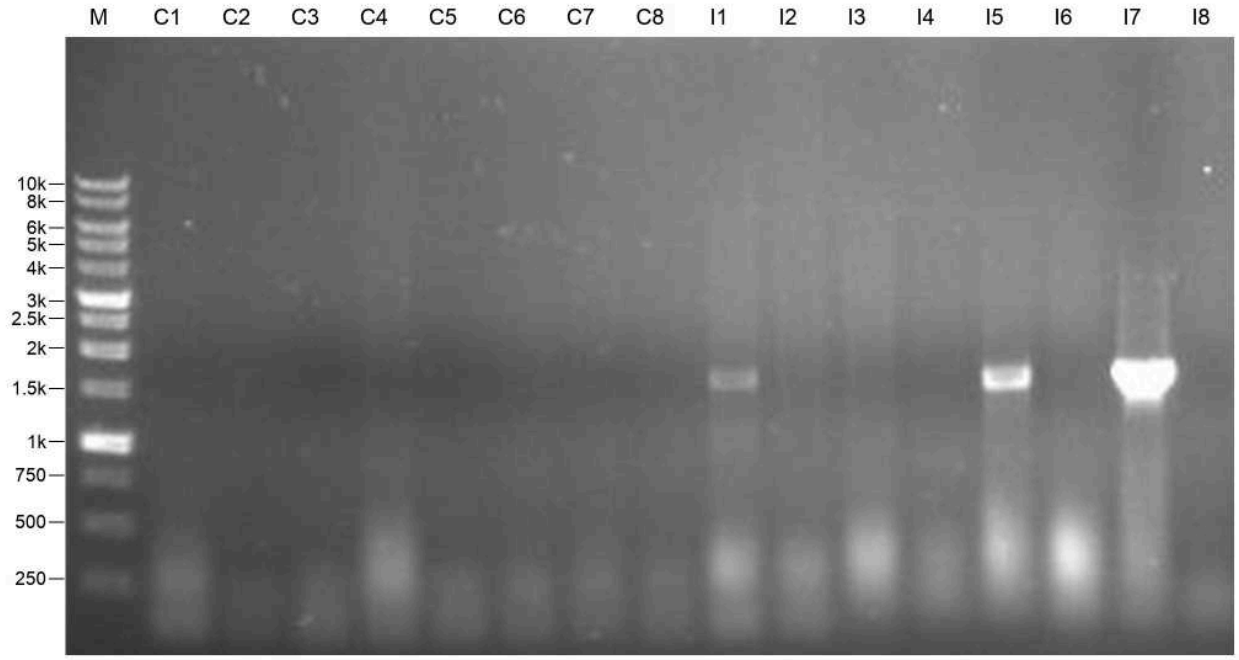
2. Circle 8 colonies each from plate
3. Pick up 8 colonies from each agar plate with tip and streak the labelled grid
4. Dispense the tips with colonies into the PCR strips and wait 5 minutes
5. Take out the tips and close the lid
6. Place into PCR machine for thermal cycle

95 ℃	2 min	1 cycle
95 ℃	3 sec	30 cycles
65 ℃	40 sec	
72 ℃	1 min 45 sec	
72 ℃	1 min	1 cycle

Done by Sophie H.

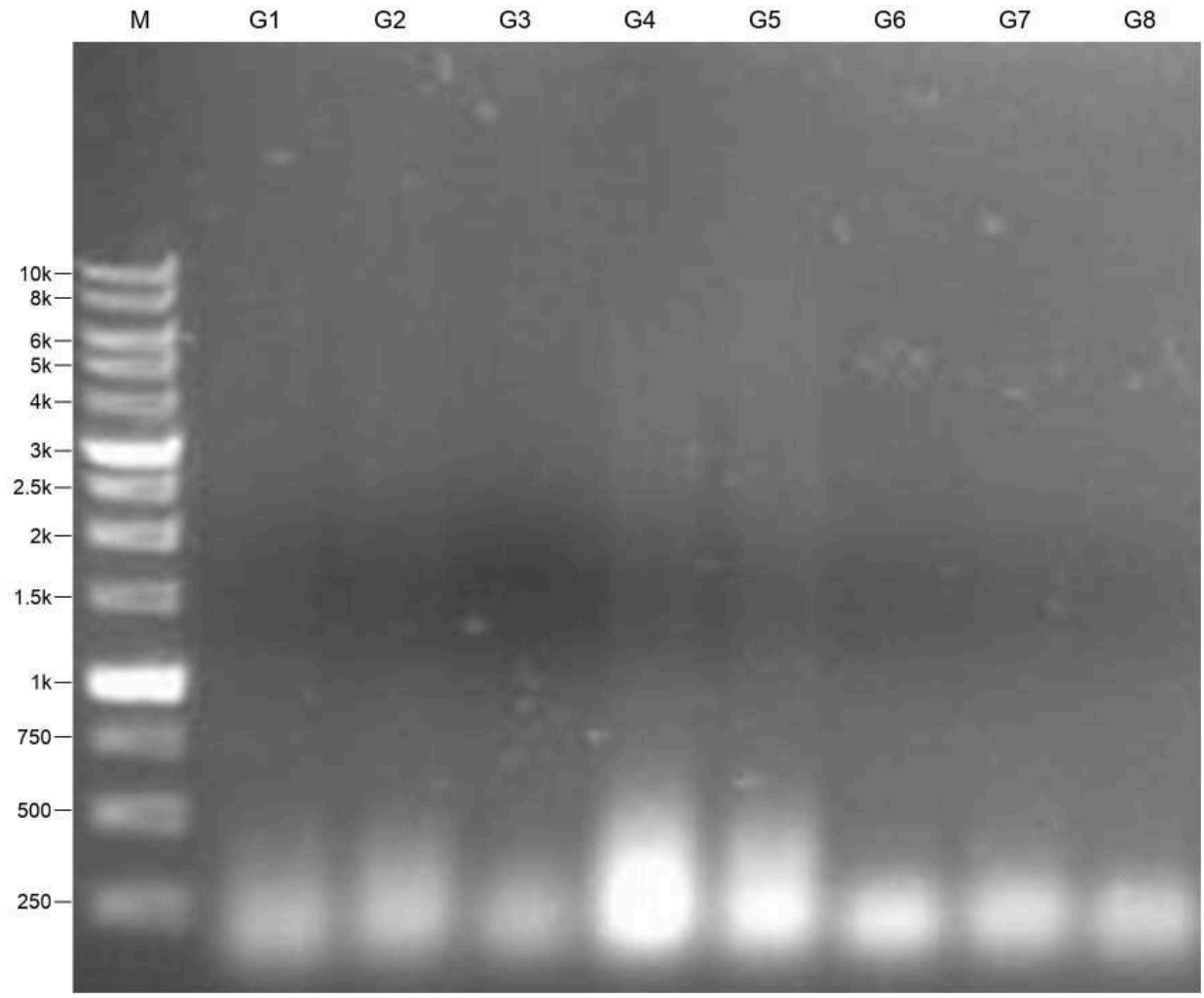
Date: 20260719
Title: Gel electrophoresis
Purpose: verify the sizes of the colony PCR product
Procedure:

1. Load the gel in the layout shown below (5 µL for M; 6 µL for C, I):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
C1-C8: pSTK-1B-PlepA-Ccasm3-B0015	biobrick prefix	CcaSm3 R	1596
I1-I8: pSTK-1D-PcpcG2-veg-Csn:ACCD-B0015	biobrick prefix	ACCD R	1667

3. Load the gel in the layout shown below (5 µL for M; 6 µL for G):
4. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
G1-G8: pSTK-1D-PcpcG2-veg-EEBD-RBS0-csn:LEA-B0015	biobrick prefix	LEA R	1135

Done by Ryan Y, Olivia Le

Date: 20260719
Title: Transformation
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:

1. Turn on the water bath and set it to 42°C

2. Prepare an ice bucket
3. Thaw the competent cells on ice until 50% thawed
4. Add 5 µl of plasmid to 100 µl of competent cell(do not pipette)
5. Place the mixture on ice for 20 minutes
6. Place the tubes in a 42°C at water bath for 45 seconds-1 minute
7. Place tubes back on ice for 3 minutes to recover
8. Add 1 mL of soc medium under a bunsen burner
9. Incubate at 37°C for 1 hour
10. Centrifuge for 5 minutes at 5000 g
11. Remove 1 mL of supernatant
12. Resuspend the cell pellet in the remaining supernatant
13. Spread the mixture on LBA plate
14. Incubate at 37°C for 16 hours

Done by Abby T, Audrey H

Date: 20260719
Title: Plant seeds for Exp set 6 and exp set 7 vernalization
Purpose: plant 4 neutral plates (27 seeds/plate)
Procedure:

1. Prepare 4 12x12cm, 1/2 MS medium, pH 5.8, 1% square agar plates (neutral)
2. Sanitize your hands and all equipments:
    a. 1/2 MS medium square agar plates
    b. Autoclaved toothpicks
    c. Sterile water
    d. Planting grid line paper then place in BSC
3. Open the aluminum foil of the autoclaved toothpicks and take one out
4. Dip the tip of the toothpick in the sterile water
5. Attach seeds one at a time and tip it on neutral plates base on the template at the bottom
6. Seal with tape and label the date of vernalization

Done by: Abby T, Audrey H
Recorded by: Audrey H

Date: 20260719
Title: Plant seeds for set A and vernalization in hydroponic system
Purpose: stabilizing plant growth
Procedure:

1. Prepare 2000 mL 1/2 MS medium using the following protocol
    a. pH: 5.72

Material	volume
1/2 MS medium powder (w/o sucrose)	4.6g

MES buffer	1.0g
Distilled water	2000mL

2. Microwave the centrifuge tubes with 1X 1/2 MS agar from 7/12 and heat it until it melts (agar is 1% agar)
3. Wait until the agar cools down a little bit
4. Prepare 48 eppendorf lids
5. Add 250 µl agar into each eppendorf lids
6. Wait until the agar solidifies
7. Add 2.2mL of medium into 48 tube rack
8. Put the lids with agar onto the tube rack with medium (hole facing up)
9. Use an autoclaved toothpick to pick up one seed and add the seed into the hole of the lid
10. Repeat step 9 until all lids have seeds
11. Wrap the tube rack with aluminium foil
12. Place the tube rack with aluminium foil into the refrigerator in 4℃
13. Wait 4 days for seeds to vernalize

Done by: Alex
Recorded by: Audrey

Date: 20260720
Title: Golden Gate Assembly of plasmid into Lv.0 vector
Purpose: to assemble the following parts to STK level 0 vector
Procedure:

A: pTwist-SamyQ-LEA-2.1
B: pTwist-SamyQ-LEA-2.2
C: pTwist-SamyQ-ACCD-2.1
D: pTwist-SamQ-ACCD-v3t1
E: pTwist-SamyQ-AmilCP-2.1
F: pTwist-SamyQ-AmilCP-v3t1

1. Prepare the reaction as follow:

A: SamyQ-LEA-2.1

Component	Amount	Dilution
Vector STK001_pSTK-0-sfGFP ● Concentration (ng/μL): 346 ● bp: 3090	0.05804913295 μL	1.16 μL (dilute 1/20) 0.5 μL vector 9.5 μL ddH_2O
Insert SamyQ-LEA-2.1 ● Concentration (ng/μL): 141 ● bp: 2840	0.2618439716 μL	1.047375886 μL (dilute 1/4) 1 μL vector 3 μL ddH_2O
T4 DNA ligase buffer	2 μL	
Sterile water	13.79 μL	
T4 DNA ligase	1 μL	
BsaI-HFv2	1 μL	
Total	20 μL	

B: SamyQ-LEA-2.2

Component	Amount	Dilution

Vector STK001_pSTK-0-sfGFP ● Concentration (ng/µL): 346 ● bp: 3090	0.05804913295 µL	1.16 µL (dilute 1/20) 0.5 µL vector 9.5 µL ddH_2O
Insert SamyQ-LEA-2.2 ● Concentration (ng/µL): 111 ● bp: 2840	0.333	1.33 µL (dilute 1/4) 1 µL insert 3 µL ddH_2O
T4 DNA ligase buffer	2 µL	
Sterile water	13.51 µL	
T4 DNA ligase	1 µL	
BsaI-HFv2	1 µL	
Total	20 µL	

C: pTwist-SamyQ-ACCD-2.1

Component	Amount	Dilution
Vector STK001_pSTK-0-sfGFP ● Concentration (ng/µL): 346 ● bp: 3090	0.05804913295 µL	1.16 µL (dilute 1/20) 0.5 µL vector 9.5 µL ddH_2O
Insert SamyQ-Accd-2.1 ● Concentration (ng/µL): 125 ● bp: 3401	0.353704 µL	1.061112 µL (dilute 1/3) 1 µL insert 2 µL ddH_2O
T4 DNA ligase buffer	2 µL	
Sterile water	13.78 µL	
T4 DNA ligase	1 µL	
BsaI-HFv2	1 µL	

Total	20 µL	

D: pTwist-SamQ-ACCD-v3t1

Component	Amount	Dilution
Vector STK001_pSTK-0-sfGFP ● Concentration (ng/µL): 346 ● bp: 3090	0.05804913295 µL	1.16 µL (dilute 1/20) 0.5 µL vector 9.5 µL ddH_2O
Insert SamQ-ACCD-v3t1 ● Concentration (ng/µL): 142 ● bp: 3401	0.311 µL	1.24 µL (dilute 1/4) 1 µL insert 3 µL ddH_2O
T4 DNA ligase buffer	2 µL	
Sterile water	13.6µL	
T4 DNA ligase	1 µL	
BsaI-HFv2	1 µL	
Total	20 µL	

E: SamyQ-AmilCP-2.1

Component	Amount	Dilution
Vector STK001_pSTK-0-sfGFP ● Concentration (ng/µL): 346 ● bp: 3090	0.05804913295 µL	1.16 µL (dilute 1/20) 0.5 µL vector 9.5 µL ddH_2O
Insert SamyQ-AmilCP-2.1 ● Concentration (ng/µL): 138 ● bp: 3029	0.2853405797	1.141362319µL (dilute 1/4) 1 µL insert 3 µL ddH_2O
T4 DNA ligase buffer	2 µL	

Sterile water	13.7 µL	
T4 DNA ligase	1 µL	
BsaI-HFv2	1 µL	
Total	20 µL	

F: SamyQ-AmilCP-v3t1

Component	Amount	Dilution
Vector STK001_pSTK-0-sfGFP ● Concentration (ng/µL): 346 ● bp: 3090	0.05804913295 µL	1.16 µL (dilute 1/20) 0.5 µL vector 9.5 µL ddH_2O
Insert SamyQ-AmilCP-v3t1 ● Concentration (ng/µL): 150 ● bp: 3029	0.263 µL	1.052µL (dilute 1/4) 1 µL insert 3 µL ddH_2O
T4 DNA ligase buffer	2 µL	
Sterile water	13.79 µL	
T4 DNA ligase	1 µL	
BsaI-HFv2	1 µL	
Total	20 µL	

Done by Naomi, Sophie H.

Date: 20260720
Title: Transfer plants from tube rack to Box 1,2,3
Purpose: to assemble the following parts to STK level 0 vector
Procedure:

1. Prepare 2000 ml of 10 % bleach and add into 2000 ml beaker
2. Place the 3D boards into the beaker and use plastic wrap to cover it to prevent the 3D board from floating too high
3. Wait for 10 minutes
4. Use ddH_2O to rinse the 3D boards until there are no smell of the bleach
5. Put one tweezer, 3 tip boxes that are autoclaved, 3 3D printer boards, and the tube rack with 41 plants into BSC
6. Put 1000 ml of 1/2 MS medium from 7/19 into the BSC and add about 300 ml medium into each tip box
7. Put the 3D printer boards onto the medium in the tip boxes
8. Use the tweezer to pick up one lid and transfer it to the 3D printer boards in the tip boxes. Remember to choose the lids with plants that are growing well.
9. Take everything out of BSC
10. Label the 3 tip boxes as Box 1, Box 2, Box 3
11. Put Box 1, 2, and 3 into the growth chamber at 22.2 %

Done by Alex, Ryan
Recorded by: Audrey H

Date: 20260721
Title: Prepare antibiotic - erythromycin
Purpose: to prepare erythromycin for growing colonies after electroporation
Procedure:

1. Add 0.01 g of erythromycin with 10 mL 100 % alcohol
2. Mix until erythromycin completely dissolve in the alcohol
3. Distribute 1000 µL to each tubes

Done by Sarah, Naomi, Sophie H.

Date: 20260721
Title: Miniprep of colour highlighted CDS in STK library
Purpose: Collect concentrated bacterial liquid
Procedure:

		Concentration (ng/ml)	A260/230	A260/280
pSTK 052	pSTK-OC-eCFP codon	108	1.200	1.964
pSTK 053	pSTK-OC-GFPmut3b	82	1.155	1.952
pSTK 054	pSTK-OC-scarlett	95	1.532	2.065
pSTK 055	pSTK-OC-ECFP codon	100	1.220	1.961
pSTK 125	PSTK-OC-mcherry	33	2.357	2.063
pSTK 127	pSTK-OC-morange	128	1.422	2.000
pSTK 158	pSTK-OC-mRFP1	108	1.113	1.929

1. Transfer 1/8 mL of liquid culture to 2 mL microtube
2. Centrifuge at 16000 g for 60 seconds. Discard the supernatant.
3. Repeat steps 1-2 until collected all medium
4. Add 200 mL MX1 buffer
5. Resuspend by vortexing or pipetting
6. Add 250 mL MX2 buffer
7. Gently mix by inverting 8-10 times. Incubate at RT for 2 minutes
8. Add 350 µL MX3 buffer
9. Invert slowly 4-5 times
10. Centrifuge for 5 minutes at 1600 g
11. Transfer ~800µL of the supernatant into assembled column (till all transferred)
12. Centrifuge at 7000 g for 30 seconds and discard flow through
13. Wash the column with 500 µL of WN buffer. Centrifuge at 7000 g for 30 seconds. discard flow through
14. Wash the column with 700µL of WS buffer. Centrifuge at 7000 g for 30 seconds. Discard flow through.

15. Centrifuge for 8 mins at 10000 g
16. Place column into new 1.5mL microtube
17. Add 50µL EB to the center of the membrane
18. Incubate for 3 mins at RT
19. Centrifuge at 10000 g for 3 minutes.

Done by: Abby K. Sophie C
Recorded by: Audrey H

Date: 20260721
Title: Preparation of LBE plates with glycerol
Purpose: Culture electroporation results
Procedure:

1. Measure 3.75 of agar powder and 6.25 g of LB powder
2. Pour them into a flask with 250 mL of ddH_2O and 2 mL of glycerol
3. Use a string bar to mix until even
4. Wrap with aluminium foil and stick indicator tape on top
5. Autoclave for 20 minutes at 127 °C.
6. Add 250 mL 1 mg/mL erythromycin
7. Use a pipette gun to fill 10cm circular agar plates, 15 mL per plate
8. Set at room temperature under the BSC

Done by: Sophie C, Abby K
Recorded by: AudreyH

Date: 20260721
Title: Transformation of level 0 SamyQ and 0720 GGA of G, H, J, K
Purpose: Level 0 assembly
Procedure:

C	PlepA-RBS0-ccasm3
G	pepcG2 veg-Csn:LEA
J	Pveg-Csn:ACCD
H	Pveg-Csn:LEA

1. Level 0 assembly products:
2. Turn on water bath to 42 ℃, prepare ice bucket
3. Get LBA with glycerol plates
4. Thaw competent cells until 50 % thawed and add 10 µL plasmid
5. Incubate on ice for 20 mins
6. Heat shock at 42 ℃ for 45 seconds
7. Incubate on ice for 3 mins
8. Add 1mL SOC medium and incubate at 37 ℃ for 1 hour
9. Centrifuge 5000 g for 5 mins and take out 1200 µL

10. Resuspend cell, spread on LBA plate and incubate at 57 ℃ overnight

Done by: Chloe and Sophie C
Recorded by: Audrey H

Date: 20260721
Title: Electrophoration
Purpose: insert the plasmid pSTK-EXP1-Pveg-EEBD-RBS0-SamyQ-TAT-HA-AmilCP-B0015 into B.sub
Procedure:

1. Prewarm SOC recovery medium in 37 ℃
2. Precold 0.2 cm electrophoration cuvette
3. Mix 4Ml plasmid (con 2050 ng/Ml) with 96Ml electrocompetent cell in 1-5 mL microfuge
4. Set up electroporator (gene pulsatory cell)
    a. Go to home page
    b. When using the 0.2 cm curvette, press 2 or 3 and enter
    c. Set voltage resistance, voltage, and capacity
        i. Voltage: 2.3 kV
        ii. Resistance: 300 Ω
        iii. capacitance: 25 F
5. Transfer mixture of cell and DNA to cold electrophoration cuvette and tap the suspension to the bottom
6. Place the cuvette into the shock pad and close the lid
7. Wait until beep sound
8. Remove the cuvette quickly from the chock pad and immediately add 1 mL SOC medium to the cuvette
    a. Purification of process should be in 10 seconds
9. Transfer the cell suspension to a polypropylene tube and incubate at 37 ℃ for three hours, shaking at 200 rpm
10. Check and record the parameter
11. Centrifuge the tubes at 5000 g for 5 mins
12. Spread the LB plate with erythromycin

Done by: Sarah Chou, Phoebe
Supervised by: Gabriel, abby K, Sophie C
Recorded by: Sophie H

Date: 20260721
Title: Redone Golden Gate Assembly of L
Purpose: Reassemble GanA-Gala
Procedure:

1. Assemble the following parts:

L: pSTK-EXP-01-Pveg-EEBD-RBS0-GanA-GalA-B0015

	Concentration	Bp length	Original (ML)	Diluted (ML)
Vector (EXP1)	162 ng/mL	6929	0.278015	1.39 (1/5)
Promoter(Pveg)	162 ng/mL	2361	0.109618	1.1 (1/10)
RBS(EEBD-RBS0)	150 ng/mL	2113	0.207306	2.1 (1/10)
CDS(GanA-GalA)	177 ng/mL	2080	0.152768	1.53 (1/10)
Terminator (B0015)	107 ng/mL	2213	0.268869	1.34 (1/5)
Spacer(pSTK-0CX-spacer)	127 ng/mL	2095	0.10722	1/07 (1/10)
Histag(pSTK-0CS-HisGX)	135 ng/mL	2402	0.11561	1.10 (1/10)
T4 DNA ligase buffer	X	X	2	2
Sterile water	X	X		8.54
T4 DNA ligase	X	X	1	1
BsaI-HFv2	X	X	1	1
Total	X	X	20	20

Done by: Olivia L
Supervised by: Gabriel
Recorded by: Audrey H

Date: 20260720
Title: Transfer Arabidopsis to salinity plates
Purpose: Start experiment set 5 using seeds from 0719
Procedure:

1. Take the stack of agar plates wrapped in aluminum foil out of the refrigerator
2. Open the BSC
3. Spray alcohol on the agar plates before placing them inside the BSC
4. Use a marker to draw a straight line along a ruler 3cm from the top of the plate using
5. Draw 8 dots at the ends of the roots
6. Place the square plates vertically at a slight angle
7. Choose 8 germinated seeds after vernalization that have similar root lengths or germinated status and align the endpoints of their roots to the line drawn on the plate

8. Seal the plates with breathable tape and label the incubation date with a marker before placing the vertical plates on the stacks in the growth chamber, and slightly tilted towards the back

Done by Audrey H, Abby K
Recorded by: Audrey H

Date: 20260722
Title:Colony PCR of C, G, and K
Purpose: To amplify the desired DNA fragments
Procedure:

1. Prepare the master mix using the following reaction:

		[K]	[G]	[C]
	1 rxn	4 rxn	2 rxn	11 rxn
FW primer	1 µL	4 µL	2 µL	11 µL
RV primer	1 µL	4 µL	2 µL	11 µL
Q-Amp master mix	10 µL	40 µL	20 µL	110 µL
ddH_2O	8 µL	32 µL	16 µL	88 µL

K: pExp1-Pveg-SamyQ-TAT-HA-AmilCP-*B.sub*
G: (STK) Module 4: pcpcG2-veg-EEBD-RBS0-Csn:LEA-B0015
C: (STK) Module 2: PlepA-RBS0-Ccasm3-B0015 (bp:1667)
[K]:FW&RV primer: Exp1
[G]:FW primer: biobrick prefix/ RV primer:LEA
[C]: Fw primer: biobrick prefix/ RV primer:Ccasm3 rv

2. Distribute 20 µL per microtube
3. Pick the circled colonies with the tip and streak the gridded plates
4. Dispense the tips into one of the microtubes
5. Repeat steps 3 & 4 until all microtubes are filled
6. Set up a 5min timer and begin after last tip
7. Remove tips and close lids after 5 min and centrifuge
8. Put the PCR tubes into PCr machine and set up thermal cycle as below:

95℃	5 min	1 cycle
95℃	20 secs	35 cycles
65℃	30 secs	//
72℃	2 min (bp 1800)	//
25℃	5 min	1 cycle

Done by Sophie C.
ELN by Sophie H.

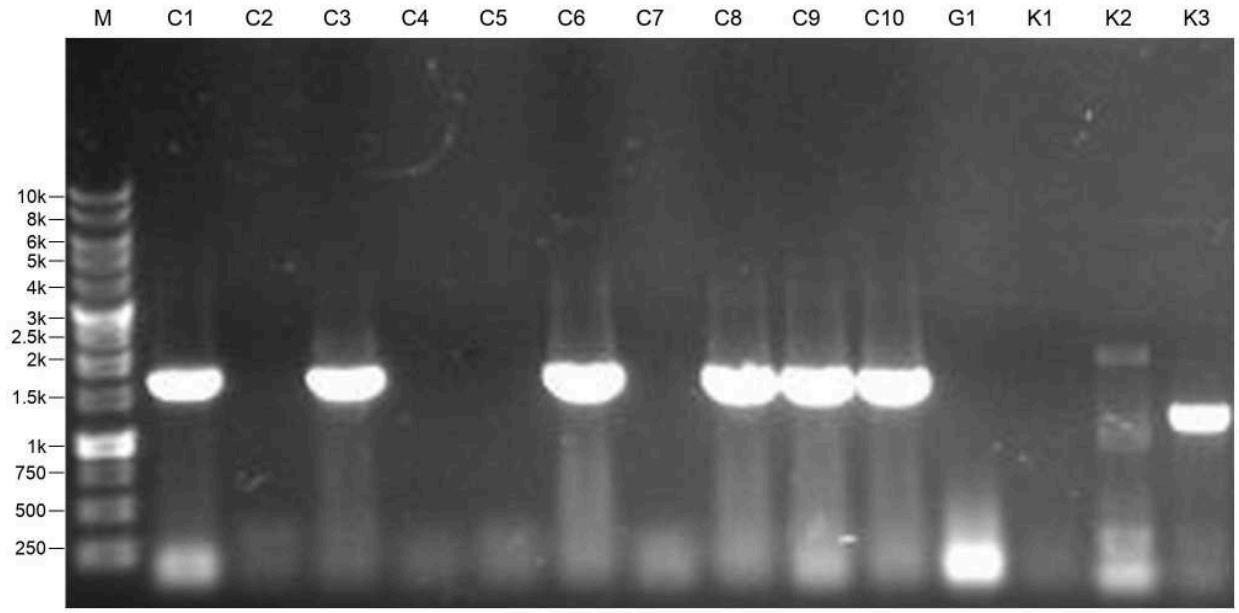
Date: 20260722
Title: 0.8% gel preparation
Purpose: Make 0.8 % gel for el ectrophoresis
Procedure:

1. Assemble mold, tray, and comb
2. Weight 0.64 g of agarose powder and prepare 80 mL of 1X TAE buffer
3. Add 1X TAE buffer and Agarose powder into conical flask
4. Cover flask with plastic wrap and use tip to puncture holes
5. Microwave for 30 sec than agitate, repeat this step until all particles were dissolved
6. When gel stops smoking but is still hot, add 2 µL of safety dye and agitate it
7. Pour the melted gel into two trays
8. Leave to solidify in room temperature

Done by Sophie C, Sophie Liu
ELN by Sophie H

Date: 20260722
Title: Electrophoresis
Purpose: Check colony PCR product of C,G,K,A,E,
Procedure:

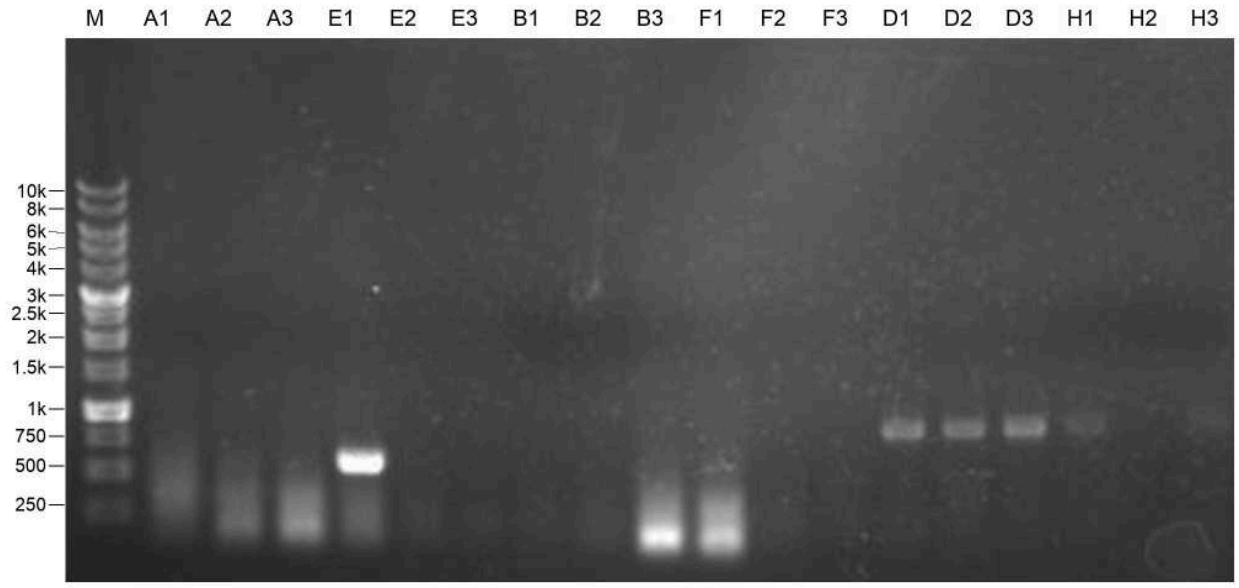
1. Load the gel in the layout shown below (4 µL for M; 5 µL for all samples):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)

C: PlepA-EEBD-RBS0-CcaSm3-B0015 (C6, C10)	biobrick prefix	CcaSm3 R	1667
G: PcpcG2-veg-EEBD-RBS0-csn::LEA-B0015	biobrick prefix	LEA R	1135
K: Pveg-EEBD-RBS0-SamyQ-TAT-HA-AmilCP-B0015	pEXP_Seq.Fw	pEXP.Seq.Rv	1825
Note: Primers: K- STK-Exp1-fw and STK-Exp1-rv; G- Biobrick fw and LEA rv; C- Biobrick fw and CcaSm3 rv			

3. Load the gel in the layout shown below (5 µL each):
4. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
A: pSTK-SamyQ:LEA-2.1	BioBrick Prefix	LEA R	652
E: pSTK-SamyQ:Lea-2.2	BioBrick Prefix	LEA R	652
B: pSTK-SamyQ-AccD-2.1	BioBrick Prefix	ACCD R	1213

F: pSTK-SamyQ-AccD-3.1	BioBrick Prefix	ACCD R	1213
D: pSTK-SamyQ-AmilCP-2.1	BioBrick Prefix	AmilCP R	841
H: pSTK-SamyQ-AmilCP-3.1	BioBrick Prefix	AmilCP R	841

Done by Sophie C, Abby K
ELN by Sophie H

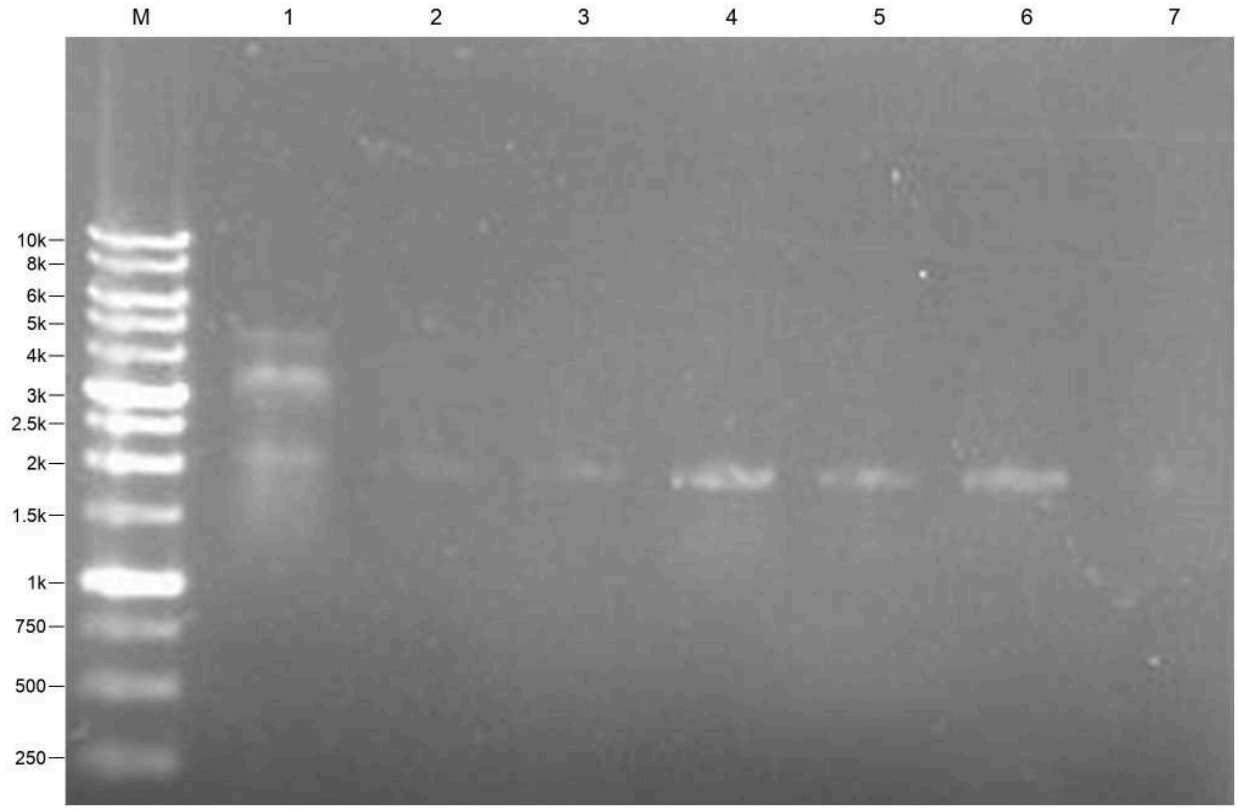
Date: 20260722
Title: Transformation
Purpose: To transform the desired bacteria in medium to agar plate
Procedure:
pSTK-0A-Pveg
pSTK-0D-B0015
pSTK-0B-EEBD
pGEM-Teasy-0C-Ho1:pcyA

1. Turn on water bath at 52 ℃ and prepare ice
2. Get LBC with glycerol plates
3. Thaw competent cell until 100 % thawed and split them into 3 tubes of 30 and 1 tube of 10
4. Add 2μL plasmid into the tubes of 30 and add 1 μL plasmid into the tube of 10
5. Incubate on ice for 20 min
6. Heat shock at 42 ℃ for 45 sec and then recover on ice for 3 min
7. Spread on LBC plates and incubate at 37 ℃ overnight

Done by Chole, Sophie C
ELN by Sophie H

Date: 20260722
Title: Gel Electrophoresis for Enzyme Digestion
Purpose: Check the approximate length of enzyme digestion product
Procedure:

1. Load the gel in the layout shown below (5 μL each):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder		
Sample	Plasmid Construct	Enzyme
1	Ho1:pcyA	Bsa1
2	pveg	Bsa1
3	spacer (old enzyme)	Bsa1
4	GanA-Gala	Bsa1
5	HIStag	Bsa1
6	B0015	Bsa1
7	spacer (new enzyme)	Bsa1

Done by Sophie L

Date: 20260722
Title: Issue of Box BI

Purpose: Summarize and analyze the problem of hydroponic box and find these problem before next round of testing

Problems:

Problem 1: The 3D printer board in Box B1 started to sink into the medium, This is because the medium leads into the 3D printer board. This caused the lids to float on medium, and not on the board.

Problem 2: Two of the lids seemed to be contaminated by red bacteria or fungi

Problem 3: The medium is contaminated and it looks turbid

Problem 4: 8 out of 13 plants are not growing, including the 2 that had been contaminated by red bacteria

Done by plant model team

Date: 20260723
Title: Liquid culture
Purpose: To transfer colony inside TB medium
Procedure:

1. Label 10 liquid culture tubes

C1, C3, C6, C8, C9, C10: PlepA-RBS0-CcaSm3-B0015
E1: pSTK-SamyQ-LEA-2.2
D1, D2, D3: pSTK-SamyQ-AmilCP-3.1

2. Turn on bunsen burner
3. Add 3 µL of ampicillin and 3 mL of TB medium into each tubes
4. Dip a pipette tip into colonies
5. Dispense tips in their respective tubes
6. Place the tubes in incubator at a slanted angle

Done by Sophie C
Supervised by Gabriel

Date: 20260723
Title: Colony PCR of Ho1:pcyA
Purpose: Screen for successful bacterial inserts from the transformation of Ho1:pcyA
Procedure:

1. Prepare the reaction as follows:
2.

A: pSTK-1A-Pveg-RBS0-Ho1:pcyA-B0015

	1 rxn	58 rxn
Forward Primer (pSTK-seqFW)	1 µL	58 µL
Reverse Primer (pcyA R)	1 µL	58 µL
Q-amp master mix	10 µL	580 µL
ddH_2O	8 µL	464 µL
total	20 µL	1160 µL

3. Circle 24 colonies each from plate A1 and A2, and 8 colonies from plate B

A1 & A2: Pveg-RBS0-Ho1-pcyA-B0015
B: pGEM-Teasy-Ho1-pcyA

4. Pick up 24 colonies from each agar plate with tip and streak the labelled grid 3 times
5. Dispense the tips with colonies into the PCR strips and wait 5 minutes
6. Take out the tips and close the lid

7. Place into PCR machine for thermal cycle

95 ℃	2 min	1 cycle
95 ℃	30 sec	30 cycles
55 ℃	20 sec	
72 ℃	2 min	
72 ℃	5 min	1 cycle

Done by Abby T, Abby K, Sophie C
Supervised by Chars

Date: 20260723
Title: Golden Gate Assembly of mCherry, mScarlet
Purpose: Assemble parts
Procedure:

1. Assemble the following parts:

R: Pveg-RBS0-mCherry-B0015	original (μL)	diluted (μL)
Vector: EXP1 (162ng/μL, 6929 bp)	0.278	1.39 (1/5)
I promoter: Pveg (168ng/μL, 2361 bp)	0.182	1.82 (1/10)
I RBS: RBS0	0.146	1.46 (1/10)
I CDS: mCherry (83ng/μL, 2795bp)	0.437	0.44
I terminator: B0015 (148ng/μL, 2213bp)	0.194	1.94 (1/10)
T4 DNA ligase buffer	2	2
Sterile water	last added, add to 20 μL	
T4 DNA ligase	1	1
BsaI-HFv2	1	1
total	20	20

Z: Pveg-RBS0-mScarlet-B0015

Vector: EXP1 (162 ng/μL, 6929 bp)	0.278	1.39(1/5)
I promoter: Pveg (168 ng/μL, 2361 bp)	0.182	1.82(1/10)
I RBS: RBS0	0.146	1.46(1/10)
I CDS: mScarlet (95ng/μL, 2780bp)	0.38	0.4
I terminator: B0015 (148ng/μL, 2213bp)	0.194	1.94(1/10)
T4 DNA ligase buffer	2	2
Sterile water	last added, add to 20μL	
T4 DNA ligase	1	1
BsaI-HFv2	1	1
total	20	20

J: (STK) Module4-Pveg-EEBD-RBS0-CSN::ACCD

Vector: EXP1 (162ng/μL, 6929 bp)	0.278	1.39(1/5)
I promoter: Pveg (168ng/μL, 2361 bp)	0.182	1.82(1/10)
I RBS: EEBD-RBS0	0.205	1.02(1/5)
I CDS: Csn::ACCD (142ng/μL, 4212bp)	0.385	1.9(1/10)
I terminator: B0015 (148ng/μL, 2213bp)	0.194	1.94(1/10)
T4 DNA ligase buffer	2	2
Sterile water	last added, add to 20μL	
T4 DNA ligase	1	1
BsaI-HFv2	1	1
total	20	20

I:(STK) Module 4-PcpcG2-veg-EEBD-RBS0-CSN::ACCD-B0015

Vector: 1D (132ng/μL, 3120 bp)	0.15	1.5(1/10)
I promoter: Pveg (65ng/μL, 3252bp)	0.6504	0.65
I RBS: RBS0 (151ng/μL, 2392bp)	0.205	1.02(1/5)
I CDS: CSN::ACCD (142ng/μL, 4212bp)	0.385	1.9(1/5)
I terminator: B0015 (148ng/μL, 2213bp)	0.194	1.94(1/10)
T4 DNA ligase buffer	2	2
Sterile water	last added, add to 20μL	
T4 DNA ligase	1	1
BsaI-HFv2	1	1
total	20	20

Date: 20260723
Title: Prepare 2.0 % Agarose Gel
Purpose: prepare gel for electrophoresis
Procedure:

1. Assemble mad, tray, and comb
2. Weight 0.40 g of Agarose powder and prepare 20 mL of 1X TAE buffer
3. Add 1X TAE and Agarose powder into conical flask
4. Cover the flask with plastic wrap and puncture holes with tip(middle one would be the best)
5. Microwave for 30sec then agitate. Repeat this step until all particles were dissolved
6. When gel stops smoking, add 1μL of safety dye and agitate it
7. Pour the melted gel into tray
8. Leave to solidify in RT

Done by Sophie L

Date: 20260723
Title: Make 1 % agarose gel*2
Purpose: Prepare agarose gel for electrophoresis
Procedure:

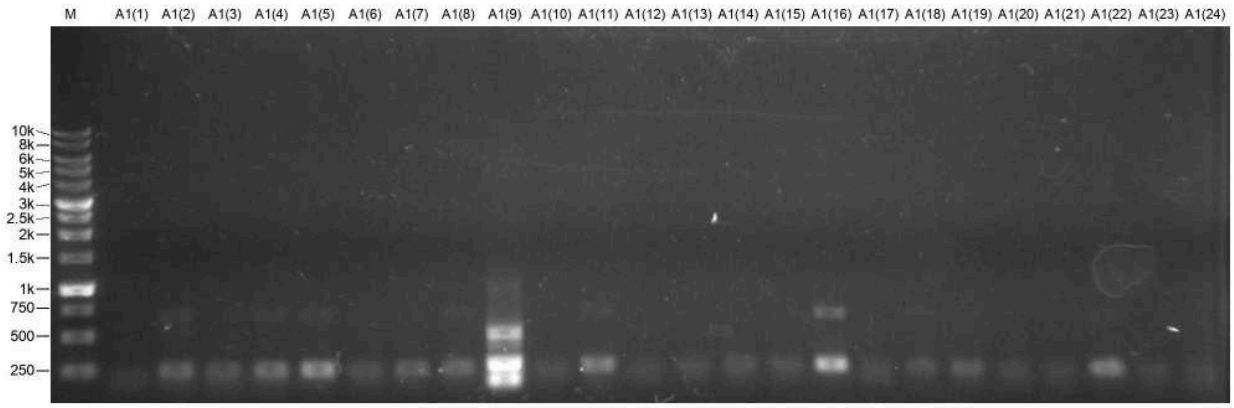
1. Assemble mad, tray, and comb
2. Weight 0.8 g of Agarose powder and prepare 80mL of 1X TAE buffer
3. Add 1X TAE and Agarose powder into conical flask

4. Cover the flask with plastic wrap and puncture holes with tip(middle one would be the best)
5. Microwave for 30 sec then agitate. Repeat this step until all particles were dissolved
6. When gel stops smoking, add 1 µL of safety dye and agitate it
7. Pour the melted gel into tray
8. Leave to solidify in RT

Done by Abby K, Sophie C

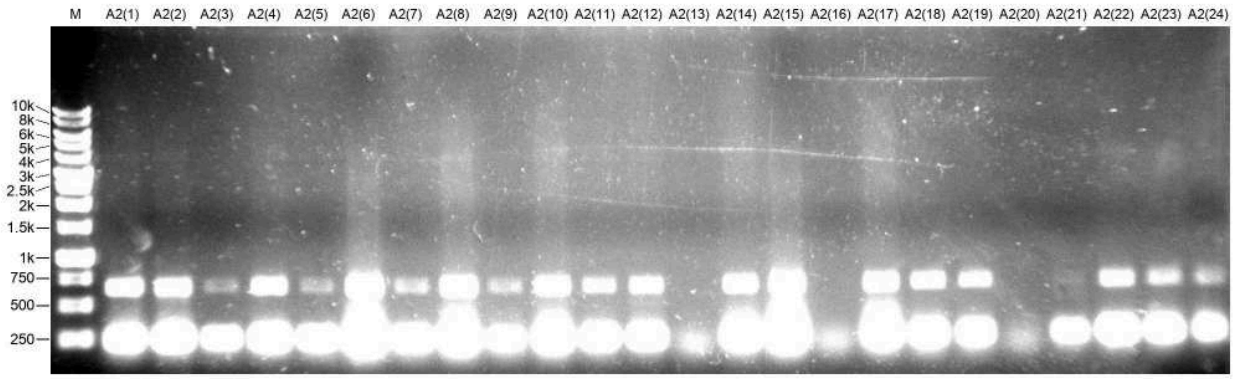
Date: 20260723
Title: Gel electrophoresis
Purpose: Verify the approximate length of basepair of Pveg-RBS0-Ho1:pcyA-B0015
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
A1(1)-A1(24): Pveg-RBS0-Ho1:pcyA-B0015	pSTK-seqFW	pcyA R	1871
Note: Primers: pSTK-seqFW and pcyA rv			

3. Load the gel in the layout shown below (5 µL each):
4. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
A2(1)-A2(24): Pveg-RBS0-Ho1:pcyA-B0015	pSTK-seqFW	pcyA R	1871

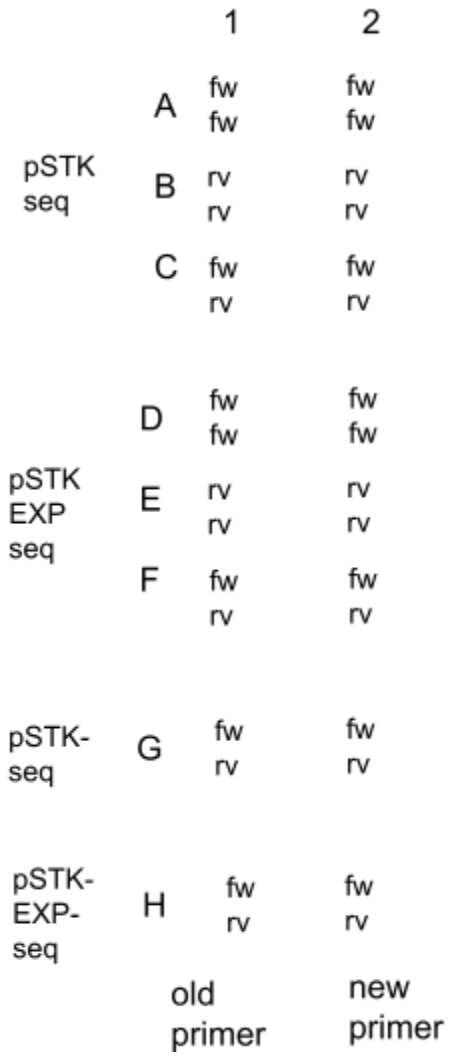
Done by Sophie C.

Date:20260724
Title:PCR for primer quality check
Purpose:Verify primer pairs for colony PCR is work
Procedure:

1. Prepare reaction master mix:

template	1.2	1.2
	pSTK-0-spGFP	pSTK-Exp2-sPEFP
ddH_2O	46.8	46.8
Q-Amp mix	60	60

2. Dispense reaction by the following layout:



3. Thermal cycle:

95℃	5 min	
95℃	20 sec	30 cycles
60℃	20 sec	
72℃	1.5 min	
72℃	5 min	
RT		

Done by: Chars
Recorded by: Audrey H

Date:20260724
Title:Make 1% agarose gel
Purpose: Make gel for electrophoresis
Procedure:

1. Assemble mold, tray, and comb
2. Weight 0.4g agarose powder and prepare 40mL 1XTAE buffer
3. Add into a flask
4. Cover with plastic wrap and poke holes
5. Place into microwave for 30 secs
6. Agitate
7. Put into microwave for 5 sec then agitate
8. Repeat step 7 until no sediments
9. When it stops smoking, add 2μLsafety dye, agitate
10. Pour into the mold
11. Let solidify at RT

Done by: Abby K, Sophie C
Supervised by: Sophie C
Recorded by: Audrey H

Date:20260724
Title:Redoing 20260723 Colony PCR
Purpose: Amplify plasmid number
Procedure:

1. Prepare the following reaction (master mix) x26 reactions and dispense solution into PCR tubes

	1 rxn	26 rxn
Fw primer (biobrick prefix)	1 μL	26 μL
Rv primer (STK reverse)	1μL	26μL
Q-amp master mix	10μL	260μL
ddH_2O	8μL	208μL

A1 & A2: Pveg-RBSO-Ho1: pcyA-B0015

2. Distribute 20 μL per microtube
3. Pick the circled colonies with the top and streak the gridded plates
4. Dispense the tops into one of the microtubes
5. Repeat steps 3 and 4 until all microtubes are filled with 1 tip
6. Set up a 5 mins timer and begin after last top
7. Remove tips and close lid after 5 min and centrifuge

8. PCR

95℃	5 mins	1
95℃	20 sec	35 cycles
65℃	20sec	
72℃	2 mins 30 sec	
72℃	5 min	1
25℃	0 min	1

Done by: AbbyK, Sophie C
Recorded by: Audrey H

Date:20260724
Title: Transformation
Purpose: insert plasmids into competent cell
Procedure:

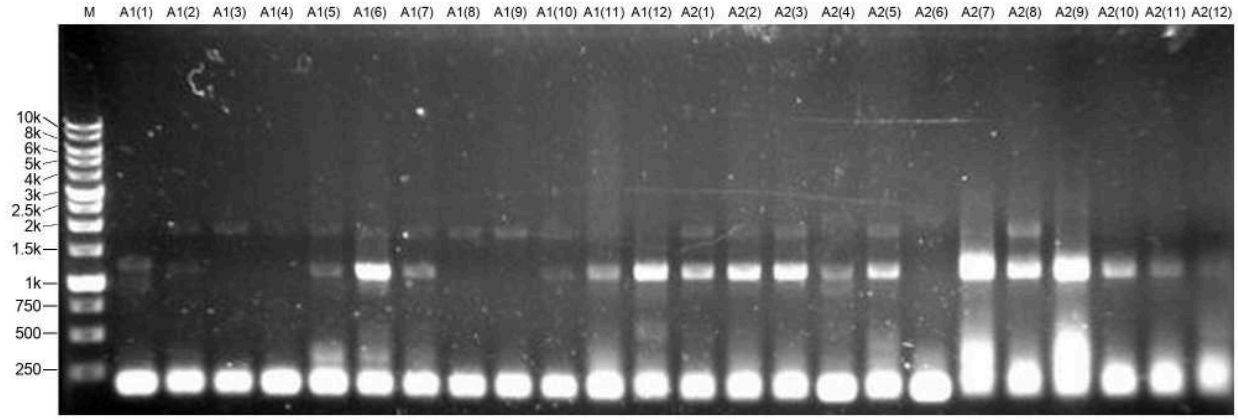
G:(STK) module 4- PcpcG2-veg-EEBD-RBS0-CSN::LEA-B0015
H:(STK) module 4- Pveg-EEBD-RBS0-Csn::LEA-B0015
I: (STK) module 4-PcpcG2-veg-EEBD-RBS0-Csn::ACCD-B0015
Y: Pveg- RBS-mcherry -B0015
J: (STK) Module 4 - Pveg - EEBD - RBSO - Csn: ACCD- B0015
Z: Pveg-RB50-mScarrlet -B0015

1. set water bath to 42°C and prepare an ice bucket
2. Prepare LBA plates (6) and place them (with soc medium) in the incubation
3. Thaw the competent cell until 50% melted, then add 10 µL of plasmid.
4. Incubate on ice for 20 minutes
5. Heat shock at 42 °C for 45 seconds.
6. Recover on ice for 8 minutes.
7. Add I mL soc medium
8. Incubate for 1 hour at 31 °C.
9. centrifuge at 5000 g for s minutes
10. Remove ImL of supernatant then vortex
11. Spread on LBA plates
12. Incubate at 31 °C for 16 hours.

Done by: Abby K, Sophie C
Recorded by: Audrey H

Date:20260724
Title:Gel electrophoresis
Purpose: Check base pair of colony PCR product
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
A1(1)-A1(12): Pveg-RBS0-Ho1:pcyA-B0015	Biobrick Prefix	STK rv	2111
A2(1)-A2(12): Pveg-RBS0-Ho1:pcyA-B0015	Biobrick Prefix	STK rv	2111
Note: A1: 3, 4, 8, 9; A2: 6			

Done by Olivia Lin, Ryan Y
Supervised by: Chars
Recorded by: Audrey H

Date:20260724
Title:GGA of A, L
Purpose: Assemble parts
Procedure:
A: (STK) Module 1(1A): Pveg–RBS0-Ho1:pcyA-B0015

1. prepare the reaction as follow:

Part	Original Amount	Dilution
Vector (1A)	0.176348 μL	1.76 μL (dilute 1/10)
Insert promoter (Pveg)	0.182 μL	1.82 μL (dilute 1/10)
Insert RBS (RBS0)	0.1461117 μL	1.46 μL (dilute 1/10)
Insert CDS(Ho1::PcyA)	0.0849916 μL	1.7 μL (dilute 1/20)
Insert Terminator (B0015)	0.194 μL	1.94 μL (dilute 1/10)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water		7.32 μL
T4 DNA ligase	1 μL	1 μL
BsaI-HFv2	1 μL	1 μL
Total	20 μL	20 μL

L: Pveg-EEBD-RBS0-GanA-Gala-B0015

Part	Original	Dilution
Vector (shuttle vector): pSTK-EXP-1-sfGFP	0.278015 μL	1.39 μL (dilute 1/5)
Insert promoter(Pveg)	0.109618 μL	1.1 μL (dilute 1/10)
Insert RBS(EEBD-RBS0)	0.205 μL	2 μL (Dilute 1/10)
Insert CDS(GanA-Gala)	0.30465 μL	1.53 (dilute 1/5) μL
• Insert Terminator (B0015)	0.194 μL	1.94 μL (dilute 1/10)
Spacer (pSTK-OCX-spacer)	0.10722440 μL	1.07 μL (dilute 1/10)
Histag (pSTK-OCZ-HisGX)	0.1156185 μL	1.16 μL (dilute 1/10)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water	μL	5.81 μL

T4 DNA ligase	1 µL	1 µL
BsaI-HFv2	1 µL	1 µL
Total	20 µL	20 µL

Done by: Chloe
Supervised by: Chars
Recorded by Audrey H

Date:20260724
Title:Liquid culture
Purpose: Transfer colony to medium
Procedure:
Pveg
RBS0
EEBD
B0015
pGEM-Teasy-Ho1:pcyA

1. Label the five liquid culture tubes
2. Turn on the bunsen burner and add 3mL LB medium. Add 3 µL ampicillin to Pveg, RBS0, EEBD, B0015 and add 3 µL chloramphenicol to Ho1:pcyA
3. Dip pipette into colonies and dispense tips into respective tubes
4. Incubate in 25 ℃ overnight

Done by: Chars
Supervised by: Chloe
Recorded by: Audrey H

Date:20260724
Title: Prepare 2 % agarose gel
Purpose: Make gel for electrophoresis
Procedure:

1. Assemble mold, tray, and comb
2. Weight 1.2 g of agarose powder and measure 60mL of 1xTAE buffer
3. Add the agarose powder and the 1x TAE buffer into conical flask
4. Microwave for 30 sec then agitate it
5. Repeat this step until no sediments
6. Wait until no smoke, add 3 µL of safety dye and agitate it as well
7. Pour the melted gel into the tray and use tip to squeeze the air and pop the bubble
8. Wait until the gel solidifies

Done by Abby K

Date: 20260725
Title: Planting Arabidopsis in soil
Purpose: To plant the Arabidopsis seed in the soil
Procedure:

1. Soak pots and water trays in 1:100 diluted bleach for 10mins, then rinse it thoroughly with ddH_2O (make sure every corner was rinsed thoroughly until no bleach smell left)
2. Mix peat soil into a 2L flask and autoclave it for 20 minutes at 127 °C.
3. it. Fill 3-inch pots until 80% full, lightly press down and thoroughly water from the surface to fully moisturize the soil before transplanting the Arabidopsis seeds.
4. Make a small hole (around one knuckle deep) in the soil using finger (the holes should be spaced evenly to prevent plants interrupting each other from growing)
5. Use tweezers to gently lift the healthy seeds (put for 7 days)
6. Place the root against the sidewall of the hole
7. Use a wash bottle to gently rinse soil over roots so they are fully covered by the soil (must avoid direct contact of the root with fingers or tweezers, support seedlings only with tweezers only from beneath the cotyledons)

Done by Alex
Record by Sophie H.

Date: 20260725
Title: Colony PCR
Purpose: To verify bacterial colonies after a transformation step in molecular cloning
Procedure:

1. Prepare the following master mix

Z: Pveg-RBS0-mScarlet-B0015
Y: Pveg-RBS0-mCherry-B0015

	1 reaction	26 reactions
Forward (Pveg-fw)	1 µl	26 µl
Reverse (EXP-rv)	1 µl	26 µl
Q-Amp master mix	10 µl	260 µl
ddH_2O	8 µl	208 µl

J: Pveg-EEBD-RBS0-Csn::ACCD-B0015
H: Pveg-EEBD-RBS0-Csn::LEA-B0015

	1 reaction	26 reactions
Forward primer (EXP fw)	1 µL	26 µL
Reverse primer (EXP seq 4 rv)	1 µL	26 µL

Q-amp master mix	10μL	260 μL
ddH_2O	8μL	208 μL

G: PcpcG2-veg-EEBD-RBS0-Csn::LEA-B0015
I: PcpcG2-veg-EEBD-RBS0-Csn::ACCD-B0015 (1836)

	1 reaction	26 reactions
Forward primer (BioBrick fw)	1 μL	26 μL
Reverse primer (STK rv)	1 μL	26 μL
Q-amp master mix	10 μL	260 μL
ddH_2O	8 μL	208 μL

1. distribute 20 μL per micro tube
2. pick the circled colonies with the tip and streak the gridded plates
3. Dispense the tips into one of the microtubes
4. Repeat steps stand for until all micro tubes are filled with one
5. Set up a five minute timer and begin after last tip
6. Remove tips and close lids after five minutes and spin down
7. Place into PCR machine for thermal cycle

95 ℃	2 min	1 cycle
95 ℃	30 sec	30 cycles
65 ℃	40 sec	
72 ℃	2 min	
72 ℃	1 min	1 cycle

Done by Sophie C, Abby K, Ryan Y
Supervised by Chars

Date: 20260725
Title: Miniprep
Purpose: Collect concentrated bacterial liquid
Procedure:

1. Transfer 1.8 mL of liquid culture to a 2 mL tube
2. Centrifuge at 16,000 g for 60 seconds
3. Repeat steps 1-2 until collected all medium
4. Add 200 μl of MX1 buffer

5. Resuspend by vortexing or pipetting
6. Add 250 µl of MX2 buffer
7. Invert 8-10 times. Incubate at RT for 2 minutes
8. Add 350 µl of Mx3 buffer
9. Invert 3-5 times
10. Centrifuge for 5 minutes at 23,444 g
11. Add 800 µl of supernatant into assembled column
12. Centrifuge at 7000 g for 30 seconds and discard the flow through
13. Add 500 µl of WN buffer and centrifuge at 7000 g for 30 seconds and discard the flow through
14. Add 700 µl of WS buffer and centrifuge at 7000 g for 30 seconds and discard the flow through
15. Centrifuge the empty column at 10,000 g for 8 minutes
16. Assemble 1.5 mL microtube w/ column
17. Add 50 µl of EB to the center of the membrane.
18. Incubate at RT for 3 minutes
19. Centrifuge at 10,000 g for 3 minutes

Biodrop results:

	concentration	A260/280
Pveg	129 ng/µl	2.016
EEBD	123 ng/µl	1.952
RBS0	216 ng/µl	2.057
B0015	139 ng/µl	1.986

Done by Olivia L and Chloe

Date: 20260725
Title: Making agarose gel (1 %) (5 gels)
Purpose: Make gel for electrophoresis
Procedure:

1. Add 2 g of agarose powder and 200 ml of 1xTAE buffer into a flask
2. Prepare the mold, tray and comb
3. Cover with plastic wrap and poke 4-5 holes
4. Microwave for 30 seconds then agitate
5. Microwave for 5 seconds then agitate
6. Repeat step 5 until no sediments left
7. When the mixture stops smoking, add 10 µl of safety dye
8. Pipette then agitate
9. Pour into the molds
10. Let the gel solidify at room temperature

Done by Abby K and Sophie C

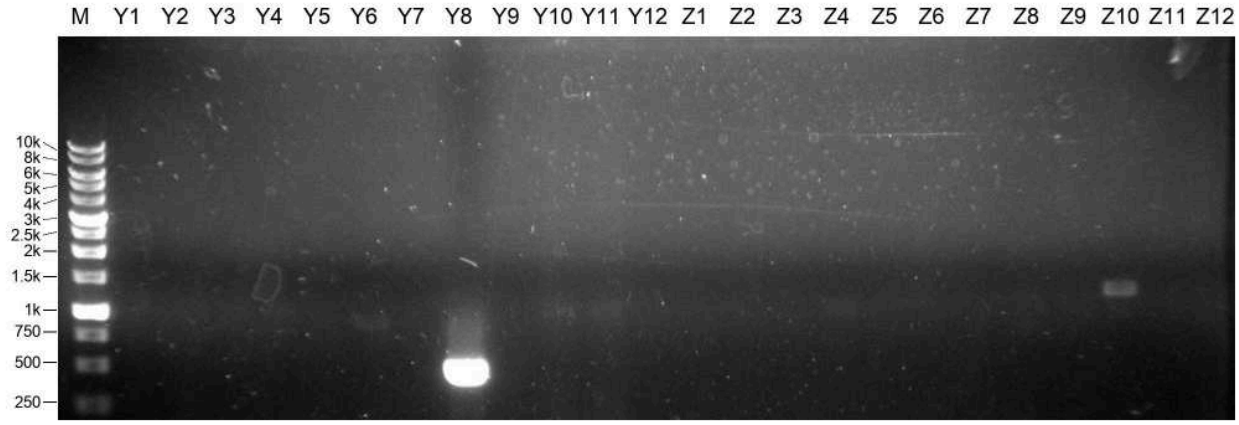
Date: 20260725
Title: Transformation
Purpose: Transform bacteria to agar plates
Procedure:

1. Set the water bath to 42 °C and prepare ice bucket
2. Get LBA plates with glycerol
3. Thaw competent cells until they're 50 % thawed and add 10µl plasmid
4. Incubate on ice for 20 minutes
5. Heat shock at 42 °C for 45 seconds
6. Incubate on ice for 3 minutes
7. Add 1 ml SOC medium
8. Incubate for 1 hour at 37 °C
9. Centrifuge at 5000 g for 5 minutes
10. Remove 1 ml of supernatant then vortex
11. Spread on LBA plates
12. Incubate at 37 °C for 16 hours

Done by Chloe, Sophie C, Abby T

Date: 20260725
Title: Gel electrophoresis of 0725 colony PCR
Purpose: Verify plasmid sizes of 0725 colony PCR
Procedure:

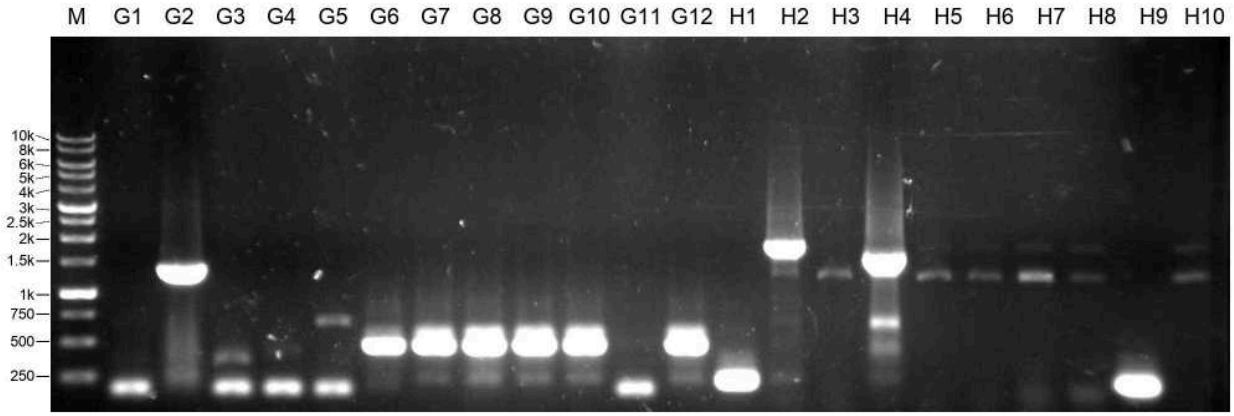
1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)

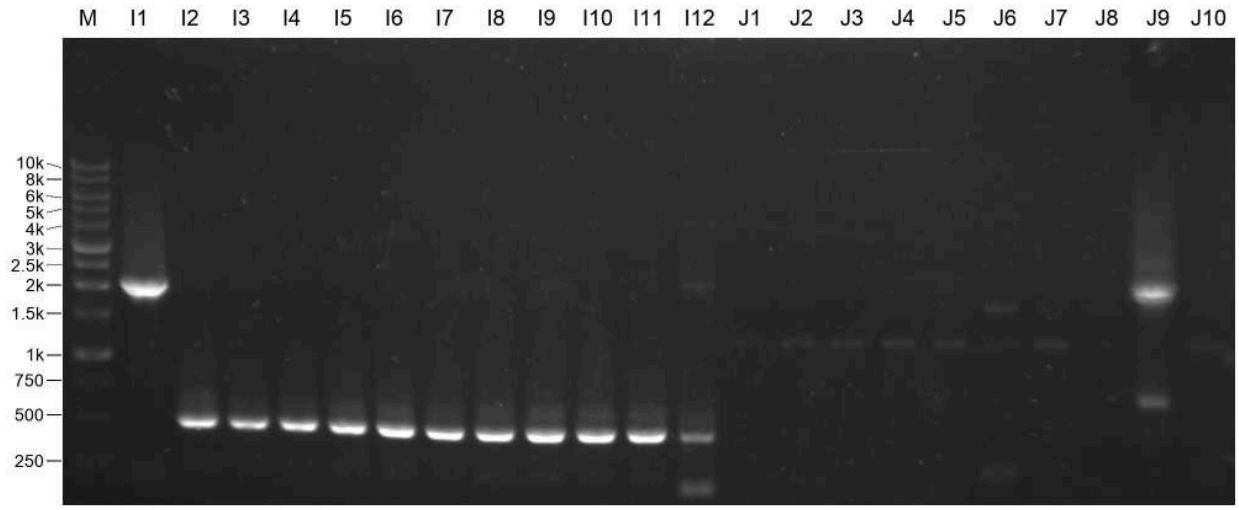
Y1-Y12: Pveg-RBS0-mCherry-B0015	pveg fw	EXP rv	1164
Z1-Z12: Pveg-RBS0-mScarlet-B0015	pveg fw	EXP rv	1149

3. Load the gel in the layout shown below (5 µL each):
4. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
G1-G12: PcpcG2-veg-EEBD-RBS0-csn::Lea-B0015	Biobrick fw	STK rv	1375
H1-H10: Pveg-EEBD-RBS0-csn::Lea-B0015	Exp fw	Exp seq 4 rv	1372

5. Load the gel in the layout shown below (5 µL each):
6. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
I1-I12: PcpcG2-veg-EEBD-RBS0-Csn::ACCD-B0015	Biobrick fw	STK rv	1836
J1-J10: Pveg-EEBD-RBS0-Csn:ACCD-B0015	Exp fw	Exp seq 4 rv	2109

Done by Sophie C and Abby K

Date: 20260725
Title: Colony PCR
Purpose: Verify level 0 parts(mutants)
Procedure:

1. Prepare the following reaction and master mix

B: pSTK-SamyQ-ACCD-2.1
F: pSTK-SamyQ-ACCD-v3t1

	1 reaction	20 reactions
Forward primer (BioBrick-fv)	1 µL	20 µL
Reverse primer (ACCD R)	1 µL	20 µL
Q-Amp master mix	10 µL	200 µL

ddH_2O	8 µL	16/0 µL

A: pSTK-SamyQ-LEA-2.1

	1 reaction	10 reactions
Forward primer (BioBrick-fv)	1 µL	10 µL
Reverse primer X(LEA R)	1 µL	10 µL
Q-amp master mix	10 µL	100 µL
ddH_2O	8µL	80 µL

H: pSTK-SamyQ-AmilCP-v3t1

	1 reaction	10 reactions
Forward primer (BioBrick-fv)	1 µL	10 µL
Reverse primer (AmilCP R)	1 µL	10 µL
Q-amp master mix	10 µL	100 µL
ddH_2O	8 µL	80 µL

2. distribute 20 µL per micro tube
3. pick the circled colonies with the tip and streak the gridded plates
4. Dispense the tips into one of the microtubes
5. Repeat steps stand for until all micro tubes are filled with one
6. Set up a five minute timer and begin after last tip
7. Remove tips and close lids after five minutes and spin down
8. Place into PCR machine for thermal cycle

95 ℃	2 min	1 cycle
95 ℃	30 sec	30 cycles
65 ℃	40 sec	
72 ℃	1 min 30 sec	
72 ℃	1 min	1 cycle

Done by Ryan Y, Olivia L, and Chloe
Supervised by Chars and Gabriel

Date: 20260725

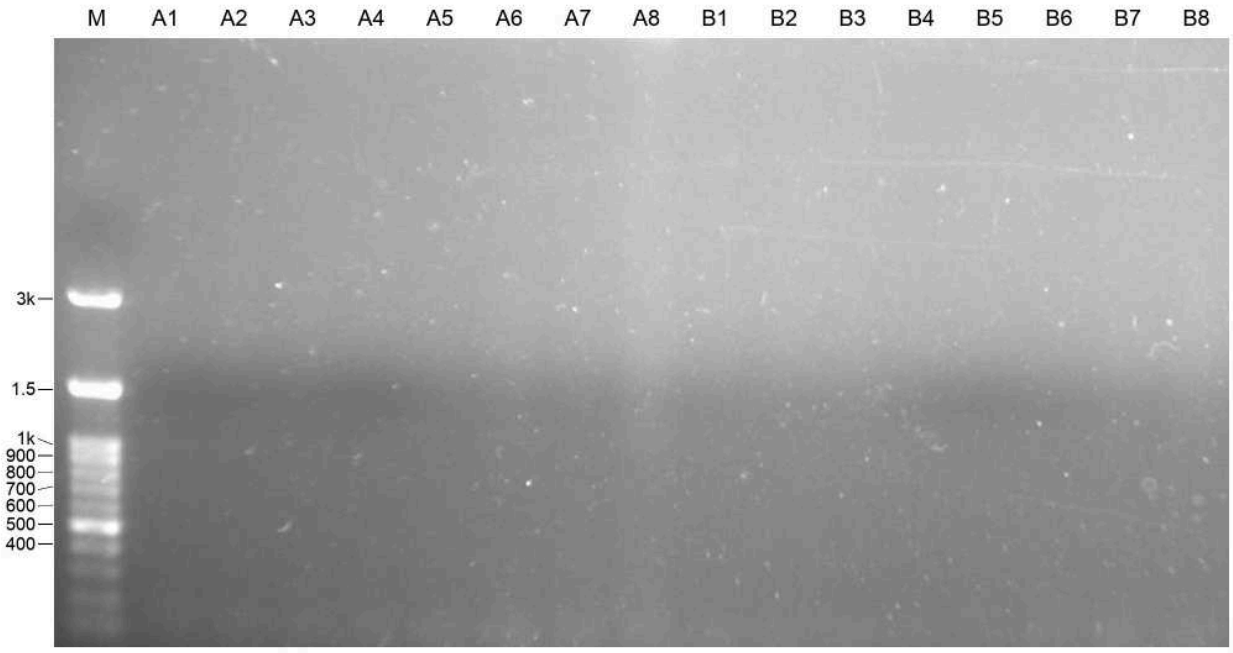
Title: 1/2 MS medium preparation (2000mL)
Purpose: Prepare 1/2 MS medium for further Arabidopsis planting(hydroponics)
Procedure:

1. Measure and add to the flask
    a. 1/2 MS medium(without sucrose): 4.6 g (2.3*2)
    b. MES buffer: 1 g (0.5*2)
    c. ddH_2O: 2 L (1*2)
2. Use the hot plate to stir(with a stirring bar) until no sediments are left in the solution
3. Determine pH of the solution and adjust the pH of the 1/2 MS medium using a pH probe to around 5.7-5.8pH
    a. Add 1M KOH to increase the pH
        i. Original pH: 4.1
        ii. Final pH: 5.7
4. Split the 2000 mL medium into 2 500mL flask and the original beaker
5. Wrap the neck with aluminum foil and place the indicator tape on top
6. Autoclave at 127 °C for 20 minutes
7. Wait for the pressure to reach 0 before opening the autoclave
8. Pour 300 mL of medium into each of the hydroponic boxes

Done by Sara and Sophie H
Supervised by Alex

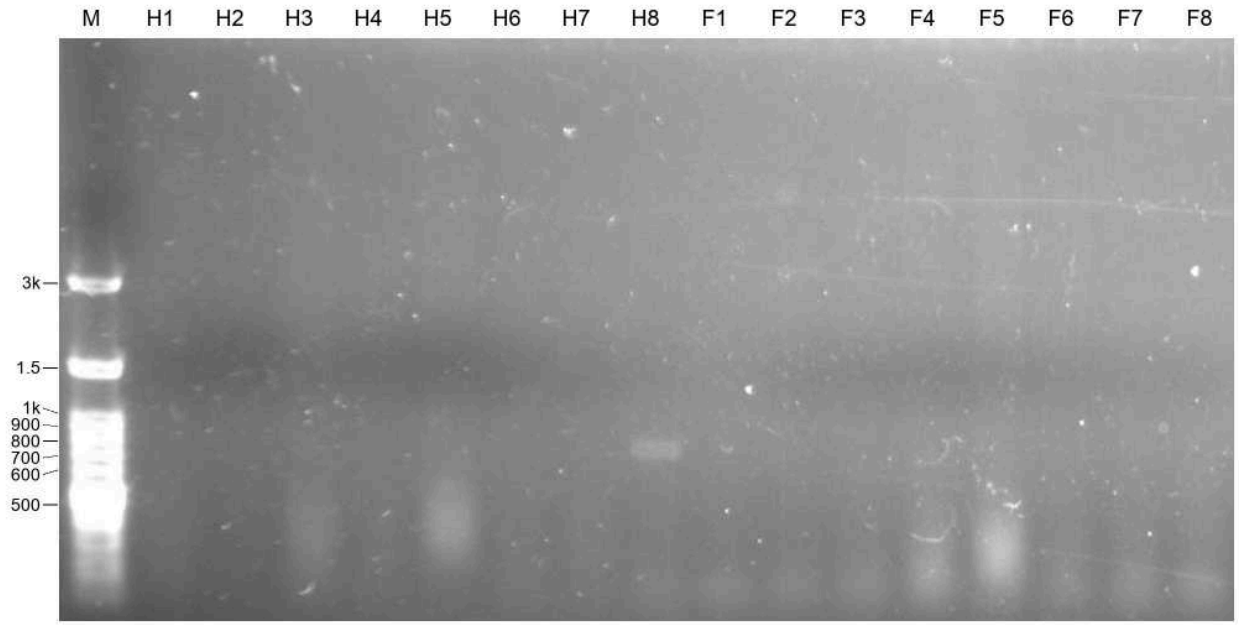
Date: 20260725
Title: Gel electrophoresis
Purpose: Check base pair of colony PCR product
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
A: pSTK-0C-SamyQ-Lea-2.1	Biobrick fv	LEA R	652
B: pSTK-0C-SamyQ-ACCD-2.1	Biobrick fv	ACCD R	1213

3. Load the gel in the layout shown below (5 µL each):
4. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA Ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
H: pSTK-0C-SamyQ-AmilCP-v3t1	biobrick-fv	AmilCP R	841
F: pSTK-0C-SamyQ-ACCD-v3t1	biobrick-fv	ACCD R	1213
Note: H8: liquid culture 7/25			

Done by Sophie C and Abby T

Date: 20260725
Title: Transformation
Purpose: Transform bacterial liquid on agar plates
Procedure:
A: Pveg-RBS0-Ho1:pcyA-B0015
L: Pveg-EEBD-RBS0-GanA-Gala-B0015

1. Set the water bath to 42 °C and prepare ice
2. Get LBA plates with glycerol
3. Thaw competent cells until they’re 50 % thawed and add 10 µl of plasmid
4. Incubate on ice for 20 min
5. Heatshock at 42 °C for 45 seconds

6. Incubate on ice for 3 minutes
7. Add 1 mL SOC medium
8. Incubate for 1 hour at 37 °C
9. Centrifuge at 5000 g for 5 minutes
10. Remove 1 ml of supernatant then vortex
11. Spread on LBA plates
12. Incubate at 37 °C for 16 hours

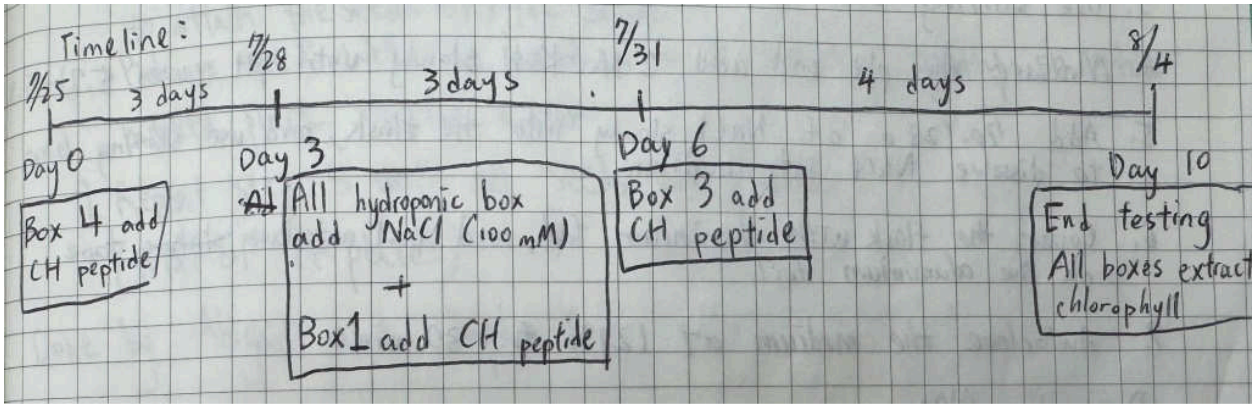
Done by Chloe and Sophie C

Date: 20260725
Title: Liquid Culture
Purpose: Transfer colony to medium
Procedure:
A1-3, A1-4, A1-8, A1-9, A2-6: Pveg-RBS0-Ho1:PcyA-B0015
H3, H7: Pveg-EEBD-RBS0-Csn:LEA-B0015
G2: PcpcG2-veg-EEBD-RBS0-Csn:LEA
Z10: Pveg-RBS0-mScarlet-B0015
J9: Pveg-EEBD-RBS0-csn:ACCD- B0015
I1: PcpcG2-veg-EEBD-RBS0-Csn:ACCD-B0015

1. Label 11 liquid culture tubes
2. Turn on Bunsen Burner, add 3 ml TB medium and 3 µl of ampicillin to all 11 tubes
3. Dip pipette into colonies and dispense tips into respective tubes
4. Incubate at 37 °C overnight

Done by Abby T and Sophie C
Supervised by Chars

Date: 20260725
Title: CH biotech protectant testing on box 1, 2, 3, 4
Purpose: test on when we should add protectants
Procedure:
1. Autoclave one tip box for 20 minutes at 127 °C and cleanse the 3D printed board with 10 % bleach
2. Add 300 mL of 1/2 MS medium into the tip box, and label this box, and label this box as box 4
3. Add 600 µL of CH biotech protectant into the medium of box 4
4. Place the 3d printed board onto the medium in box 4
5. Transfer 3 plants each from box 1 and box 3 to box 4
6. Transfer 4 plants from box 2 to box 4



Done by Alex
Supervised by Gina

Date: 20260725
Title: Prepare 3M NaCl with 1/2 MS medium
Purpose: hydroponics testing with CH biotech protectant
Procedure:

1. Prepare 400ml 1/2 MS medium with the following protocol

1/2 MS powder (w/o sucrose)	0.92 g
MES buffer	0.2 g
Distilled water	400 ml

2. Add the components into 50 ml flask
3. Use stirring bar to dissolve the powder into water
4. Measure the pH and add 1 M KOH slowly until pH reaches 5.7-5.8
5. Add 70.128 g of NaCl slowly into the flask, and use stirring bar to dissolve NaCl into medium
6. Cover the flask with aluminum foil, and put autoclave tape on the aluminum foil
7. Autoclave for 20 minutes at 127 °C

Done by Alex
Supervised by Gina

Date: 20260726
Title: Miniprep
Purpose: extract plasmid DNA from liquid culture of colony PCR products(0725)
Procedure:
A1-3, A1-4, A1-8, A1-9, A2-6: veg-RBS0-Ho1:PcyA-B0015
H3, H7: Pveg-EEBD-RBS0-Csn:LEA-B0015
G2P: cpcG2-veg-EEBD-RBS0-Csn:LEA
Z10: Pveg-RBS0-mScarlet-B0015
J9: Pveg-EEBD-RBS0-Csn:ACCD- B0015
I1: PcpcG2-veg-EEBD-RBS0-Csn:ACCD-B0015

1. Pellet bacterial overnight culture by centrifugation at 23444 g for 30 seconds at RT
2. Resuspend pelleted bacterial cells in 250 µl Buffer P1 and transfer to a microcentrifuge tube
3. Add 250 µl Buffer P2 and mix thoroughly by inverting the tube 4-6 times until the solution becomes clear(turn blue)
4. Add 350 µl Buffer N3 and mix immediately and thoroughly by inverting the tube 4-6 times(turn colorless)
5. Centrifuge for 10 min at 17900 g in a table top microcentrifuge
6. Apply supernatant from step 5 to QIA prep 2.0 spin column by pipetting. Centrifuge for 30 seconds and discard the flow through
7. Wash by adding 0.5ml Buffer PB. Centrifuge for 30 seconds(17900 g) and discard the flow through
8. Wash by adding 0.75 ml Buffer PE. Centrifuge for 30 seconds(17900 g) and discard the flow through
9. Centrifuge for 1 min to remove residual wash buffer
10. Place QIA prep 2.0 column in a clean 1.5 ml centrifuge tube. Add 50 µl sterilized water to the center of membrane
11. Stand for 1 min
12. Centrifuge at 17900 g for 1 min

Biodrop results

	concentration(ng/ µl)	A260/A280
A1-3	312.0	1.914
A1-4	420.0	1.918
A1-8	464.0	1.909
A1-9	467.0	1.914
A2-6	377.0	1.914
G2	133.0	1.928

H3	143.0	1.882
H7	108.0	1.929
I1	162.0	1.884
J9	410.0	1.898
Z10	274.0	1.903

Done by Abigail and Ryan W
Supervised by Chars

Date: 20260726
Title: Colony PCR of A, L
Purpose: screen for successful bacterial inserts from 7/25 transformation
Procedure:
A: Pveg-RBS0-Ho1:pcyA-B0015
L: Pveg-EEBD-RBS0-GanA-Gala-B0015

1. Prepare the reaction as follows

A. pSTK-1A-Pveg-RBS0-Ho1:pcyA-B0015 (pcr product bp: 2111)

	1 reaction	12 reactions
Forward primer (Biobrick Prefix)	1 µL	12 µL
Reverse primer (pSTK_Seq.Rv)	1 µL	12 µL
Q-Amp master mix	10 µL	120 µL
ddH_2O	8 µL	96 µL

L: pSTK-EXP1-Pveg-EEBD-RBS0-Spacer-GanA-Gala-B0015 (pcr product: 2993)

	1 reaction	6 reactions
Forward primer (Exp-seq-fw)	1 µL	6 µL
Reverse primer (Exp-seq-rv)	1 µL	6 µL
Q-amp master mix	10 µL	60 µL
ddH_2O	8 µL	48 µL

1. Circle 11 colonies from A, 5 colonies from L

2. Turn on Bunsen Burner and Pick up 11 colonies from A, 5 colonies from L with tip and streak the labelled grid
3. Dispense the tips with colonies into the PCR strips and wait 5 minutes
4. Take out the tips and close the lid
5. Place into PCR machine for thermal cycle

95 ℃	2 min	1 cycle
95 ℃	30 sec	30 cycles
55 ℃	20 sec	
72 ℃	3 min (largest bp: 2993)	
72 ℃	5 min	1 cycle

Done by Sophie L and Abby T
Supervised by Chars

Date: 20260726
Title: Prepare 0.8% Agarose Gel
Purpose: make gels for gel electrophoresis
Procedure:

1. Assemble mold, tray and comb
2. Measure 0.32 g of agarose powder and 40 ml of 1xTAE buffer
3. Pour agarose powder and 1XTAE buffer into conical flask
4. Microwave for 30 seconds then agitate it. Repeat this step until no sediments remain
5. Wait until no longer smoking then add 2 µl of safety dye then agitate
6. Pour melted gel into mold
7. Store in RT until solidified

Done by Sophie L

Date: 20260726
Title: Enzyme Digestion
Purpose: Check plasmid size before DNA sequencing
Procedure:
A1-3, A1-4, A1-8, A1-9, A2-6: Pveg-RBS0-Ho1:PcyA-B0015
H3, H7: Pveg-EEBD-RBS0-csn:LEA-B0015
G2: PcpcG2-veg-EEBD-RBS0-csn:LEA
Z10: Pveg-RBS0-mScarlet-B0015
J9: Pveg-EEBD-RBS0-Csn:ACCD- B0015
I1: PcpcG2-veg-EEBD-RBS0-Csn:ACCD-B0015

1. Prepare the following reaction:

A1-3 (full length base pair: 4032):

10x CutSmart	2 µL
DNA	1.603 µL
ddH_2O	15.397 µL
EcoRI (1 site)	0.5 µL

A1-4 (full length base pair: 4032):

10x CutSmart	2 µL
DNA	1.190 µL
ddH_2O	15.81 µL
EcoRI (1 site)	0.5 µL

A1-8 (full length base pair: 4032):

10x CutSmart	2 µL
DNA	1.078 µL
ddH_2O	15.922 µL
EcoRI (1 site)	0.5 µL

A1-9 (full length base pair: 4032):

10x CutSmart	2 µL
DNA	1.071 µL
ddH_2O	15.929 µL
EcoRI (1 site)	0.5 µL

A2-6 (full length base pair: 4032):

10x CutSmart	2 µL
DNA	1.326 µL
ddH_2O	15.674µL
EcoRI (1 site)	0.5 µL

H3 (full length base pair: 7030):

NE Buffer	2 µL
DNA	3.497 µL
ddH_2O	13.503 µL
BsmbI (1 site)	0.5 µL

H7 (full length base pair: 7030):

NE Buffer	2 µL
DNA	4.630 µL
ddH_2O	12.37 µL
BsmbI (1 site)	0.5 µL

G2 (full length base pair: 3296):

NE Buffer	2 µL
DNA	3.759 µL
ddH_2O	13.241 µL
BsmbI (2 sites)	0.5 µL

Z10 (full length base pair: 2780):

10x CutSmart	2 µL
DNA	1.825 µL
ddH_2O	15.175 µL
EcoRI (1 site)	0.5 µL

J9 (full length base pair: 7767):

10x CutSmart	2 µL
DNA	1.220µ L
ddH_2O	15.78 µL
BbsI (1 site)	0.5 µL

I1 (full length base pair 3757):

10x CutSmart	2 µL
DNA	3.086 µL
ddH_2O	13.914 µL
EcoRI (1 site)	0.5 µL

2. Prepared the reaction according to the formula: add water, DNA, buffer first before enzymes
3. Incubated the formula in PCR machine

55 °C	15 hours
80 °C	20 min

Done by Abby T, Ryan W, and Abigail
Supervised by Chars

Date: 20260726
Title: PCR
Purpose: Amplify plasmid number
Procedure:

1. Prepare the following reaction for:

L3SIPS2-L3S2P56

MF001

MF002

Reagent	1 rxn (50 µL)	3 rxn (150 µL) *3 (master mix)
10x ExTaq Buffer	5 µL	15 µL
2.5 µM dNTP mix	4 µL	12 µL
10 µM forward primer	1 µL	3 µL
10 µM reverse primer	1 µL	3 µL
DNA temp	1 µL	3 µL

PCR grade water	37.75 µL	113.25 µL
Ex Taw polymerase	0.25 µL	0.75 µL

2. Distribute 50 µL of the 3 rxn mixture into each tube for the corresponding part
3. Place into PCR machine, applying the thermalcycle below:

98 °C	30 seconds	1x
98 °C	10 seconds	25x
65 °C	30 seconds	
72 °C	30 seconds	
72 °C	2 minutes	1x
25 °C	0 minutes	

Done by Chloe

Date: 20260726
Title: SOC Medium Preparation
Purpose: prepare soc medium for transformation
Procedure:

1. Prepare 3.17 g of SOC broth powder.
2. Prepare 100 mL ddH_2O
3. Mix together by using a stirring rod.

Done by Abby T
Supervised by Chars

Date: 20260726
Title: Transformation
Purpose: redo transformation of 0725
Procedure:
A: Pveg-RBS0-Ho1:pcyA-B0015
L: Pveg-EEBD-RBS0-GanA-Gala-B0015

1. Set the water bath to 42 °C and prepare ice
2. Get LBA plates with glycerol
3. Thaw competent cells until they’re 50% thawed and add 10 µl of plasmid
4. Incubate on ice for 20 min
5. Heatshock at 42 °C for 45 seconds

6. Incubate on ice for 3 minutes
7. Add 1 mL SOC medium
8. Incubate for 1 hour at 37 °C
9. Centrifuge at 5000 g for 5 minutes
10. Remove 1 ml of supernatant then vortex
11. Spread on LBA plates
12. Incubate at 37 °C for 16 hours

Done by Chloe and Abigail
Supervised by Chars

Date: 20260727
Title: Miniprep
Purpose: Purify desired plasmid from liquid culture
Procedure:
D10: pSTK-0AB-PtagC+RBS
H2: pSTK-0D-bkdB Cold double terminator
G12: pSTK-0D-L3S2P21
E8: pSTK0-B-RBS2
D11: pSTK-0B-3P promoter
D7: pSTK-0A-Phyperspank

1. Transfer 1 mL of liquid culture to a 2mL tube
2. Centrifuge at 16000 g for 60 seconds
3. Discard the supernatant
4. Repeat step 1-3 until all medium are collected
5. Add 200 µl Mx1 buffer
6. Resuspend by vortexing or pipetting
7. Add 250 µl Mx2 buffer
8. Invert slowly 8-10 times, incubate at room temperature for 2 min
9. Add 350 µl Mx3 buffer
10. Invert slowly 4-5 times
11. Centrifuge at 16000 g for 5 min
12. Transfer around 800 µl supernatant into the column
13. Centrifuge at 7000 g for 30 min
14. Discard the flow through
15. Add 500 µl of WN buffer, centrifuge 7000 g for 30 seconds, discard the flow through
16. Add 500 µl of WS buffer, centrifuge 7000 g for 30 seconds, discard the flow through
17. Centrifuge 1000 g for 8 min
18. Assemble 1.5 mL microtube
19. Add 50 µl of EB buffer on the center of the membrane
20. Incubate in RT for 3 min
21. Centrifuge 10000 g for 3 min

Biodrop results

	Concentration (ng/µl)	A260/A280	A260/A230
D10	119	1.725	0.967
H2	39	1.857	1.114
G12	119	1.653	0.902
E8	101	1.683	1.01
D11	123	1.662	0.932

D7	110	1.774	0.887

Done by Sarah, Naomi, Sophie L

Date: 20260727
Title: Make 0.8 % Agarose gel (small+big)
Purpose: electrophoresis for 0726 enzyme digestion
Procedure:

1. Assemble mold, tray and comb
2. Measure 0.48 g of agarose powder and 60 mL of 1x TAE buffer
3. Pour agarose powder and 1xTAE buffer into conical flask
4. Microwave for 30 seconds then agitate it
5. Repeat this step until no sediments
6. Wait until no smoke and add 3 µl of safety dye
7. Agitate it
8. Pour the melted gel into the mold
9. Wait until the gel solidifies in RT

Done by Sophie C

Date: 20260727
Title: Colony PCR of I, L
Purpose: Amplify plasmid number
Procedure:
I: pSTK-PcpcG2-veg-EEBD-RBS0-csn:ACCD-B0015
L: pSTK-Exp1-Pveg-GanA:Galactosidase DH5a Assembly

1. Prepare the reaction as follows

I: pSTK-PcpcG2-veg-EEBD-RBS0-csn:ACCD-B0015

	1 reaction	12 reactions
Forward primer (PcpcG2-veg fw)	1 µL	12 µL
Reverse primer (ACCD rv)	1 µL	12 µL
Q-amp master mix	10 µL	120 µL
ddH_2O	8 µL	96 µL

L: pSTK-Exp1-Pveg-GanA:Galactosidase DH5a Assembly(x26)

	1 reaction	26 reactions
Forward primer (Pveg Fw)	1 µL	26 µL

Reverse primer (Exp seq4 rv)	1 µL	26 µL
Q-amp master mix	10 µL	260 µL
ddH_2O	8 µL	208 µL

1. Circle 10 colonies from I, 24 colonies from L
2. Turn on Bunsen Burner and pick up 11 colonies from A, 5 colonies from L with tip and streak the labelled grid
3. Dispense the tips with colonies into the PCR strips and wait 5 minutes
4. Take out the tips and close the lid
5. Place into PCR machine for thermal cycle

95 ℃	2 min	1 cycle
95 ℃	30 sec	30 cycles
55 ℃	20 sec	
72 ℃	3 min (largest bp: 2993)	
72 ℃	5 min	1 cycle

Done by Sophie C
Supervised by Gabriel

Date: 20260727
Title: Electrophoresis and Gel extraction
Purpose: Isolate desired DNA fragment from enzyme digestion
Procedure:

1. Mix 50 µL DNA with 10 µL 6X loading dye.
2. Loading sample by the following layout.

Gel 1

M	A	
5	60	volume (µL)

M= 1kb DNA ladder (blue)
A= MF001

Gel 2

M	B	
5	60	volume (µL)

M= 1kb DNA ladder (blue)
B= MF002

Gel 3

M	C	
5	60	volume (µL)

M= 1kb DNA ladder (blue)
C= L3SIPS

3. Resolve the DNA by 100 v 30 minutes.

Done by Abby T, Chloe
Supervised by Gabriel

Date: 20260727
Title: Gel Extraction
Purpose: DNA fragment isolation and purification.
Procedure:

1. Use a sanitized sharp knife to cut the gel band containing the desired DNA fragment under the UV light box.
2. Place each 3 gel bands into separated and labeled 2 ml tubes (Mf002, Mf001, and L3SIPS).
3. Add 700 µL of GP Buffer into each of the tubes.
4. Place the tubes in a dry bath on 60°C to dissolve the gel slices, check every 2 minutes until entirely dissolved.
5. Load no more than 700 microliter of melted gel into the collection tube
6. Centrifuge for 30 seconds at 7000 g. Discard the flow-through. Repeat step 5-6 step for the rest of the mixture.
7. Wash the column once with 0.5 ml of WN Buffer by centrifuging for 30 seconds. Discard the flow through.
8. Wash the column once with 0.5 ml of WS Buffer by centrifuging for 30 seconds. Discard the flow through.
9. Centrifuge the column at full speed 23444 g for 3 minutes to remove residual ethanol.
10. Place the column onto a new centrifuge tube. Add 15 µL of EB buffer onto the center of the membrane.
11. Stand the column for 2 minutes and centrifuge at full speed 23444g for 2 minutes to elute DNA.

12. Turn on BioDrop, adjust the setting to pathlength unit (concentration).
13. Clean the sample port with 1 µL of Nucleus free water.
14. Calibrate.
15. Load 1 µL of DNA from the 2 tubes each time and record down the results.
16. Clean the port with Nucleus Free water and close BioDrop.

Done by Abby T, Chloe
Supervised by Gabriel

Date: 20260727
Title: Plant seeds in circular 1/2 MS medium (neutral) plates
Purpose: plant seeds to vernalize for 4 days before testing on 正瀚 protectants
Procedure:

1. Distribute (columbia-0 kyle seeds into individual tubes
2. Open bsc and put the circular 0.8% 1/2 ms medium plates from 0726, see template (horizontal 5 seeds x vertical 5 seeds = 25 seeds / plate), seeds in tubes sterilized water and bamboo sticks in the bsc
3. Place the plate on the seed template
4. Stick the bamboo stick in sterilized water so it is easier to attach the seeds on the stick
5. Pick a seed with the bamboo stick and tap lightly on the marked place on the agar
6. Repeat step 5 until all seeds are planted (total of 14 plates)

Done by Abigail and Sophia Y.

Date: 20260727
Title: Make experiment set 6 salinity plates
Purpose: to prepare salinity plates for experiment set 6
Procedure:

1. Melt the solidified 1/2 MS agar (in tubes 30mL) by putting the 2 tubes into microwave

For 300 mL of agar:
6 agar plates — only made 2 agar plates (control X control and 150 mM NaCl x control with the stored 1/2 MS solidified agar, 3M NaCl liquid and 100 mM trehalose liquid)

- Control (no trehalose) × 75 mM NaCl
- Control (no trehalose) × 100 mM NaCl
- 5 mM tre × control (no NaCl)
- 5 mM tre × 75 mM NaCl
- 5 mM tre × 100 mM NaCl
- 5 mM tre × 150 mM NaCl

1. Prepare the following
    a. 1/2 MS medium powder (w/o sucrose): 0.6 g
    b. MES buffer: 0.15 g
    c. Distilled water: 300 mL
    d. Sucrose (1%): 3 g

    e. Agar powder (1%): 3 g
2. Add the contents into a 1000 mL flask
3. Use a hot plate to stir (with a magnetic stirring rod) until no sediments are left in the solution
4. Use the pH probe and measure the pH of the solution, make sure the pH is 5.7–5.8
    a. If pH too low, add 0.1 M KOH slowly to increase it
5. Pour 300 mL of mixture out into another flask to separate into six flasks, label with:
    a. [0 mM] NaCl, 5 mM trehalose
    b. [75 mM] NaCl, 5 mM trehalose
    c. [100 mM] NaCl, 5 mM trehalose
    d. [150 mM] NaCl, 5 mM trehalose
    e. [75 mM] NaCl, 0 mM trehalose
    f. [100 mM] NaCl, 0 mM trehalose
6. Add 219.2 mg of NaCl into the flask labelled [75 mM NaCl]
7. Add 292.2 mg of NaCl into the flask labelled [100 mM NaCl]
8. Add 438.3 mg of NaCl into the flask labelled [150 mM NaCl]
9. To their respective trehalose concentrations, add:

Trehalose concentration	Amount of trehalose
0 mM	0 mg
5 mM	85.575 mg

10. Wrap the necks of the flasks with aluminum foil and stick a piece of indicator tape on top
11. Autoclave for 20 min at 127 °C
12. Wait for pressure to reach 0 before retrieving flasks

Done by: Abby, Abigail
Supervised by: Gabriel
ELN by Naomi

Date: 20260728
Title: Prepare 1/2 MS medium (Hydroponic medium)
Purpose: prepare medium for hydroponic testing (正瀚protectant + NaCl)
Procedure:

1. The following protocol is for 2000mL of 1/2 MS medium
    a. 1/2 MS medium powder without sucrose: 4.6 g
    b. MES buffer: 1 g
    c. Distilled water: 2000 mL
2. Add the contents in the beaker
3. Use a hot plate to stir (with a magnetic stir bar) until no sediments are left in the solution
4. Use the pH probe and measure the pH of the solution, ensure the pH is 5.7~5.8
    a. Actual pH: 4.1, pH after adjustment: 5.7
5. If the pH is too low, add 0.1 M KOH slowly to increase it
6. Wrap the beaker with aluminum foil and stick a piece of autoclave indicator tape on the top
7. Autoclave for 20 minutes at 127 °C.

Done by Abigail, AudreyH, Sophie H

Date: 20260728
Title: PCR
Purpose: To amplify the number of DNA circuit
Procedure:

1. Prepare the following reaction for L3S1PS2-L3S2P56, MF001, MF002

Reagent	1 rxn	3 rxn (150μL)
10x EX Taq buffer	5 μL	15 μL
2.5 μm dNTP mix	4 μL	12 μL
10 μL fw	1 μL	3 μL
10 μL rv	1 μL	3 μL
DNA temp	1 μL	3 μL
PCR-grace H_2O	37.75 μL	113.25 μL
Ex taq polymerase	0.25 μL	0.75 μL

2. Distribute 50μL of the 3rxn mixtures into each tube for the corresponding parts
3. Place in PCR machine and assemble the thermalcycle as below:

98 ℃	30 sec	1 cycle
98 ℃	10 sec	25 cycles

65 ℃	30 sec	
72 ℃	30 sec	
72 ℃	2 min	1 cycle
25 ℃	0 sec	

Done by Sophie C, Chloe

Date: 20260728
Title: Making 2% agarose gel x3
Purpose: Make gel for electrophoresis
Procedure:

1. Assemble mold, tray, and comb
2. Weight 1.2 g of agarose powder and measure 60 mL of 1xTAE buffer
3. Add the agarose powder and the 1x TAe buffer into conical flask
4. Microwave for 30 sec then agitate it
5. Repeat this step until no sediments
6. Wait until no smoke, add 3μL of safety dye and agitate it as well
7. Pour the melted gel into the tray and use tip to squeeze the air and pop the bubble
8. Wait until the gel solidifies

Done by Sophie C, Chloe

Date: 20260728
Title: Gel extraction
Purpose: recollect PCR product from gel after electrophoresis
Procedure:

1. Use a sanitized sharp knife to cut the gel band containing the desired DNA fragment under the UV light box
2. Place each 3 gel bands into separated 2 ml tubes (MF001, MF002, L3S1PS)
3. Add 300μL GP buffer into each tube
4. Place tubes in a 60 ℃ dry bath to dissolve the slices(check every 2min until fully dissolved)
5. Load no more than 700μL of melted gel in the collection tube
6. Centrifuge 7000 g for 30secs, then discard the flow through and repeat steps 5-6 for the rest of the mixture
7. Wash column with 500 μL WN buffer and centrifuge 7000 g for 30 secs, then discard the flow through
8. Wash column with 500 μL WS buffer and centrifuge 7000 g for 30 secs, then discard the flow through
9. Centrifuge empty column at 23,444 g for 3min
10. Place column on a new 1.5 mL tube and add 15 μL Eb buffer to the center of the membrane
11. Stand the column for 2 min and centrifuge at 23,444 for 2 min to elute DNA

12. Measure concentration and A260/280 of plasmid on biodrop.

Done by Chloe, Sophie C.

Date: 20260728
Title: GGA of A, L
Purpose: To assemble DNA construct
Procedure:

A: (STK) Module 1 (1A): Pveg-RBS-ho1:pcyA-L3S2P21

1. prepare the reaction as follow:

Part	Original Amount	Dilution
Vector (1A) 115 ng/μL, 3120 bp	0.18 μL	1.76 μL (dilute 1/10)
PtagC+RBS 725 ng/μL, 2184 bp	0.23 μL	2.27 μL (dilute 1/10)
Ho1:pcyA 238 ng/μL, 1556 bp	0.08 μL	1.7 μL (dilute 1/20)
L3S2P21 170 ng/μL, 2145bp	0.16 μL	1.64 μL (dilute 1/10)
T4 DNA ligase buffer	2 μL	2 μL
Sterile water	8.63 μL	//
T4 DNA ligase	1 μL	1 μL
BsaI-HFv2	1 μL	1 μL
Total	20 μL	20 μL

L: (STK) Module 5: 3P-EEBD-RBS0-spacer-GanA-Gala-spacer-B0015

Part	Original Amount	Dilution
Vector (Exp1) 162 ng/μL, 6929 bp	0.28 μL	1.39 μL (dilute 1/10)
3P 106 ng/μL, 2909 bp	0.36 μL	1.8 μL (dilute 1/10)
EEBD-RBS0 123 ng/μL, 2392 bp	0.25 μL	1 μL (dilute 1/20)
GanA-Gala 177 ng/μL, 4148bp	0.30 μL	1.5 μL (dilute 1/10)
Spacer CX	0.107 μL	1.07 μL
Spacer CZ	0.116 μL	1.16 μL
L3S2P21	0.16 μL	1.64 μL
T4 DNA ligase buffer	2 μL	2 μL
Sterile water	6.44 μL	6.44 μL
T4 DNA ligase	1 μL	1 μL
BsaI-HFv2	1 μL	1 μL
Total	20 μL	20 μL

Use the thermocycler to run the program below:

Temperature	Time	Cycles
37 °C (Restriction enzyme digestion)	5 minutes	30 cycles

16 °C (Ligation)	5 minutes	
37 °C (Restriction enzyme digestion)	10 minutes	1 cycle
80 °C (Heat inactivation of BsaI restriction enzyme)	20 minutes	1 cycle

Done by Sophie C, Chloe

Date: 20260728
Title: Liquid culture
Purpose: To transfer the colony on agar plate in medium
Procedure:
D7: pSTK-0A-Phyperspank
D10: pSTK-0AB-PtagC+RBS
D11: pSTK-0A-P3P
E8: pSTK-0B-RBS2
H2: pSTK-0D-bkaB-Co1D-double terminator
G12: pSTK-0D-L3S2P21

1. Label 6 liquid culture tubes
2. Turn on bunsen burner and add 3 mL TB medium and 3 µL of chloramphenicol in each tube
3. Add glycerol stock into each tube
4. Incubate in shaker overnight

Done by Sophie C, Gabriel

Date: 20260728
Title: Miniprep
Purpose: Purify desired plasmid
Procedure:
A11: Pveg-RBS0-Ho1:pcyA-B0015

1. Transfer 1.8 mL of A11 liquid culture to a 2 mL microtube
2. Centrifuge at 23444 g for 30 sec, then discard the supernatant
3. Repeat steps 1-2 until all medium is collected, remember to empty the medium in the tips too
4. Add 200 µL of MX1 buffer, vortex or pipette to resuspend it
5. Add 250 µL of MX2 buffer, vortex or pipette to resuspend it
6. Invert slowly 8-10 times, Incubate at RT for 2 mins
7. Add 350µL of MX3 buffer, and invert slowly 4-5 times
8. Centrifuge for 5min at 16000 g

9. Transfer about 900 µL of supernatant into assembled column
10. Centrifuge at 7000 g for 30secs and discard the flow through
11. Wash the column with 500 µL WN buffer, then centrifuge at 700 g for 30sec
12. Discard the flow through
13. Wash the column with 500 µL WS buffer, then centrifuge at 700 g for 30sec
14. Discard the flow through
15. Centrifuge for 8 min at 10000 g
16. Place the new column into new 1.5 ml microtube
17. Add 50 µL Eb to the center of th membrane
18. Incubate for 30 min at RT
19. Centrifuge at 10000 g for 30 sec

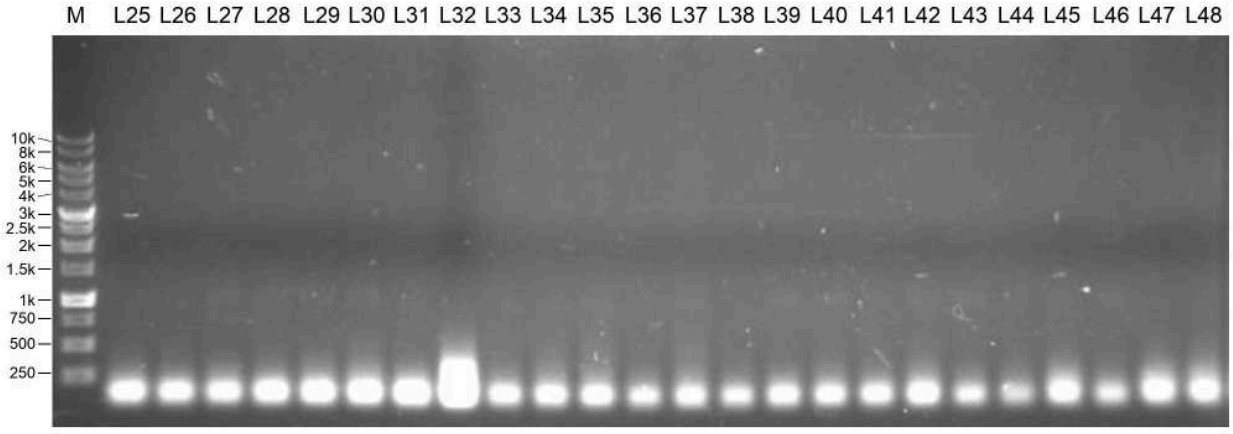
Biodrop Results:

A11	**Concentration**	**A260/230**	**A260/280**
	75 ng/µL	1.230	2.032

Done by Sophie C.

Date:20260728
Title:Gel electrophoresis
Purpose: To check the results of DNA circuit after GGA and PCR
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)

L: Pveg-EEBD-RBS0-GanA-Gala-B0015	Pveg fw	EXP-seq4 rv	2946
Note: All fail			

Done by Sophie C

Date: 20260728
Title: Transfer plants from incubator to salinity plates
Purpose: start salinity stress testing of experiment 6 (0, 75, 150, 150mM NaCl, 0, 5mM trehalose)
Procedure:

1. Take plates (incubated for 4 days) out of growth chamber
2. Open BSC
3. Place 2 vertical plate holders, tweezers, [unclear/crossed out], the salinity plates and plates that incubated for 4 days into the BSC
4. Choose plants that have similar root lengths to transfer after visually comparing the incubated plants
5. Lightly lift the chosen plant using a tweezer below the leaves
6. Place the plant on the salinity plates
7. Use a marker to draw a straight line along a ruler using the seed template
8. Align the root endpoints with the drawn line and use a marker to indicate the endpoint for further image analysis
9. Seal the plates with breathable tape and label the test group and transferred date
10. Place the vertical plates on the racks in the growth chamber at a slanted angle towards the back (front faces forward)

Done by: Sara, Sophie H, Abby T

Date: 20260728
Title: Change medium for hydroponic box 1, 2, 3, 4
Purpose: change medium to test on when we should add protectant
Procedure:

1. Take 4 new tip boxes, 3M NaCl, CH protectant, and 1000 µL pipette, P100 pipette gun, and serological pipettes to BSC
2. Add 290 mL of medium to Box 1, 2, 3 and 4 and use the beaker with a label of 290 mL to measure the amount of medium
3. Add 600 µL of CH protectant into Box 1
4. Add 10 mL of 3M NaCl to Box 1, 2, 3, and 4
5. Take the 3D printed rack from the corresponding boxes and put them into the corresponding boxes
6. Label the new boxes

7. Wrap your phone with plastic wrap and spray alcohol on it and use it to take test pictures of the 4 boxes

Done by: Abby T, Audrey H

Date: 20260729
Title: Gel extraction
Purpose: to isolate desired DNA fragment
Procedure:

1. Use a sanitized sharp knife to cut the gel band containing the desired DNA fragment under the UV light box.
2. Place each 3 gel bands into separated and labeled 2ml tubes
   Mf002 bp: 89
   Mf001 bp: 90
   L3SIPS bp: 175
3. Add 700 µL of GP Buffer into each of the tubes.
4. Place the tubes in a dry bath on 60 °C to dissolve the gel slices, check every 2 minutes until entirely dissolved.
5. Load no more than 700 microliter of melted gel into the collection tube
6. Centrifuge for 30 seconds at 7000 g. Discard the flow-through. Repeat step 5-6 step for the rest of the mixture.
7. Wash the column once with 0.5 ml of WN Buffer by centrifuging for 30 seconds. Discard the flow through.
8. Wash the column once with 0.5 ml of WS Buffer by centrifuging for 30 seconds. Discard the flow through.
9. Centrifuge the column at full speed 23444 g for 3 minutes to remove residual ethanol.
10. Place the column onto a new centrifuge tube. Add 15 µL of EB buffer onto the center of the membrane.
11. Stand the column for 2 minutes and centrifuge at full speed 23444 g for 2 minutes to elute DNA.
12. Turn on BioDrop, adjust the setting to pathlength unit (concentration).
13. Clean the sample port with 1 µL of Nucleus free water.
14. Calibrate.
15. Load 1 µL of DNA from the 2 tubes each time and record down the results.
16. Clean the port with Nucleus Free water and close BioDrop.

Done by Abby T

Date: 20260729
Title: Making 1% Agarose gel
Purpose: Make Agarose gel for electrophoresis
Procedure:

1. Add 0.2 g of Agarose powder and 20 mL of TAE buffer into a flask
2. Assemble mold, tray, and comb
3. Cover the flask with plastic wrap and poke few holes on it with tip
4. Microwave it for 30 sec then agitate it, repeat this step until no sentiment
5. Wait until the flask doesn’t smoke but is still hot, add 1 µL of diluted safety dye into the flask and gently mix it
6. Pour the melted gel into the mold
7. Wait for it to solidify

Done by Ryan Yuan

Date:20260729
Title:Electroporation of K
Purpose: Transfer bio-engineered plasmid into *Bacillus subtilis* common cell
Procedure:
K: Pveg-EEBD-SamyQ-TAT-HA-B0015 con (ng/ul): 354 ng/μL

1. Prewarm SOC medium in 37 degree celsius
2. Precold 0.2 cm electroporation cuvette
3. Mix 3 μL plasmid (con 354 ng/μL) with 100 μL electrocompetent cell in a 1.5 mL microfuge
4. Set up electroporator
    A. Go to homepage
    B. when using the 0.2 cm corvette, press 2 or 3 and enter
    C. set voltage resistance, voltage and capacity
    a. Voltage: 2.3 KV
    b. Resistance: 300 Ω
    c. Capacitance: 25 F
5. Transfer mixture of cells+DNA to a cold electroporation corvette and tap the suspension to the bottom
6. Place the corvette in the shock pad and close the lid
7. Wait until beep sound
8. Remove the corvette quickly form the shock pad and immediately add 1 mL SOC medium to the cuvette
9. Transfer the cell suspension to a polypropylene tube and incubate at 37 ℃ for 3 hr, shaking at 200 rpm
10. Check & record the parameter
11. Centrifuge the tubes at 5000 g for 5min
12. Spread the LB plate without erythromycin

TC(ms): 6.6
V0Hs(V): 2284

Done by Grabriel

Date:20260729
Title: PCR
Purpose: Amplify DNA plasmid sequence number
Procedure:
1. Prepare the reaction as follows for MF001, MF002, L3S1P2

	1 rxn	5 rxn (master mix)

Ex taq buffer	5 µL	25 µL
2 uM dNTP mix	4 µL	20 µL
10 uM fw primer	1 µL	5 µL
10 uM rv primer	1 µL	5 µL
DNA Temp	1 µL	5 µL
PCR grade water	37.75 µL	188.75 µL
Ex taq polymerase	0.25 µL	1.25 µL

2. Distribute 50µL of the 5rxn mixture into the corresponding tube of the part
3. Put into thermal cycle

98 ℃	30 sec	1x cycle
98 ℃	10 sec	35 cycle
55 ℃	30 sec	
72 ℃	40 sec	
72 ℃	2 min	
25 ℃	0 sec	1 cycle

Done by Sophie C, Abby T
Supervised by Gabriel

Date:20260729
Title: Make 2 % gel (x3)
Purpose: Make gel for electrophoresis
Procedure:

1. Weight 1.2 g of agarose powder and 60 mL 1X TAE buffer
2. Add the components into a conical flask
3. Wrap a plastic wrap and poke 3 hole
4. Microwave for 30 sec and agitate. Repeat this step until all mixture dissolved
5. Take the flask out and wait for it to cool down
6. When there's no more smoke but still hot, add 3 µL of safety dye and agitate it
7. Pour the melted gel into the assembled mold and tray
8. Wait 20 mins until the gel solidify in room temp

Done by Sophie C

Date:20260729

Title:Transformation
Purpose: Transfer bacterial liquid to agar plates
Procedure:
A: (STK) 1A:PtagC+RBS-Ho1:pcayA-L3S2P21
L: (STK) Exp1:3P-EEBD-RBS0-spacer-GanaA-Gala-spacer-B0015

1. Turn on the water bath at 42 ℃
2. Prepare an ice bath
3. Throw competent cells on ice until 50% thawed
4. Add 10 μL of plasmid
5. Incubate on ice for 20 min
6. Heat shock in the water bath for 1 min
7. Recover on ice for 3 min
8. Add 1 mL of SOC medium
9. Incubate at 37 ℃ for 1 hr
10. Centrifuge at 5000 g for 5 min
11. Remove 1 mL of supernatant
12. Resuspend the pellet in remaining supernatant and spread on LBA plate
13. Incubate at 37 ℃ for 16 hours

Done by SophieC

Date: 20260730
Title: Miniprep
Purpose: Collect the concentrated DNA plasmid
Procedure:
D7: pSTK-0A-Phyperspank
D10: pSTK-0AB-PtagC+RBS
D11: pSTK-0B-3P promoter
E8: pSTK0-B-RBS2
G12: pSTK-0D-L3S2P21
H2: pSTK-0D-bkdB Cold double terminator

1. Transfer 1000 µl liquid culture into 1.5mL tube
2. Put in centrifuge and centrifuge for 30 seconds at 23444 g
3. Discard the supernatant
4. Repeat step 1-3 until all medium are collected
5. Add 200 µl of MX1 buffer
6. Resuspending by vortexing or pipetting
7. Add 250 µl of MX2 buffer
8. Invert slowly 8-10 times and incubate at RT for 2 mins
9. Add 350 µl of MX3 buffer and invert slowly 4-5 times
10. Centrifuge at 16000 g for 5 min
11. Transfer around 800 µl supernatant into the column
12. Centrifuge at 7000 g for 30 min
13. Discard the flow through
14. Add 500 µl of WN buffer, centrifuge 7000 g for 30 seconds, discard the flow through
15. Add 700 µl of WS buffer, centrifuge 7000 g for 30 seconds, discard the flow through
16. Centrifuge 1000 g for 8min
17. Assemble 1.5 mL microtube
18. Add 50 µl of EB buffer on the center of the membrane
19. Incubate in RT for 3 min
20. Centrifuge 10000 g for 3min

Biodrop results:

	concentration(ng/ µl)	A260/230	A260/280
D7	187.0	1.345	2.055
D10	69.0	1.255	1.974
D11	134.0	1.149	1.925
E8	108.0	1.091	1.929
G12	138.0	1.155	1.443

H2	110.0	0.846	1.914

Done by Sarah and Sophie H

Date: 20260730
Title: Make *B. sub* electrocompetent cells(x4)
Purpose: restock for 0730 electroporation
Procedure:

1. Dilute an overnight culture of *B. sub* 16-fold in growth medium(LB medium containing 0.5M sorbitol) by adding 312.5 µl of B sub in 5mL GM and grow at 37°C to OD600 of 0.944
2. Cool the cells on ice and put on shaker for 10 minutes
3. Centrifuge at 4°C at 5000 g for 5 minutes to harvest cells
4. Add 1mL of ice cold electroporation medium to resuspend cells in the 1.5mL microtube by washing it along the side of the microtubes. Discard the medium. Repeat this step for 2 times to wash the cells
5. Pipette out the residue medium
6. Suspend the cells in 1/40 of the culture volume of the electroporation solution(132.8 µl for 3 tubes and 82.8 µl to the testing tube) with a cell concentration of 1-1.3 x 10^10 cfu/mL
7. Store the competent cells at -80°C

Done by Abigail and Ryan W
Supervised by Gabriel

Date: 20260730
Title: Colony PCR
Purpose: To verify bacterial colonies after a transformation step in molecular cloning
Procedure:
A: PtagC+RBS-Ho1:pcyA-L3S2P21
K: Pveg-SamyQ-TAT-HA-AmilCP(*B.sub*)
L:3P-GanA-Gala-B0015

1. Prepare the following reaction(master mix) and distribute solutions into PCR tubes

A: 13 reactions

Forward primer: Biobrick prefix	13 µl
Reverse primer: STK reverse (seq)	13 µl
Q-Amp master mix	130 µl
ddH_2O	104 µl

K: 9 reactions

Forward primer: SamyQ F	9 µl
Reverse primer: AmilCP R	9 µl
Q-Amp master mix	90 µl
ddH_2O	72 µl

L: 13 reactions

Forward primer: EXP fw	13 µl
Reverse primer: EXP-seq4 rv	13 µl
Q-amp master mix	130 µl
ddH_2O	104 µl

2. Circle 12 colonies each(from A and L) and 8 from K
3. Pick up those colonies with tip and dispense in PCR tubes
4. Take out tips and close the lid after incubating in RT for 5 min
5. Place into the PCR machine for the thermal cycle below:

95°C	5 min	x1
95°C	20 sec	x35
65°C	20 sec	
72°C	3 min	
72°C	5 min	x1

Done by Sophie C, Ryan Y, and Sophia

Date: 20260730
Title: Make 1% agarose gel
Purpose: Do electrophoresis for colony PCR
Procedure:

1. Assemble mold, tray, and comb
2. Weigh 0.6g of agarose powder and 60 mL of 1xTAE buffer
3. Add the components into a conical flask
4. Cover with plastic wrap and poke holes
5. Place into microwave for 30 secs
6. Agitate
7. Put into the microwave for 5 sec then agitate. Repeat until no sediments
8. When it stops smoking, add 3µL of safety dye, agitate

9. Pour the melted gel into the assembled mold
10. Let solidify at RT

Done by Chloe and Sophie C

Date: 20260730
Title: Make LB, LBA with glycerol, and LBE with glycerol 5µg/mL
Purpose: Restock plates
Procedure:

1. Measure and add the following into flasks:

LBA with glycerol

	Mass (g)
Broth powder	6.25 g
Agar powder	3.75 g
ddH_2O	250 mL
Glycerol	2 mL
Antibiotic(ampicillin)	500 µl

LBE with glycerol

	Mass (g)
Broth powder	6.25 g
Agar powder	3.75 g
ddH_2O	250 mL
Glycerol	2 mL
Antibiotic(Erythromycin)	1250 µl

LB

	Mass (g)
Broth powder	6.25 g
Agar powder	3.75 g
ddH_2O	250 mL

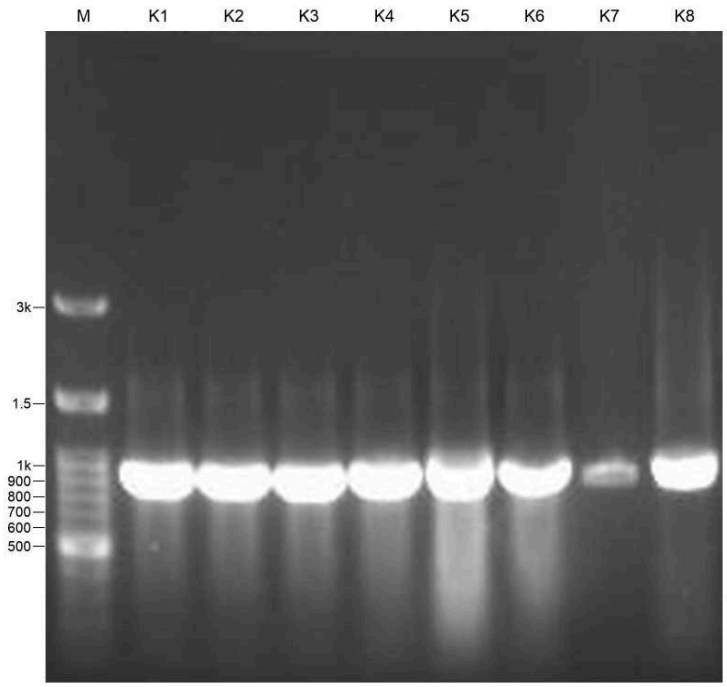
2. Add one magnetic stirring bar to the flask

3. Place the flask containing the mixture on the hot plate and stir until no sediments
4. Cover the flask with aluminum foil and apply an indicator tape
5. Autoclave for 20 minutes at 127 °C.
6. Place the new petri dishes in BSC and equipments
7. Set up water bath at 55 °C
8. Spray the working surface of the BSC and equipments with 75 % ethanol
9. Add 500 µl of Ampicillin/1250 µl of Erythromycin into flask
10. Use pipette to add 20 mL of melting LB into the petri dish
11. Let the plates dry on the BSC for 1 hour.

Done by Sarah and Sophie H

Date: 20260730
Title: Gel electrophoresis
Purpose: check bps of 0730 colony pcr
Procedure:

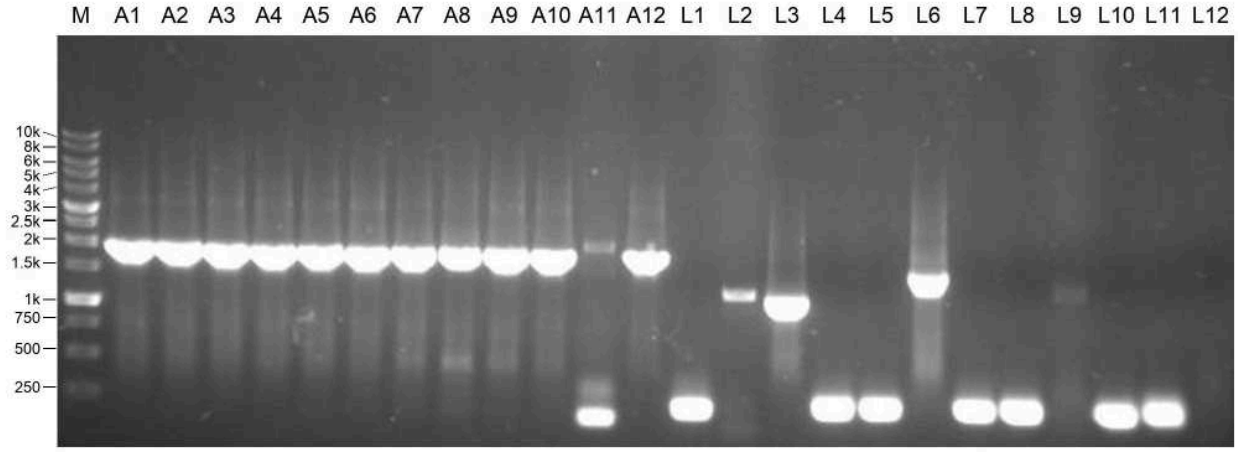
1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: [DM2300] ExcelBand™ 100 bp+3K DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
K: Pveg-SamyQ-TAT-HA-AmilCP	SamyQ F	AmilCP R	842

Note: K1, K2, K3: Liquid culture 7/30

3. Load the gel in the layout shown below (5 µL each):
4. Resolve at 100 v for 30 min



M: 1kb DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
A: PtagC+RBS-Ho1:pcyA-L3S2P21	Biobrick prefix	STK-seq rv	1833
L: 3P-GanA-Gala-B0015	EXP-seq fw	EXP-seq4 rv	3626
Note: A1, A2, A3 Liquid culture 7/30 -> sequencing all positive; L all fail			

Done by Chloe and Sophie C

Date: 20260730
Title: Transformation
Purpose: Transform liquid bacteria on plates
Procedure:
J: Pveg-Csn:ACCD

1. Turn on the water bath and set it to 42°C and prepare ice
2. Thaw competent cells on ice until 50% thawed
3. Add 10 µl of plasmid
4. Incubate on ice for 20minutes
5. Heat shock in the water bath for 1 minute

6. Recover on ice for 3 min
7. Spread on LBA plate
8. Incubate at 37°C for 16 hours

Done by Sarah and Sophie H
Supervised by Chloe and Gabriel

Date: 20260730
Title: Electroporation
Purpose: Use electric pulse to make temporary holes in a cell membrane
Procedure:
J: Pveg-EEBD-RBS0-Csn:ACCD-B0015

1. Prewarm soc medium at 37°C
2. Precold 0.2 cm electroporation cuvette
3. Mix 2.2 µl of plasmid(concentration 410 ng/ µl) with 86 µl of electrocompetent cell in a 1.5 mL microtube
4. Set up electroporator and set up voltage resistance
    a. Voltage: 23kv
    b. Resistance: 300 Ω
    c. Capacitance: 35F
5. Transfer mixture of cells and DNA to electroporation cuvette and tap suspension to bottom
6. Place in shock pad and close lid
7. Wait until beep
8. Remove from shock pad and immediately add 1 mL soc medium
9. Transfer to a polypropylene tube and incubate at 37°C for 3 hours. Shaking at 200 rpm
10. Check and record the parameter
11. Centrifuge tubes at 5000 g for 5 minutes
12. Spread the LB plate with erythromycin

TC(ms): 6.9
Volts(V): 2285

Done by Chloe and Sophie C

Date: 20260730
Title: Golden Gate Assembly(G, I, M)
Purpose: assemble parts
Procedure:

L: PtagC+RBS-Spacer CX-GanA-Gala-Histag CZ-L3S2P21

Part	Dilution

Vector (shuttle vector): pSTK-EXP-1-sfGFP	1.39 µL (dilute 1/5)
Insert promoter(Pveg)	1.1 µL (dilute 1/10)
Insert RBS(EEBD-RBS0)	2 µL (Dilute 1/10)
Insert CDS(GanA-Gala)	1.53 (dilute 1/5) µL
• Insert Terminator (B0015)	1.94 µL (dilute 1/10)
Spacer (pSTK-OCX-spacer)	1.07 µL (dilute 1/10)
Histag (pSTK-OCZ-HisGX)	1.16 µL (dilute 1/10)
T4 DNA ligase buffer	2 µL
Sterile water	5.81 µL
T4 DNA ligase	1 µL
BsaI-HFv2	1 µL
Total	20 µL

G: (STK) Module 4-PcpcG2-veg - EEBD- RBS0 - CSN::LEA - B0015

Vector (1D) • concentration(ng/µL): 132 • bp: 3120	1.5 µL (dilute 1/10)
Insert promoter(PcpcG2-veg) • concentration(ng/µL): 168 • bp: 235	1.175 (dilute 1/50)
Insert RBS(EEBD-RBS0) • concentration(ng/µL): 150 • bp: 2392	1.04 µL (Dilute 1/5)
Insert CDS(CSN::LEA) • concentration(ng/µL): 77 • bp:634 bp	1.08 µL (dilute 1/20)
Insert Terminator (B0015) • concentration(ng/µL): 107	1.34 µL (dilute 1/5)

• bp: 2213	
T4 DNA ligase buffer	2 µL
Sterile water	10.05 µL
T4 DNA ligase	1 µL
BsaI-HFv2	1 µL
Total	20 µL

M: (STK) Module 4-PcpcG2-veg - EEBD- RBS0 - spacer CX - GanAGala - histag CZ - dkdB colD

Vector (1D) • concentration(ng/µL): 132 • bp: 3120	1.5 µL (dilute 1/10)
Insert promoter (PcpcG2-veg) • concentration(ng/µL): 65 • bp: 235	1.175 (dilute 1/50)
Insert RBS (EEBD-RBS0) • concentration(ng/µL): 150 • bp: 2392	1.04 µL (Dilute 1/5)
Insert CDS(Csn::LEA) • concentration(ng/µL): 77 • bp:634 bp	1.08 µL (dilute 1/20)
Insert Terminator (B0015) • concentration(ng/µL): 107 • bp: 2213	1.34 µL (dilute 1/5)
T4 DNA ligase buffer	2 µL
Sterile water	10.05 µL

T4 DNA ligase	1 µL
BsaI-HFv2	1 µL
Total	20 µL

Part

T4 DNA ligase buffer	2 µL
Sterile water	8.28 µL
T4 DNA ligase	1 µL
BsaI-HFv2	1 µL
Total	20 µL

Done by Sophi C, Chloe, Abigail

Date: 20260730
Title: Liquid culture
Procedure:
A:Ptag+RBS-Ho1:PcyA-L3S2P21
K:pExp1-Pveg-SamyQ-TAT-HA-AmilCP

1. Label 7 liquid culture tube(A1,A2,A3,K1,K2,K3,K8)
2. Turn on Bunsen Burner
3. Add 3 mLof TB medium and 3 µL of Ampicillin
4. Pip a pipette tip into the colonies
5. Dispense tips in their respective tubes
6. Place the tubes in the incubator, slant around 45℃

Done by Ryan Wei
Supervised by Gabriel

Date: 20260730
Title: Plant seeds for set C and vernalize in hydroponic system
Purpose: heat stress testing
Procedure:

1. Create lids with 3 mm diameter holes, and autoclave for 20 min at 127°C
2. Eppendorf
3. Add 0.8% agar into 48 lids (270 µL each)
4. Wait until agar solidifies
5. Add 2.2 mL of 1/2 MS medium into 48 tube racks
6. Put lids onto the tube rack (hole facing up)

7. Use an autoclaved toothpick to pick one seed, and add the seed into the hole of the lid
8. Repeat step 6 until all lids have seeds
9. Close the lid of the tube rack
10. Wrap the tube rack with aluminium foil
11. Place the tube rack into the refrigerator at 4°C

Done by: Alex

Date: 20260730
Title: Prepare 50 mL 0.8% 1/2 MS agar
Purpose: add agar into lids for new set of seeds
Procedure:

1. 1/2 MS powder (w/o sucrose): 0.115 g
2. MES buffer: 0.025 g
3. Distilled water: 50 mL
4. Sucrose: 0.5 g
5. Agar: 0.5 g 0.4 g
6. Add 1/2 MS powder, MES buffer, sucrose, and distilled water into a flask
7. Shake the flask with hands until no sediments are in the solution
8. Use the pH probe to measure the pH of the solution, ensure the pH is between 5.7–5.8
9. If pH is too low, add 0.1 M KOH slowly to increase it
10. Add agar into the flask
11. Wrap the beaker with aluminium foil and stick a piece of indicator tape on top
12. Autoclave for 20 min at 127°C

Done by: Alex

Date: 20260730
Title: Prepare 2 1500 mL 1/2 MS medium
Purpose: prepare medium for changing medium and salinity test's medium
Procedure:

1. 1/2 MS powder (w/o sucrose): 3.45 g
    a. MES buffer: 0.75 g
    b. Distilled water: 1500 mL
2. Add the contents into a beaker
3. Use hot plate to stir (using a magnetic stirring bar) until no sediments are left in the solution
4. Use the pH probe and measure the pH of the solution, ensure the pH is between 5.7 and 5.8
5. If pH is too low, add 1 M KOH to slowly increase it
6. Wrap the beaker with aluminium foil and stick a piece of autoclave indicator tape on top
7. Repeat steps 1–6 to prepare the second 1500 mL 1/2 MS medium
8. Autoclave for 20 min at 127°C

Done by: Alex

Date: 20260731
Title: Colony PCR of J
Purpose: To verify bacterial colonies after a transformation step in molecular cloning
Procedure:
J:Pveg-EEBD-RBS0-Csn:ACCD-B0015(*B.sub*)

1. Prepare the following reaction (master mix) and distribute solutions to PCR tubes

JX9

Forward primer Csn fw	9 μL
Reverse primer ACCD R	9 μL
Q-Amp master mix	90 μL
ddH_2O	72 μL

2. Circle 8 colonies from plate J
3. Pick up those colonies with tips and dispense in PCR tubes
4. Take out the tips and close the lid after incubating in RT for 5 min
5. Place into PCR machines

95℃	5 min	x1
95℃	20 sec	x35
55℃	20 sec	
72℃	2 min	
72℃	5 min	x1

Done by Sophie C, Ryan Y

Date: 20260731
Title: Make 1% agarose gel
Purpose: for electrophoresis of colony PCR
Procedure:

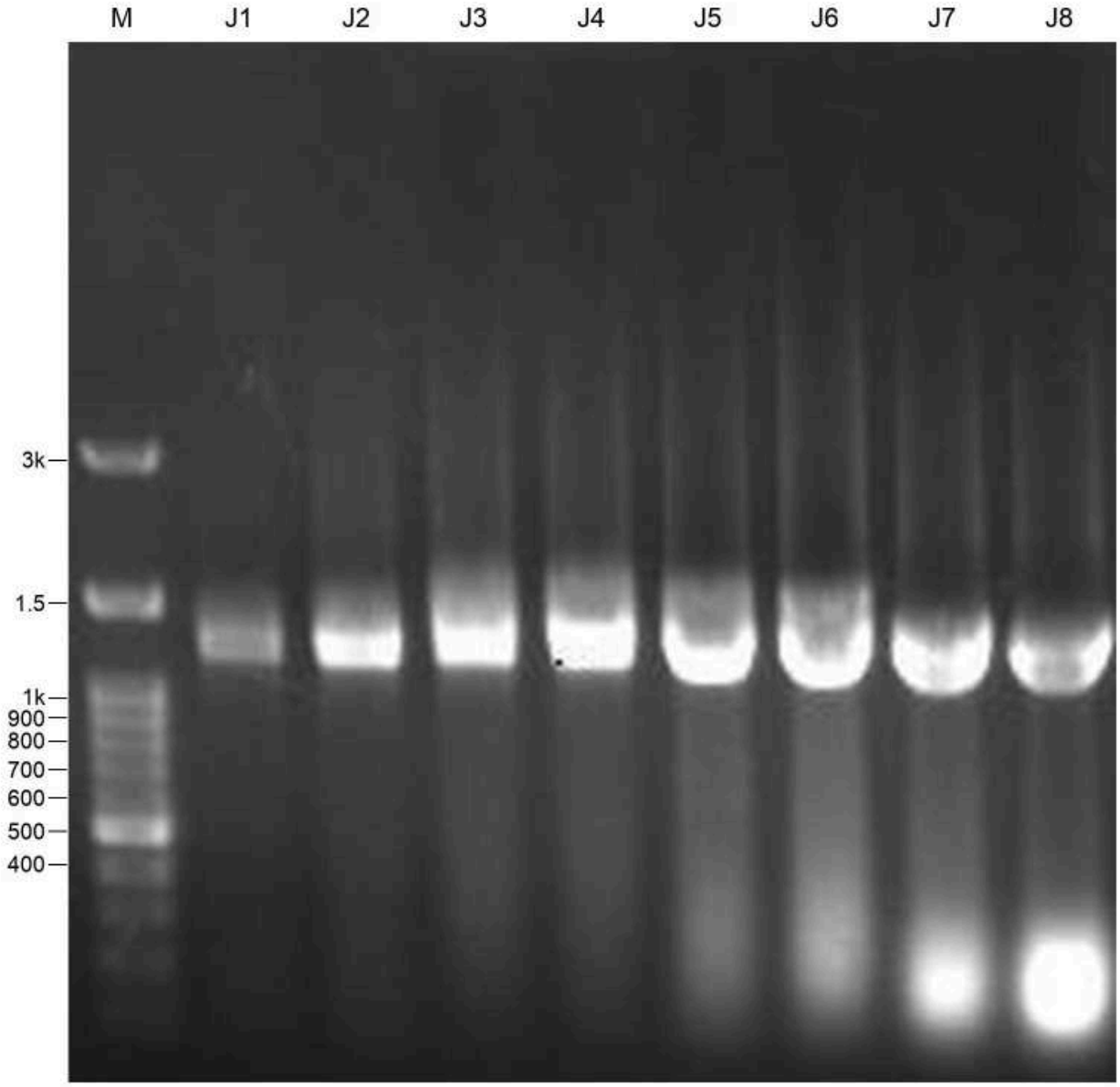
1. Assemble tray, madd, comb
2. Weigh 0.2 g of agarose powder and measure 20 mL 1xTAE buffer
3. Add all into a conical flask
4. Cover with plastic wrap&poke holes
5. Microwave for 30 sec agitate
6. Repeat until no sediments
7. When no smoke , add 1 μL safety dye agitate

8. Pour melted gel into mold
9. Let it solidify in RT

Done by Ryan Y, Sophie C

Date: 20260731
Title: Gel electrophoresis
Purpose: Check base pair of colony PCR product
Procedure:

1. Load the gel in the layout shown below (5 µL each):
2. Resolve at 100 v for 30 min



M: DNA ladder			
Plasmid Construct	Forward Primer	Reverse Primer	Expected Size (bp)
J: Pveg-EEBD-RBS0-Csn:ACCD-B0015	Csn F	ACCD R	1145(?)

Date: 20260731
Title: Transformation
Purpose: Transform and cultivate engineered bacteria on agar plate
Procedure:
G:PcpcG2-veg-EEBD

1. Prepare ice bucket and set the water bath to 42℃
2. Get LBA plates
3. Thaw competent cells until 50% thawed and add 10 μL plasmid
4. Incubate on ice for 20 min
5. Heat shock at 42 ℃ for 45 sec-1 min
6. Recover on ice for 3 min
7. Add 1mL SOC medium
8. Incubate at 37 ℃ for 1 hr
9. Centrifuge at 5000 g for 5min
10. Take out 1000 μL supernatant&resuspend pellet
11. Spread on LBA plate and incubate at 37 ℃ overnight

Done by Chloe